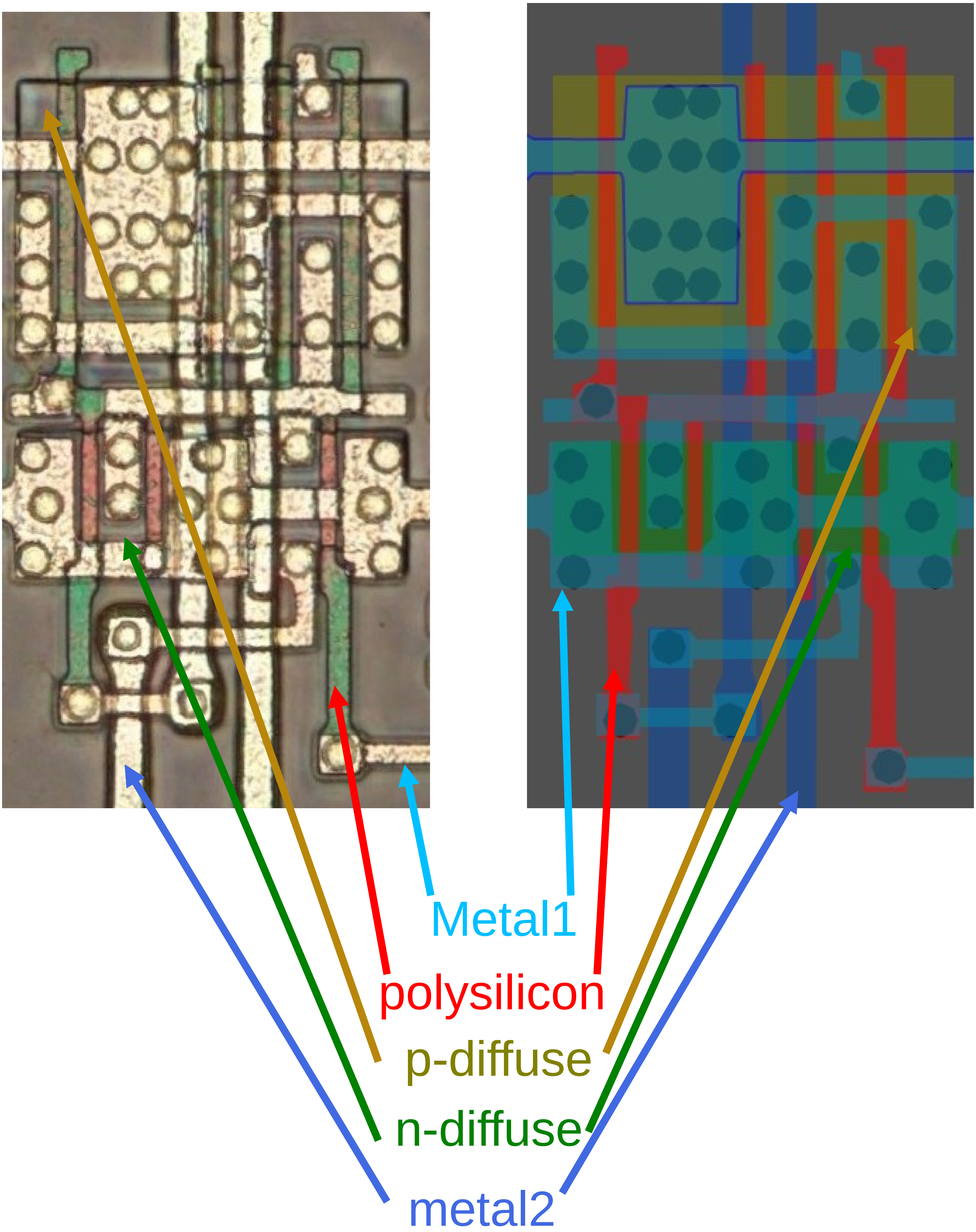
Document subject to change (for now)



Inferred rules for this 1986/87, 1.8 micron standard cell process called 'AV-OEBB'

(It is entirely possible the node size was something other than 1.8 micron)

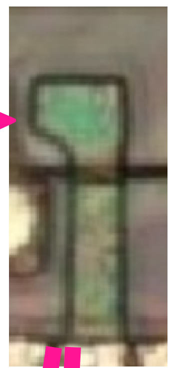
AV-OEBB Series **1.8u CMOS standard cell**

Fujitsu Micro-electronics	AV-OEBB Si-gate CMOS 1.8 µm	P1 6 µm M1 7 µm M2 9 µm	100	n/s	1.2	140 digital	RAM, ROM, PLA, multipliers, ALU	•	•	•	•	•	None

- Cells can be flipped and mirrored with minimal adjustments.
- Gate height can extend up/down y-axis for the purposes of routing.
- Cell via routing seems unique to each cell type. Good for finding them easily
- These polysilicon ends are where m1 *can be* connected

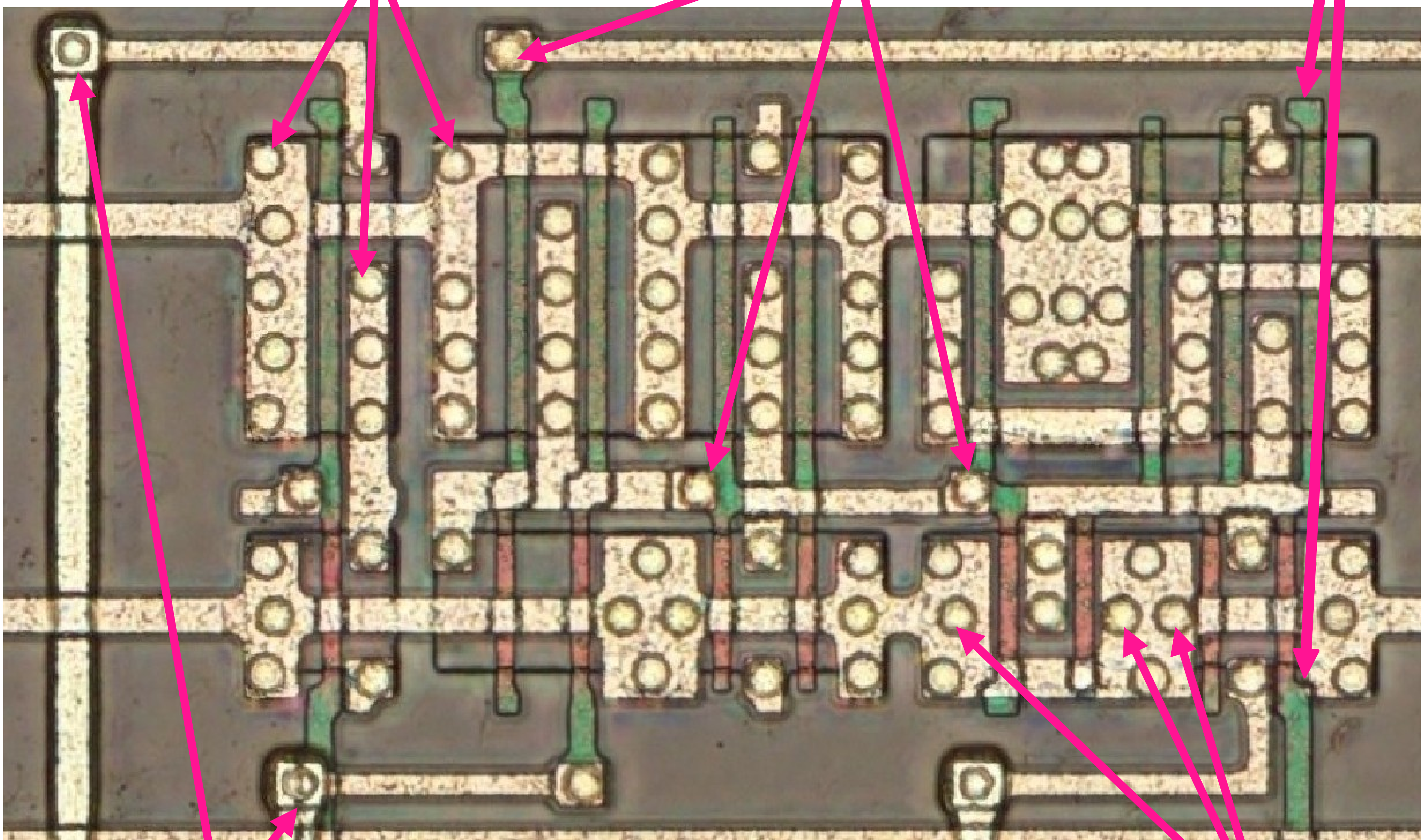
***Tip**: To understand whether the diffuse is PMOS or NMOS, you need to identify the polysilicon gate and its color:*

The gate is green in PMOS, but red in NMOS.



m1 to P-MOS diffuse via:

m1 to polysilicon via:

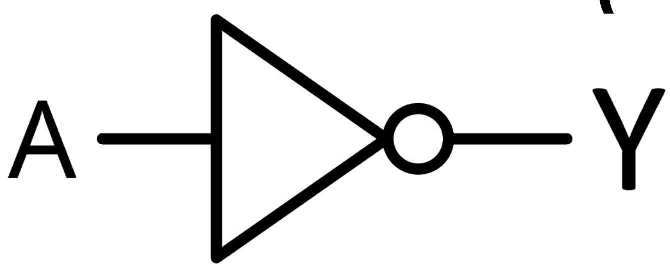


m1 to m2 via:
Generally has smaller vias

m1 to N-MOS diffuse via:

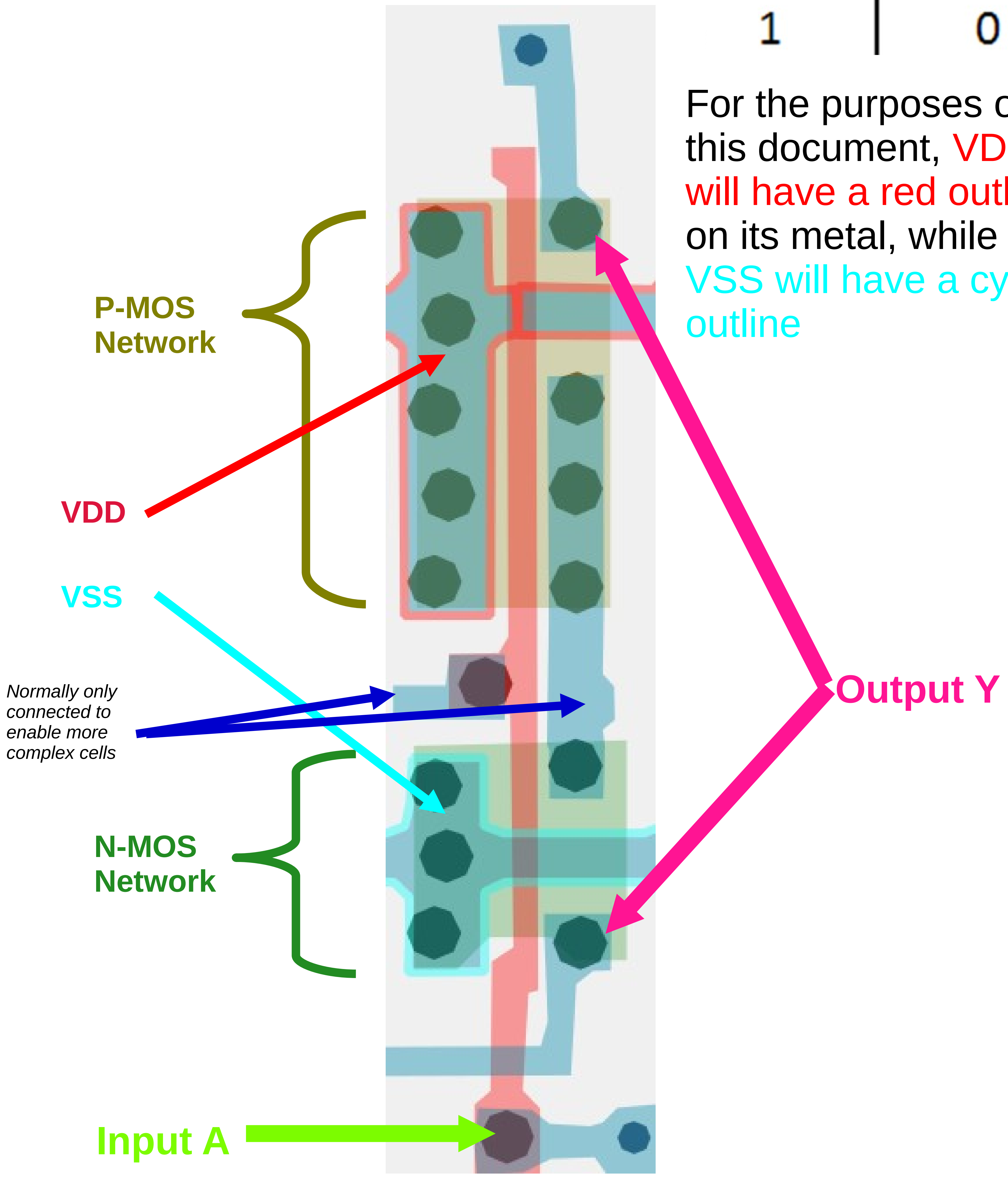
How it all works

(Inverter/NOT example)



Input	Output
A	Y
0	1
1	0

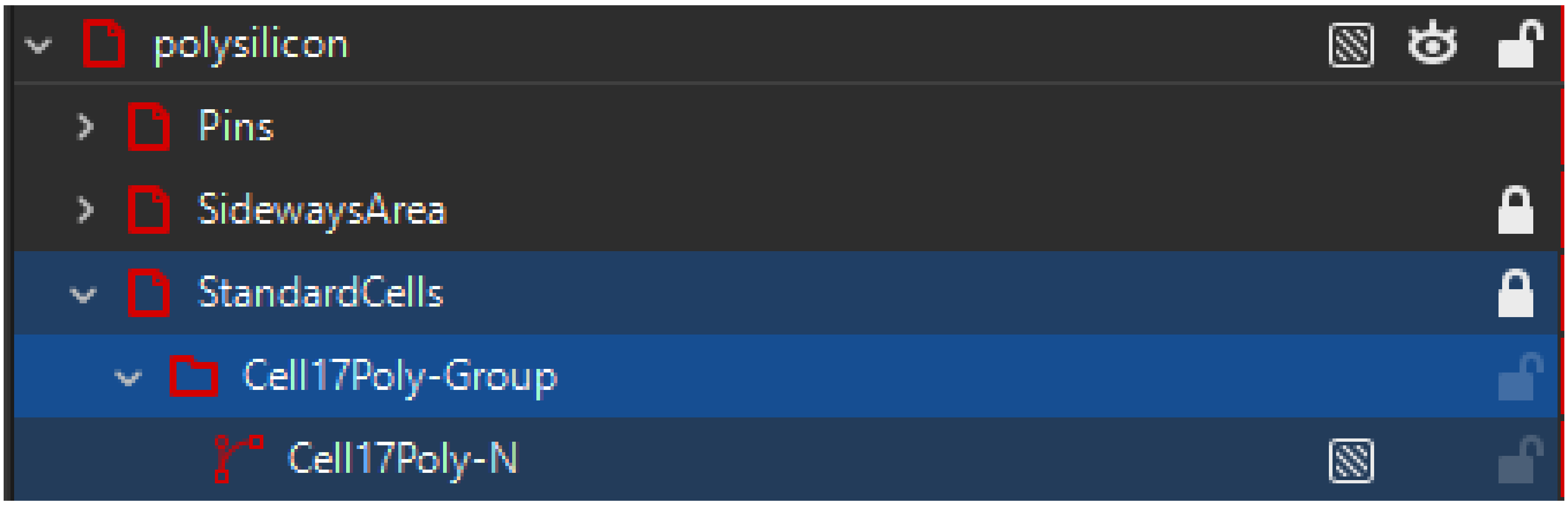
For the purposes of this document, VDD will have a red outline on its metal, while VSS will have a cyan outline



Object naming and how to understand

I have done my best to make object naming as simple as possible while keeping object names practical.

PDi – P-Diffusion
NDi – N-Diffusion
Poly – Polysilicon



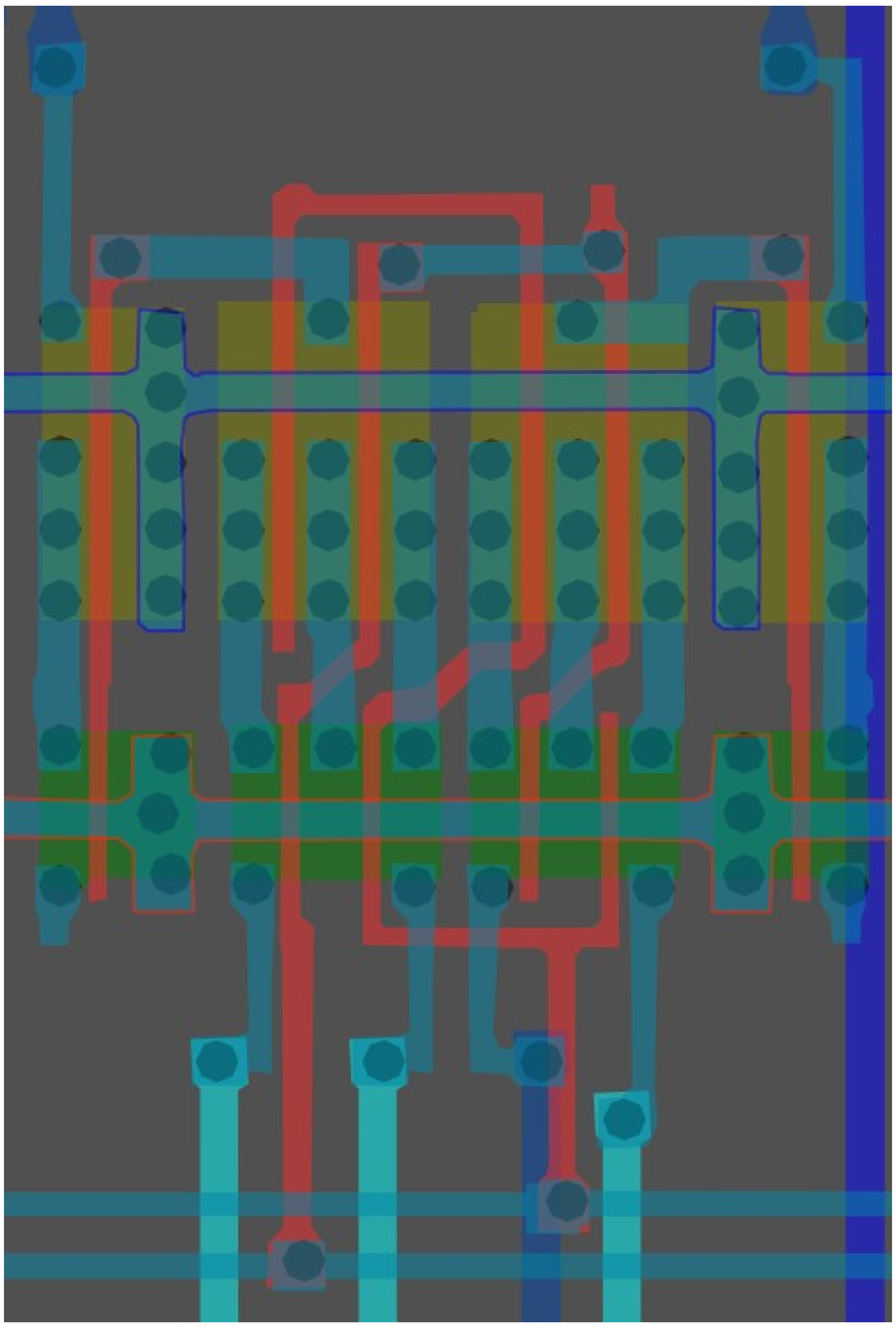
Cell 17 in standard cell area, Polysilicon layer, sub-object N.

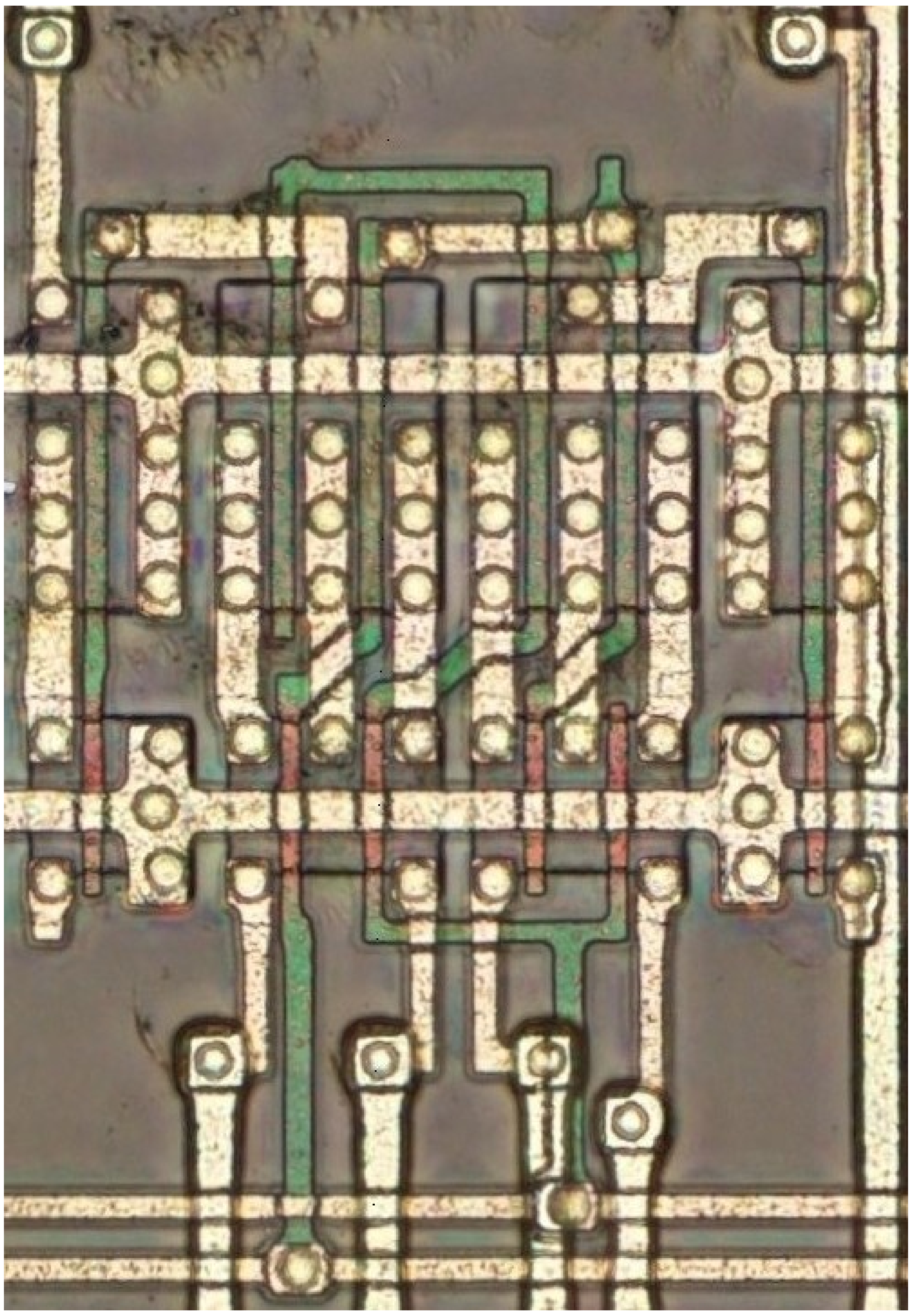
Cell number index

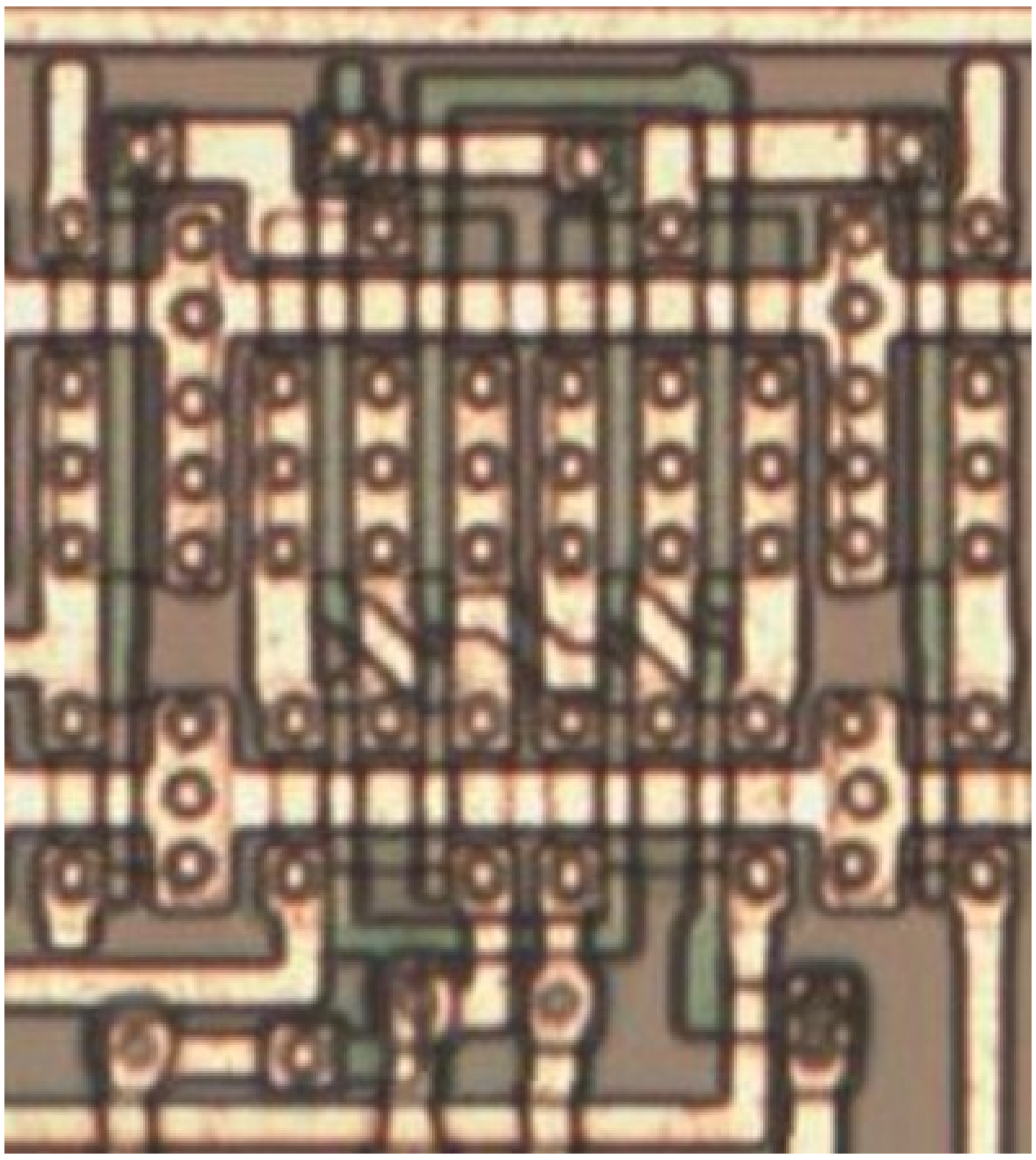
Green shaded cells mean they have been verified

1	**2**	**3a**	**3b**	**4**	**5**	**6**
7	**8**	**9**	**10**	**11**	**12**	**13a**
13b	**13c**	**13d**	**13e**	**13f**	**14**	**15**
16	**17**	**18**	**19**	**20**	**21a**	**21b**
24a	**24b**	**25**	**26**	**27a**	**27b**	**27c**
27d	**28**	**29**	**30**	**31**	**32**	**33a**
33b	**34**	**35**	**36a**	**36b**		

#f321a5ff Cell type 1



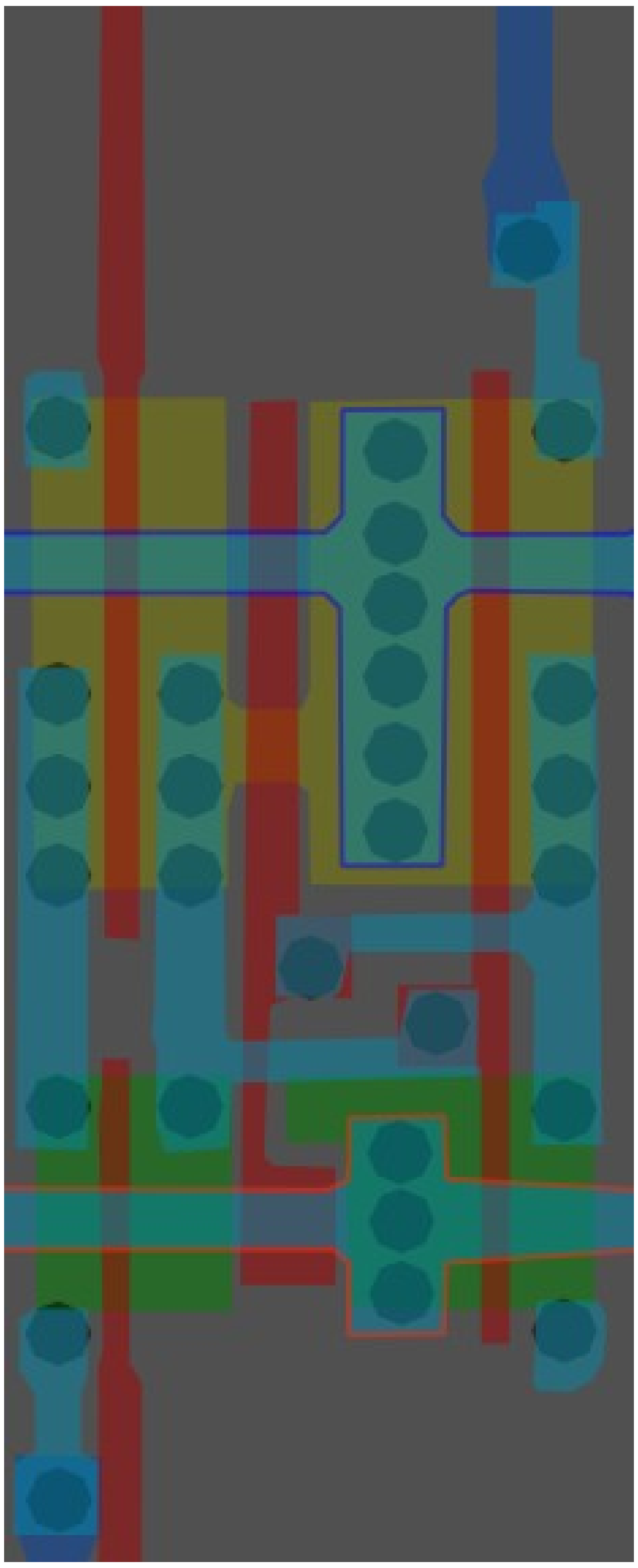




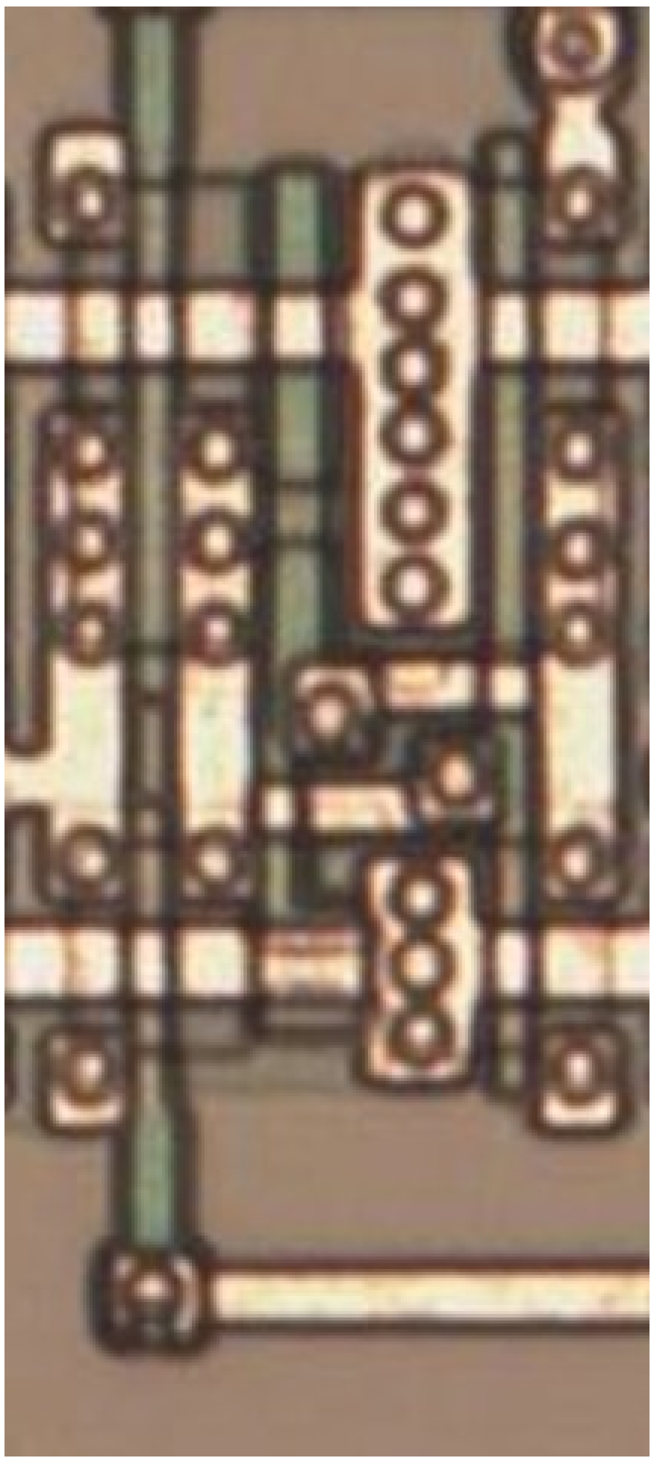
Same cell on LA32 (mirrored) (MB87136A)

#f34f00ff

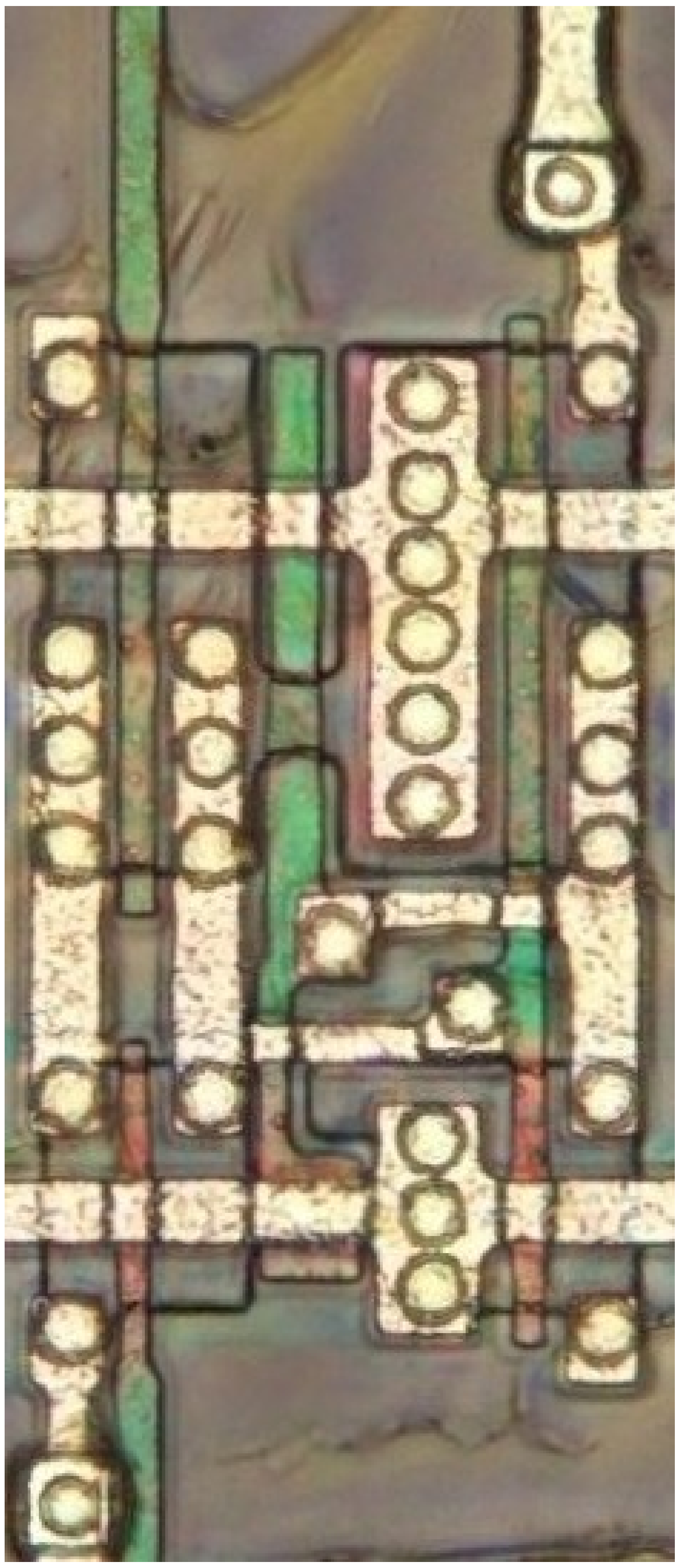
Cell 2



Maybe an SR latch?

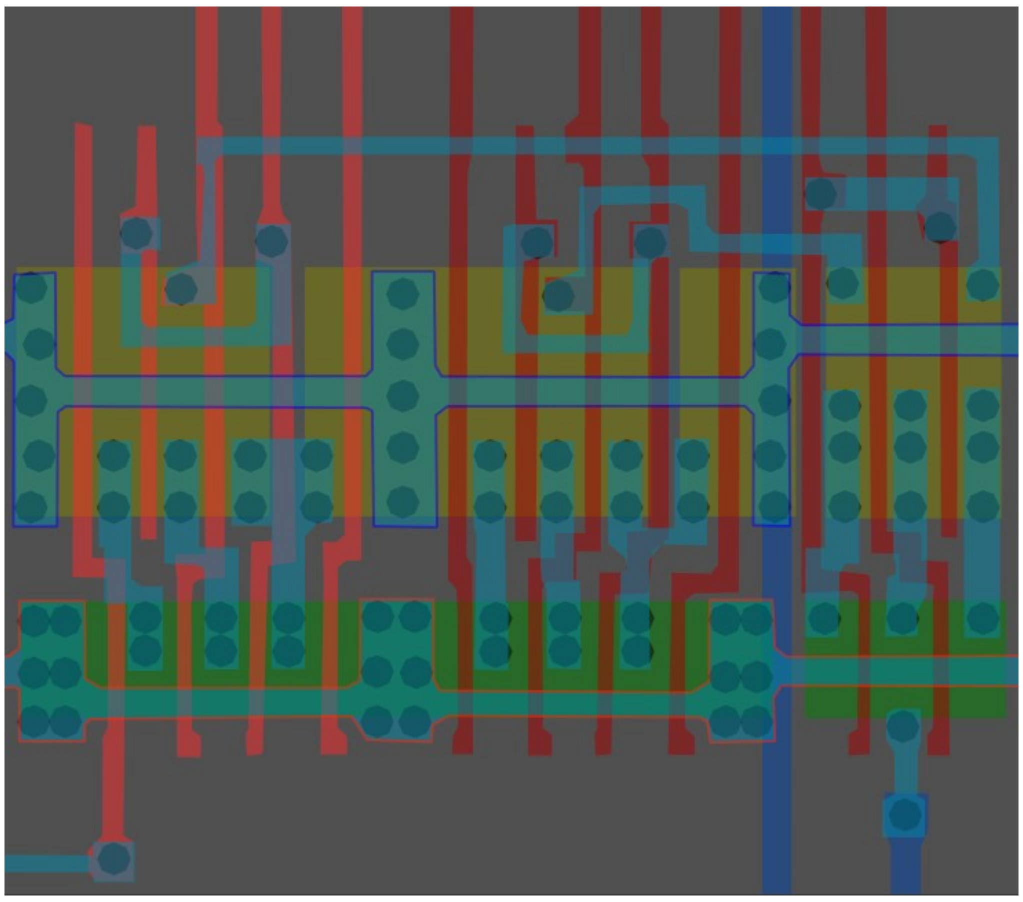


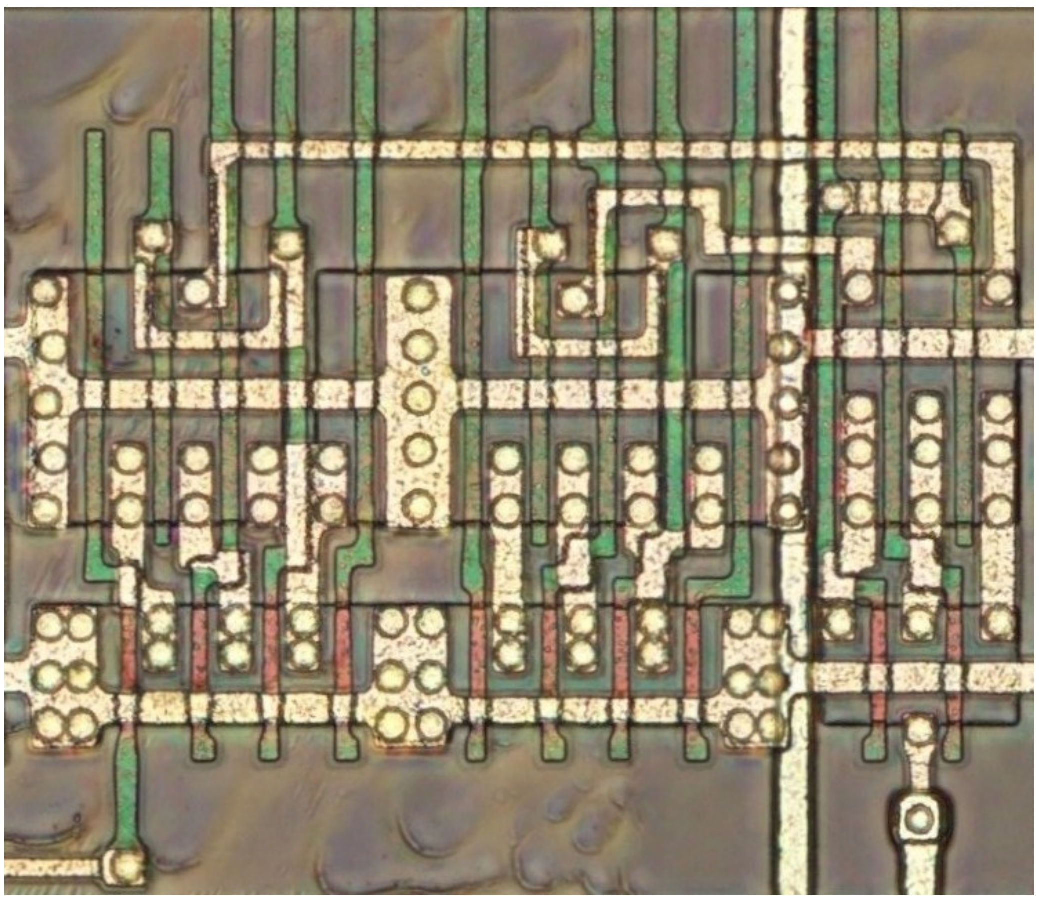
Same cell on LA32 (MB87136A)



#ffff00ff

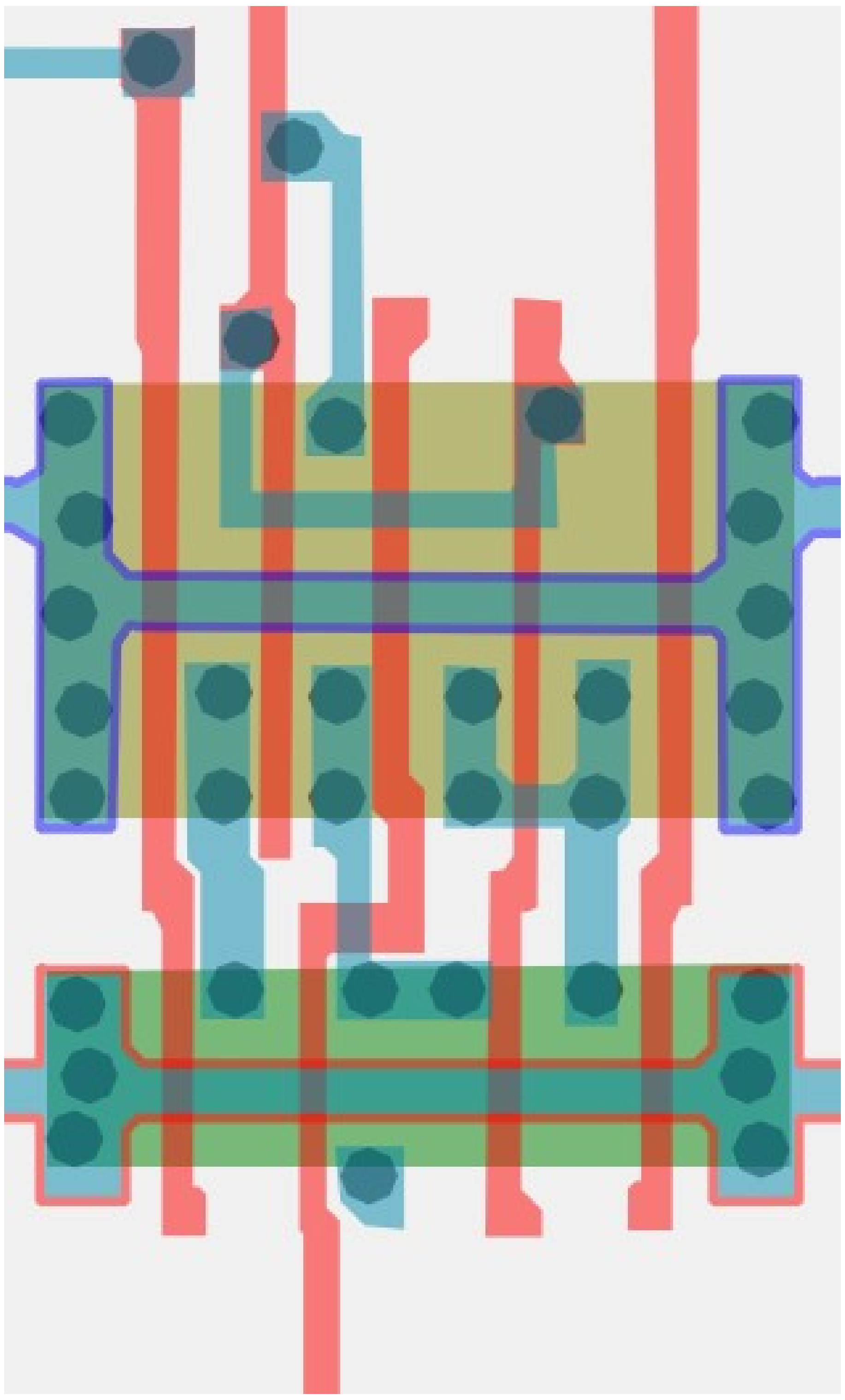
Cell type 3a

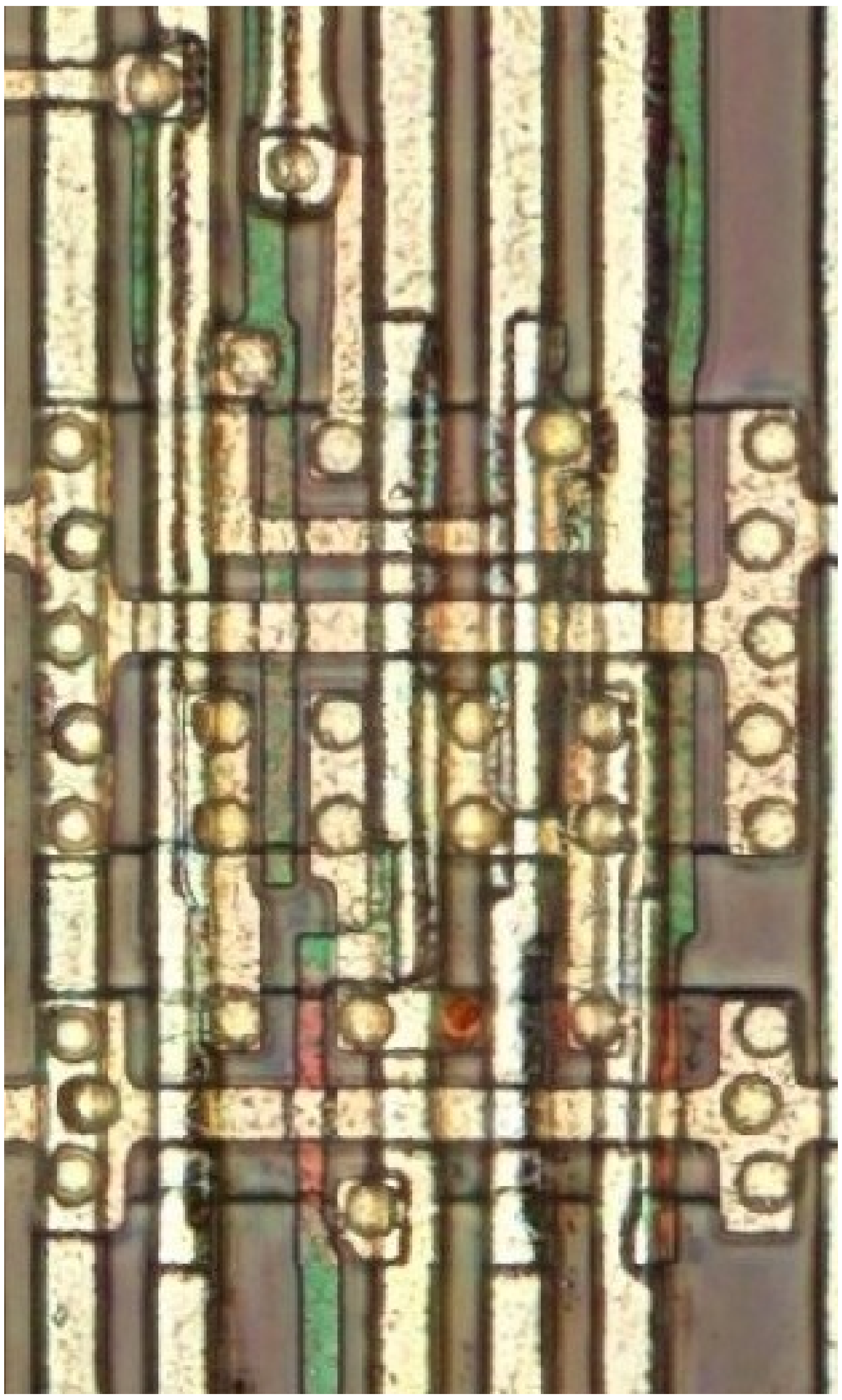




Unable to find on LA32

#b29e31ff Cell type 3b

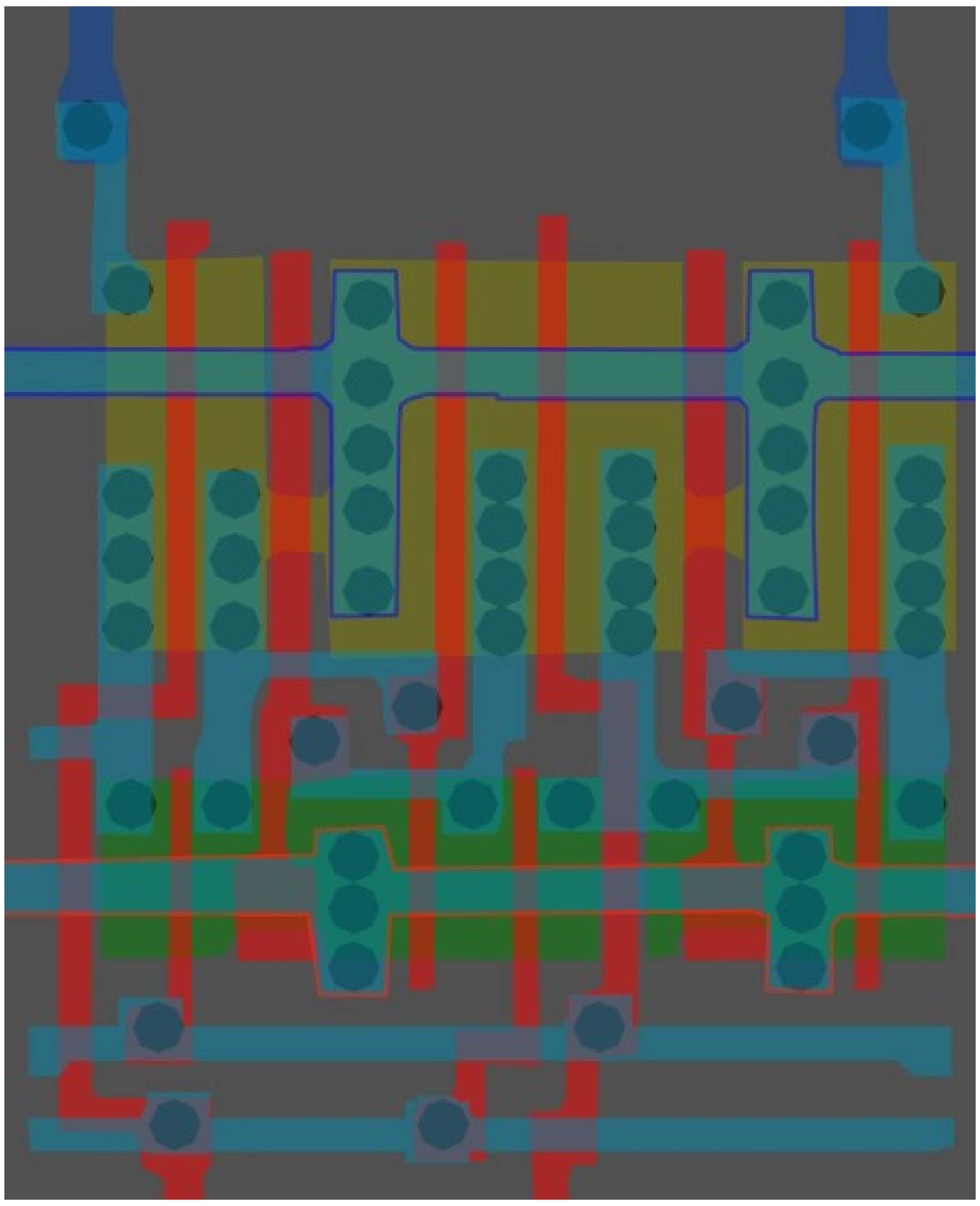


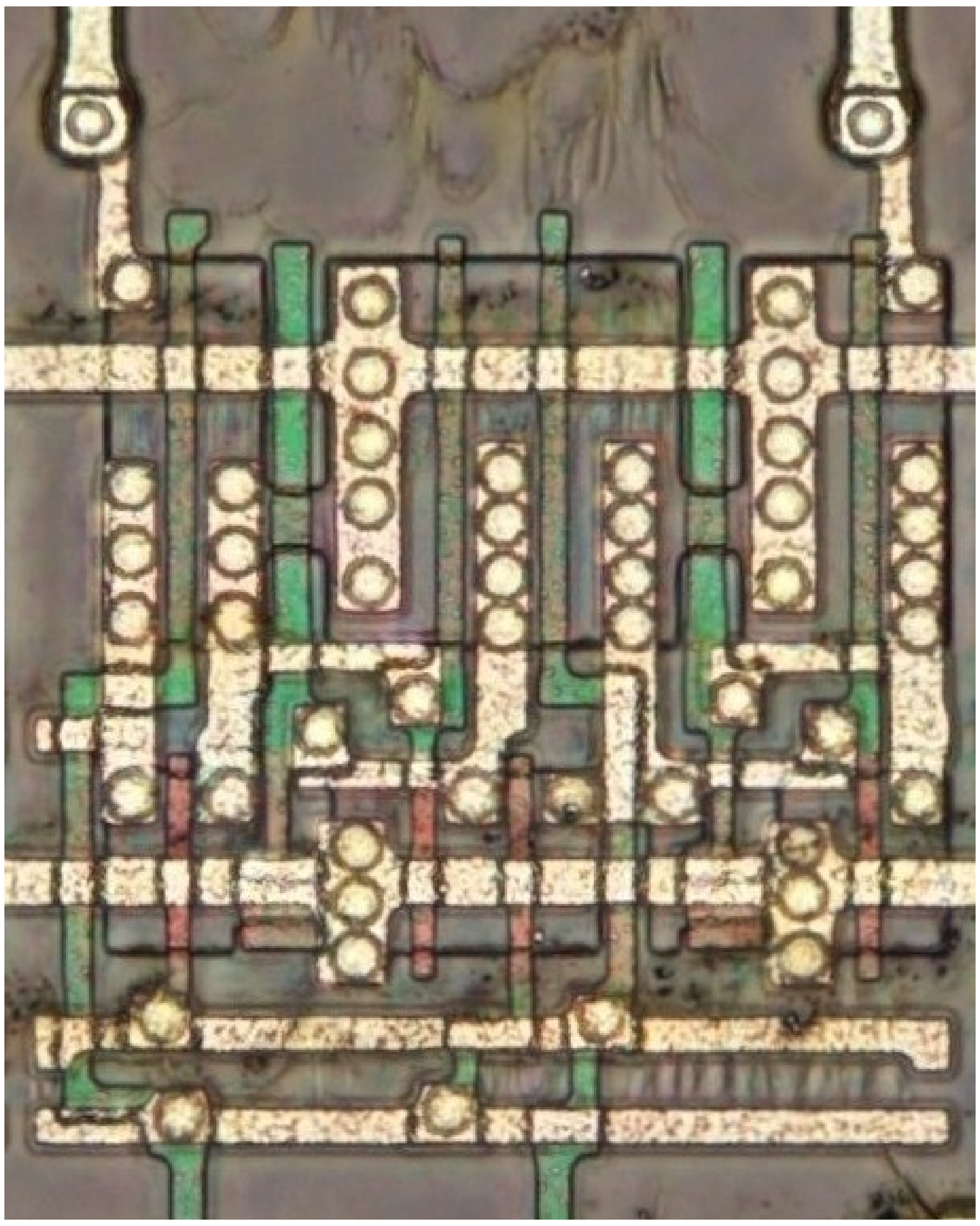


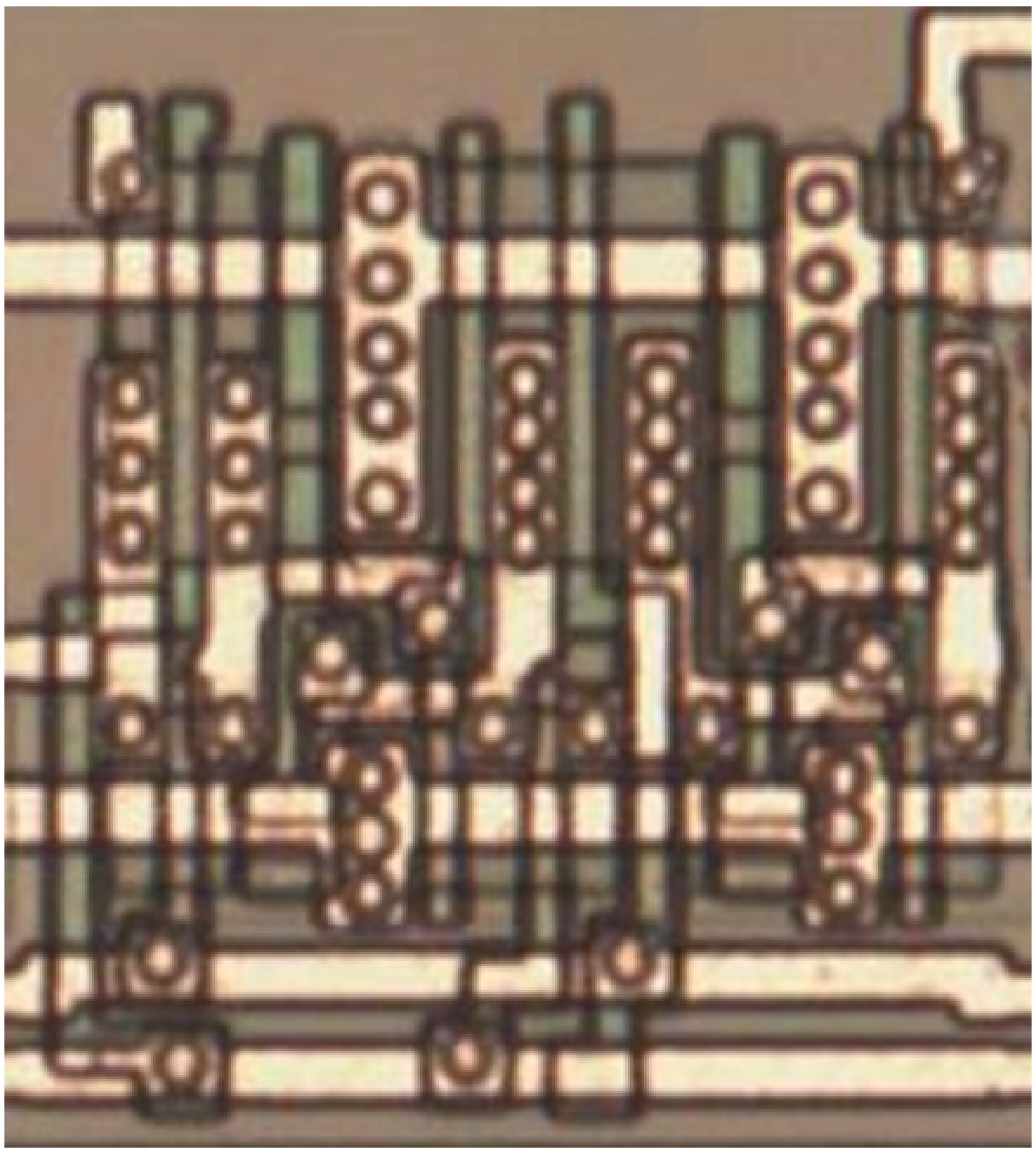
Looks like a small segment of 3a,
but altered slightly

#00ff00ff Cell 4

Double or ‘x2’ drive
version of Cell 2





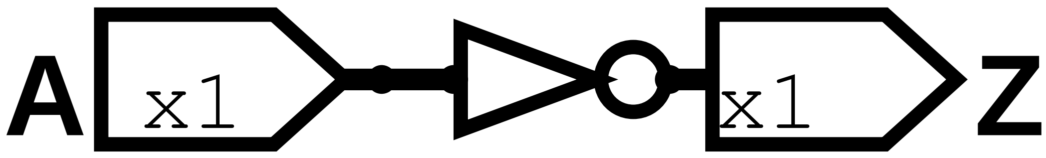


Same cell on
LA32
(MB87136A)

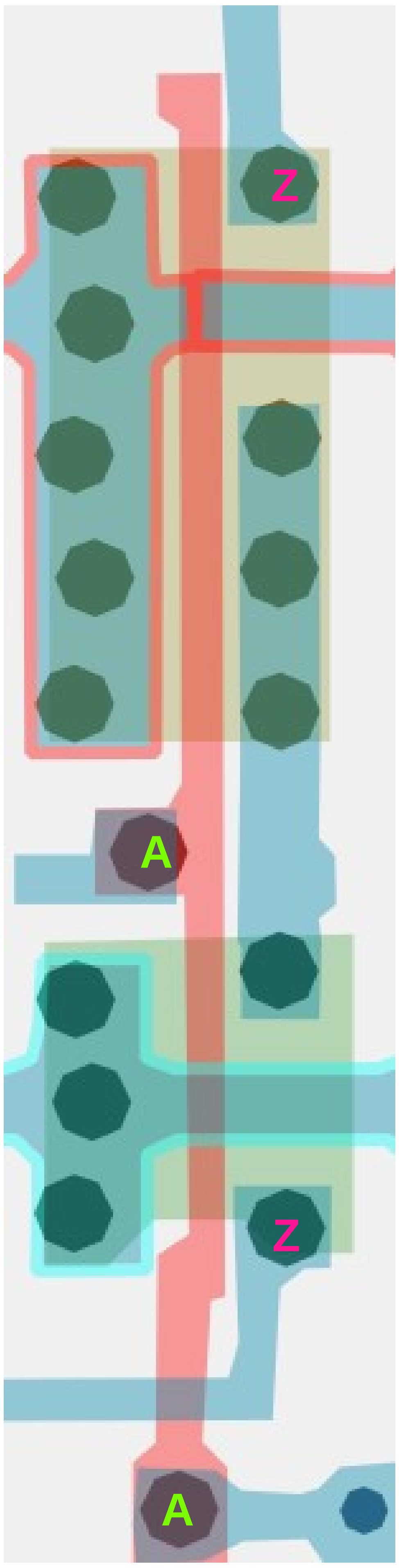
#51215aff

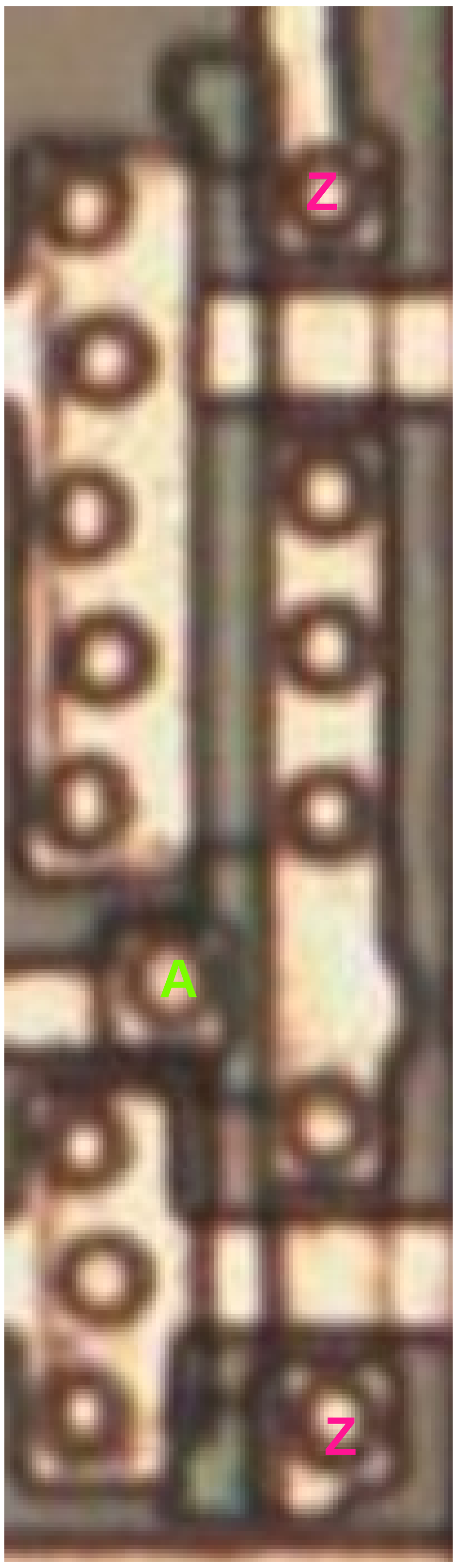
Inverter/NOT x1

Cell 5



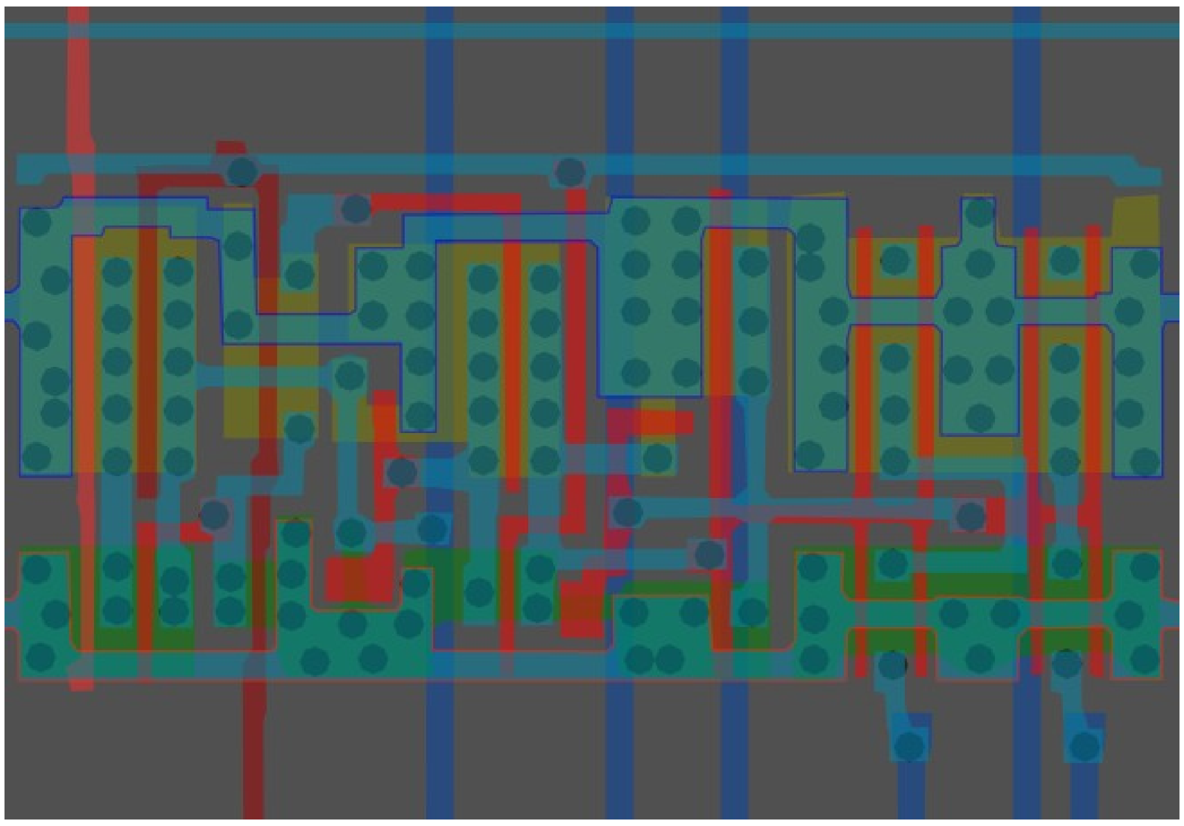
A	Z
0	1
1	0

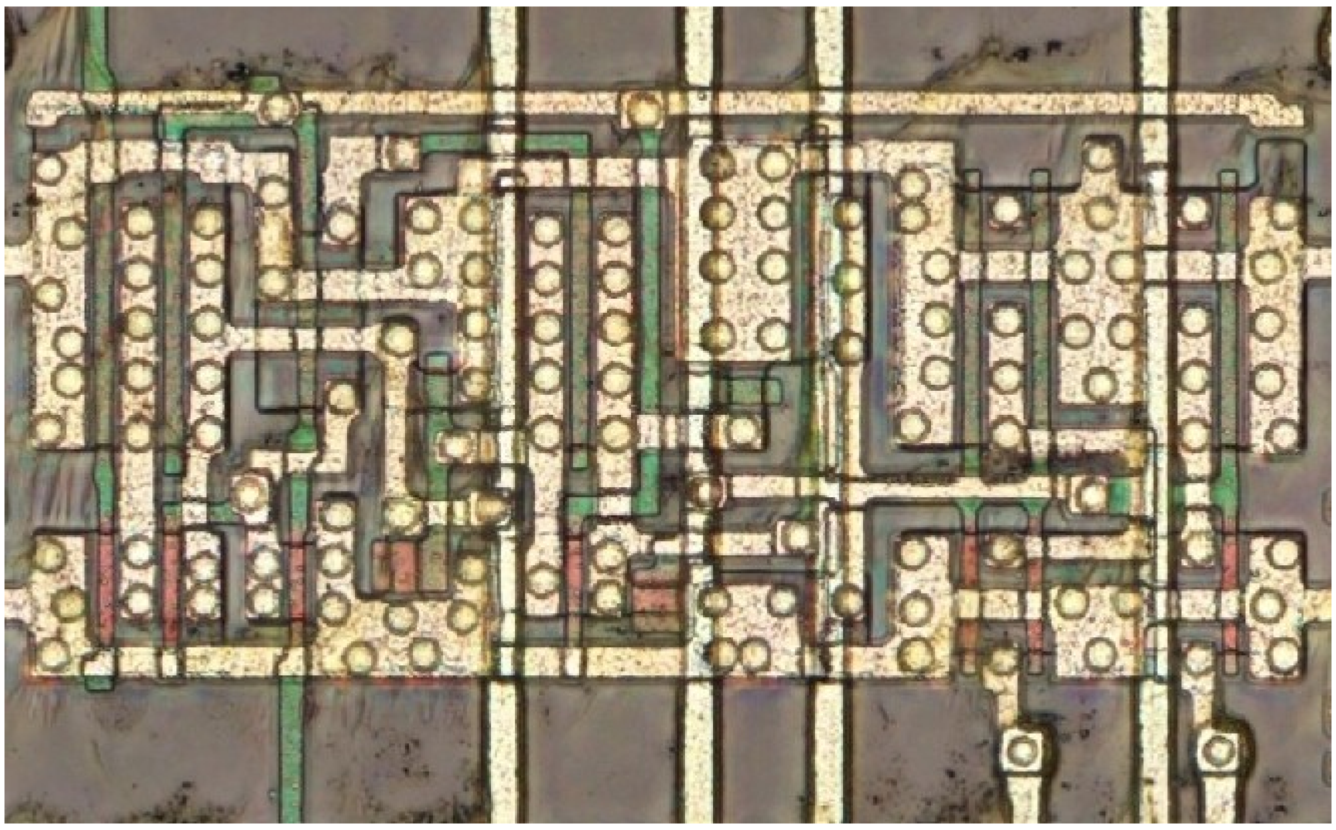




Same cell on LA32 (MB87136A)

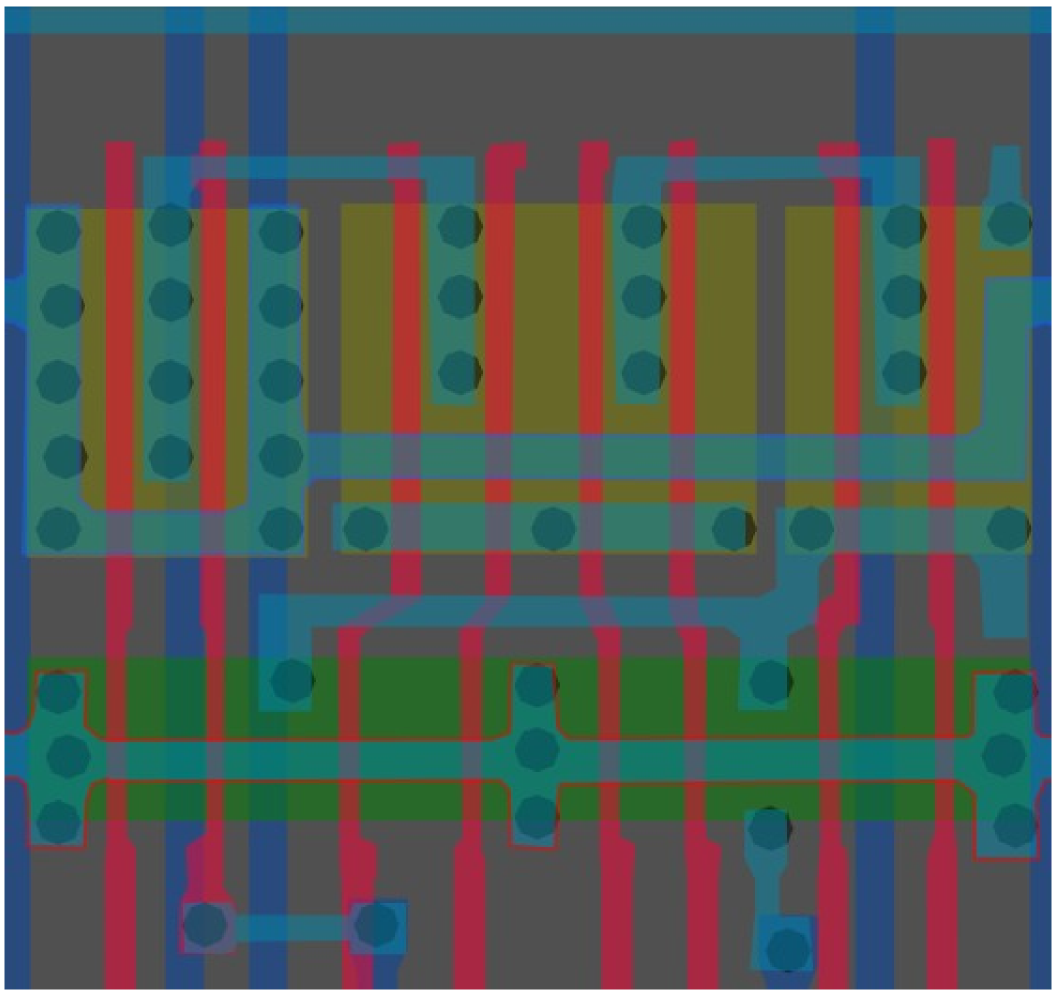
Cell type 6

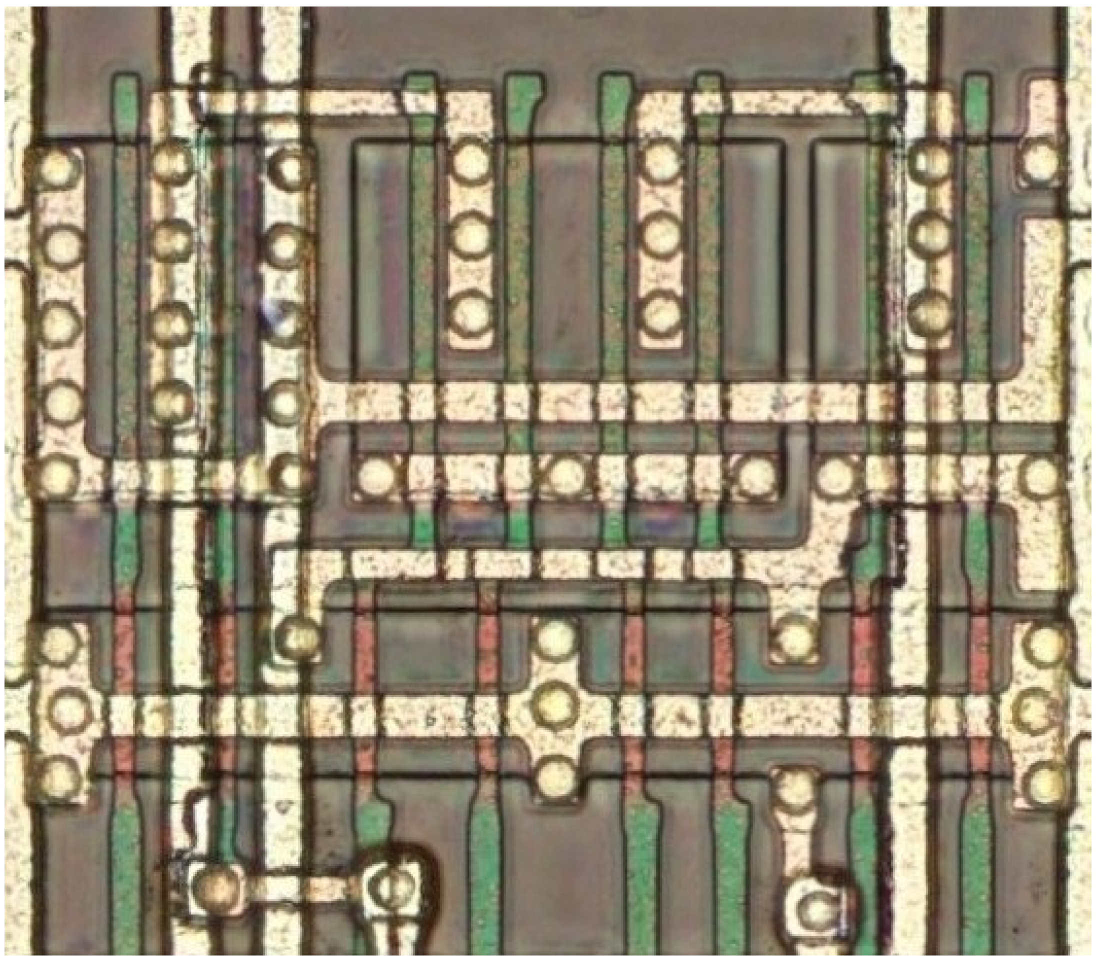


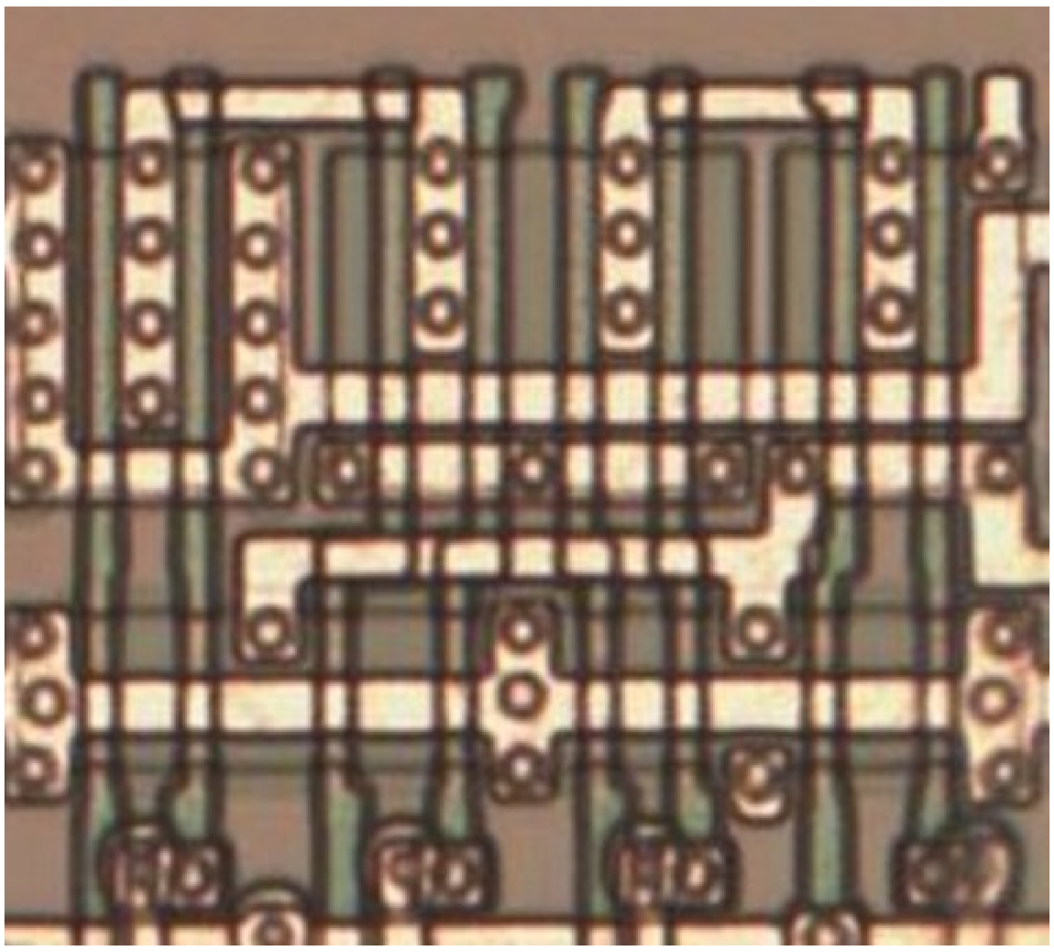


#13215aff

Cell type 7

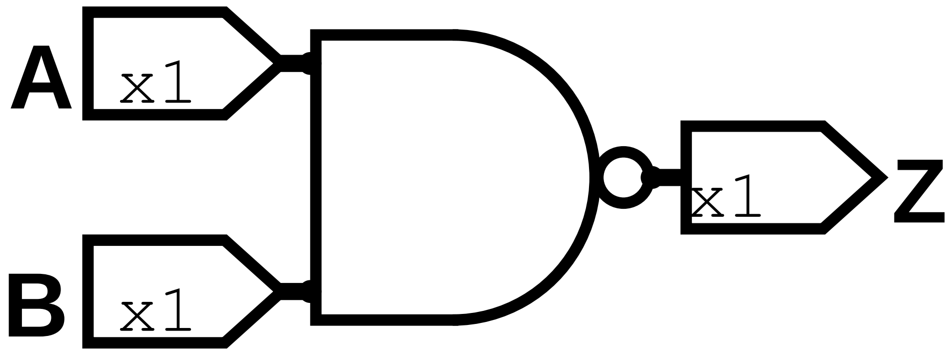




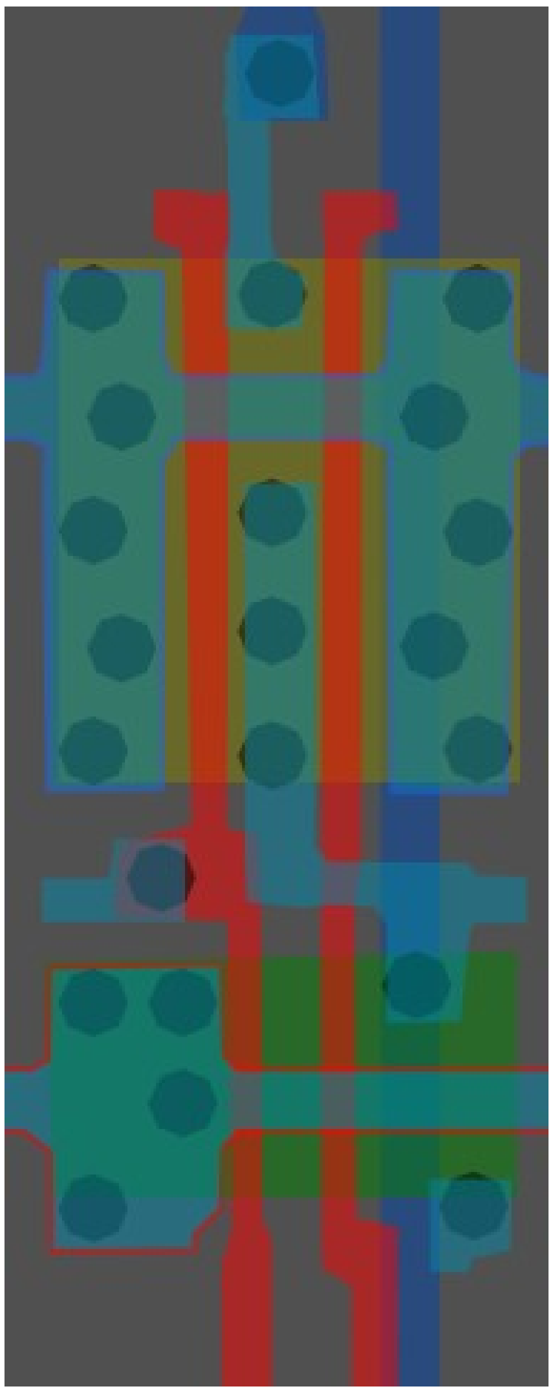


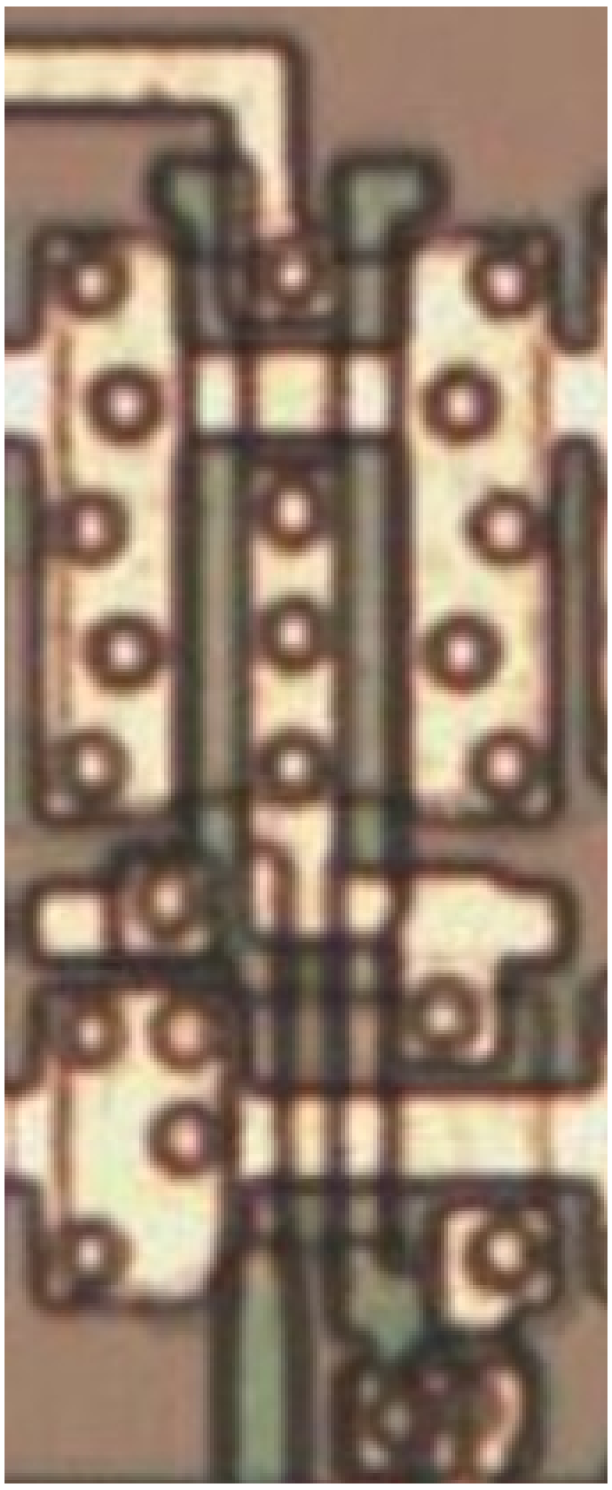
Same cell on
LA32
(MB87136A)

#89ff86ff 2-Input NAND Cell 8

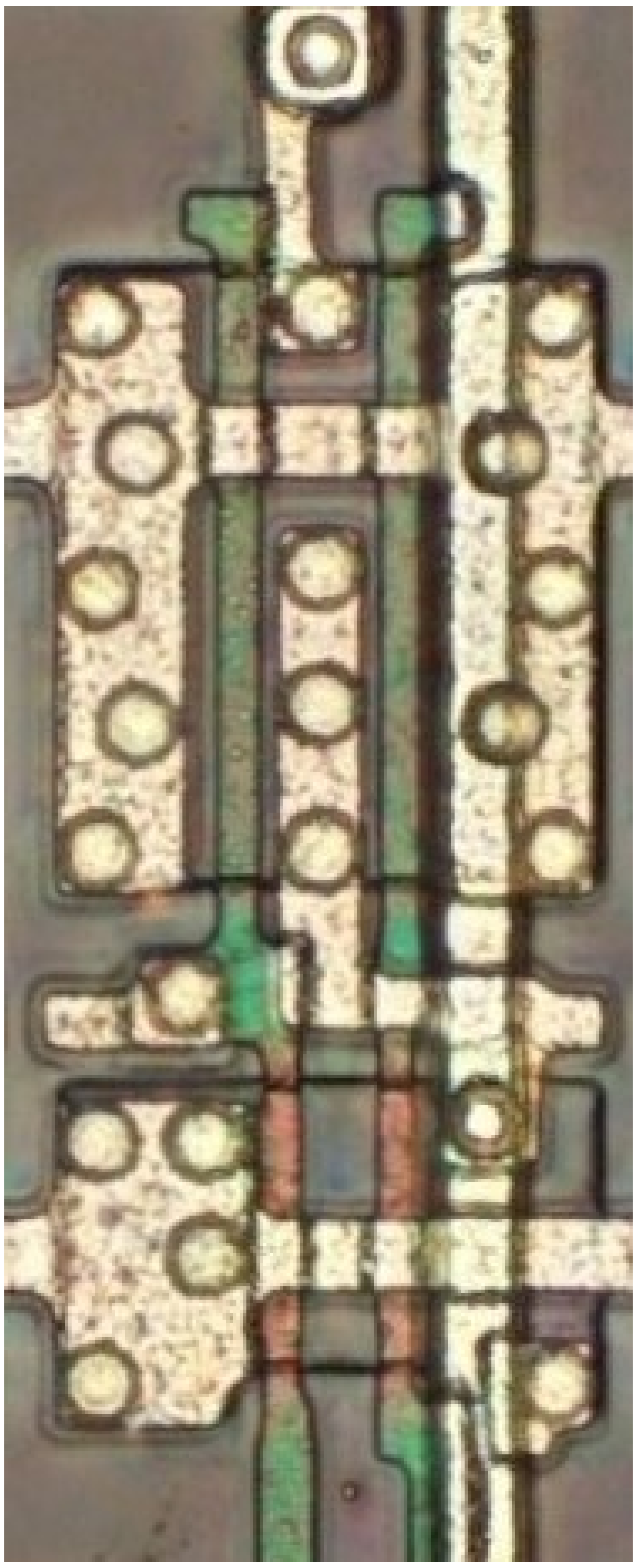


A	B	X
0	0	1
0	1	1
1	0	1
1	1	0





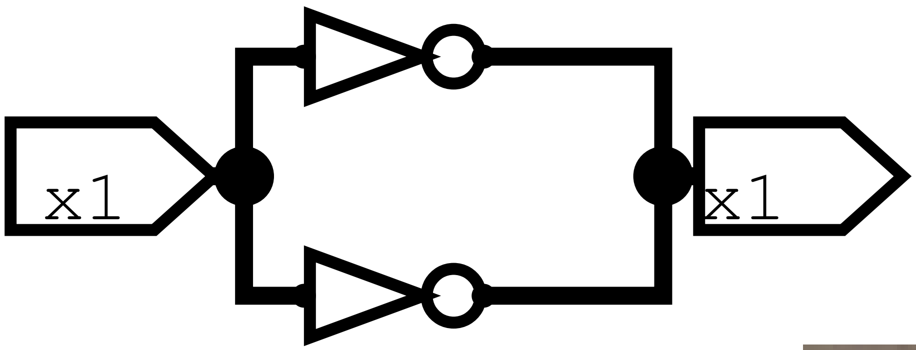
Same cell on
LA32
(MB87136A)

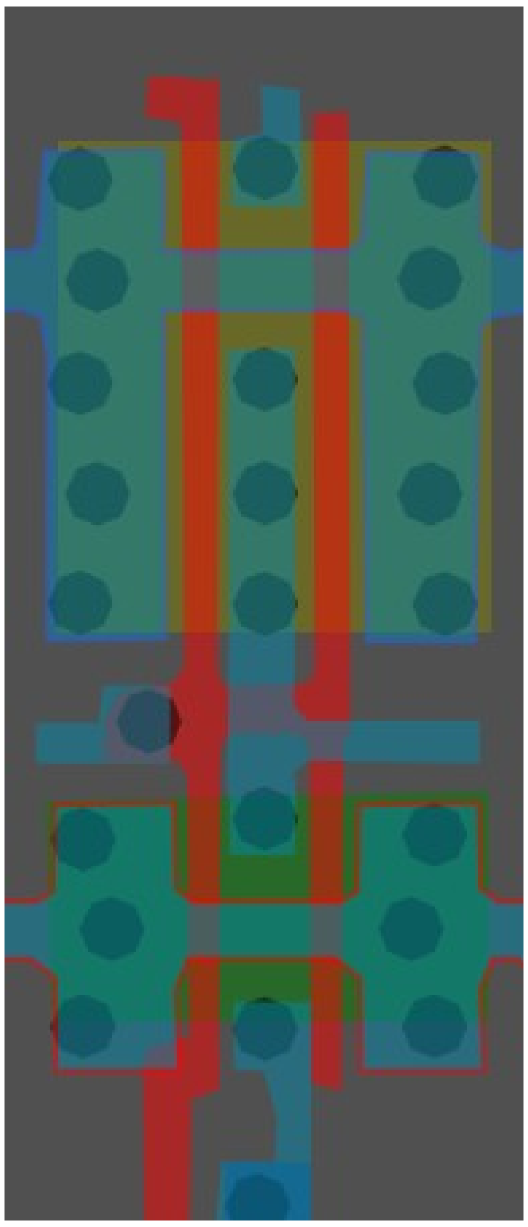


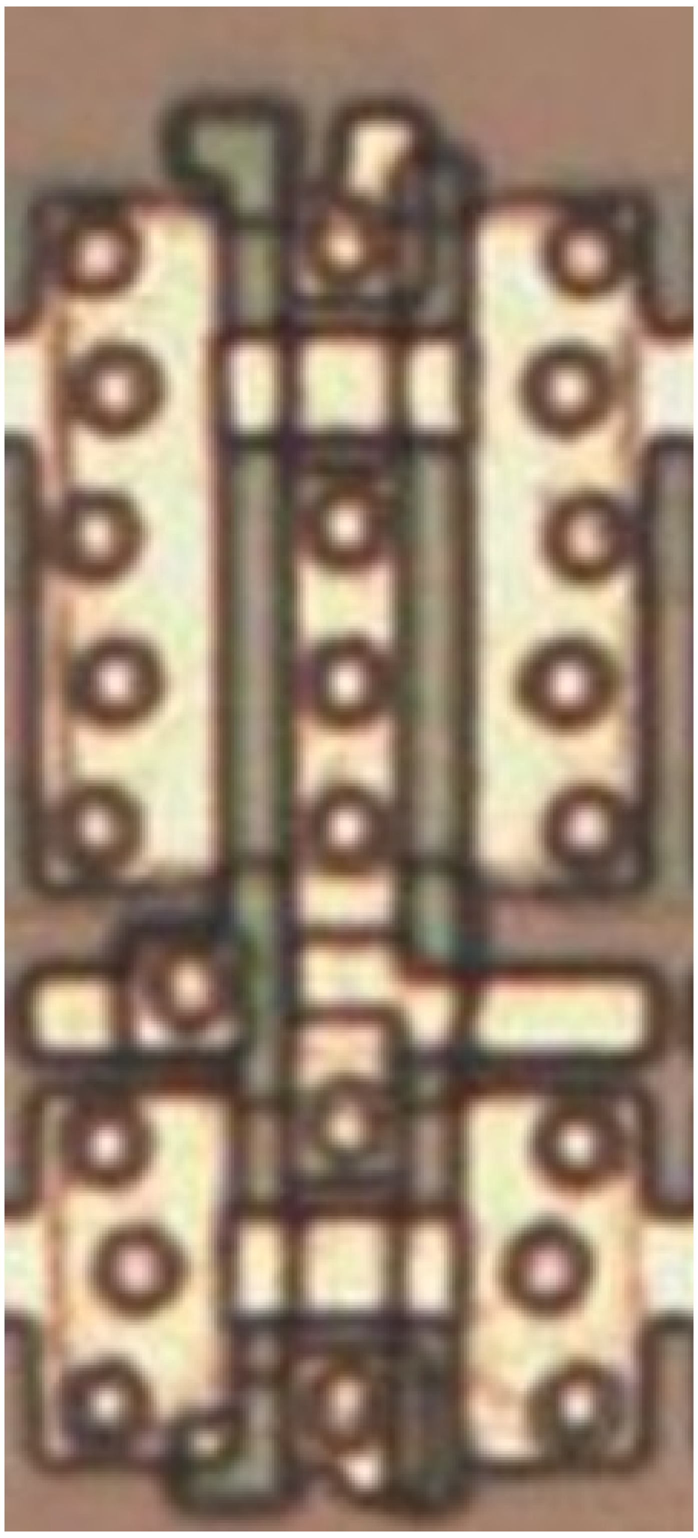
#393c3cff

Inverter/NOT X2

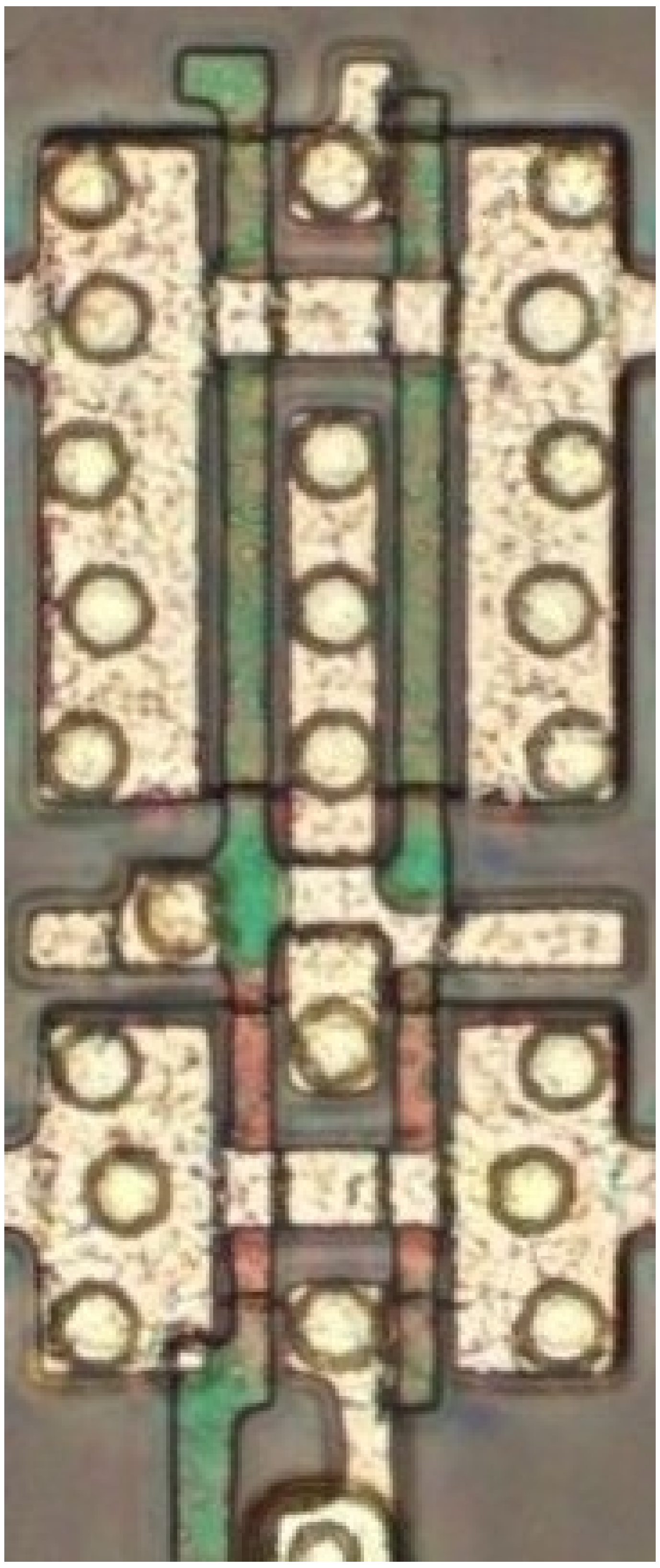
Cell 9



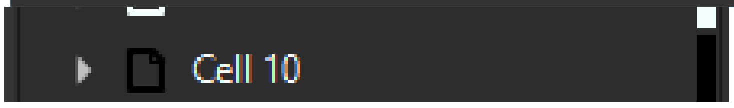


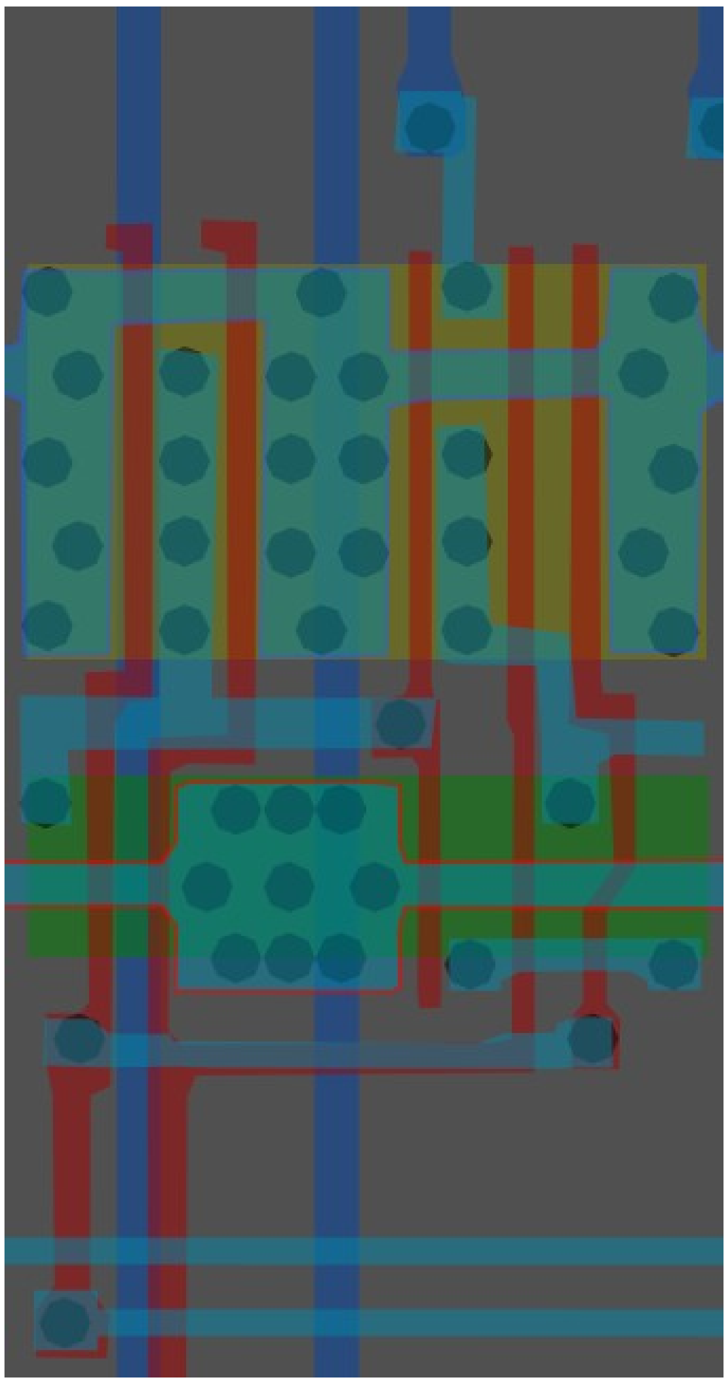


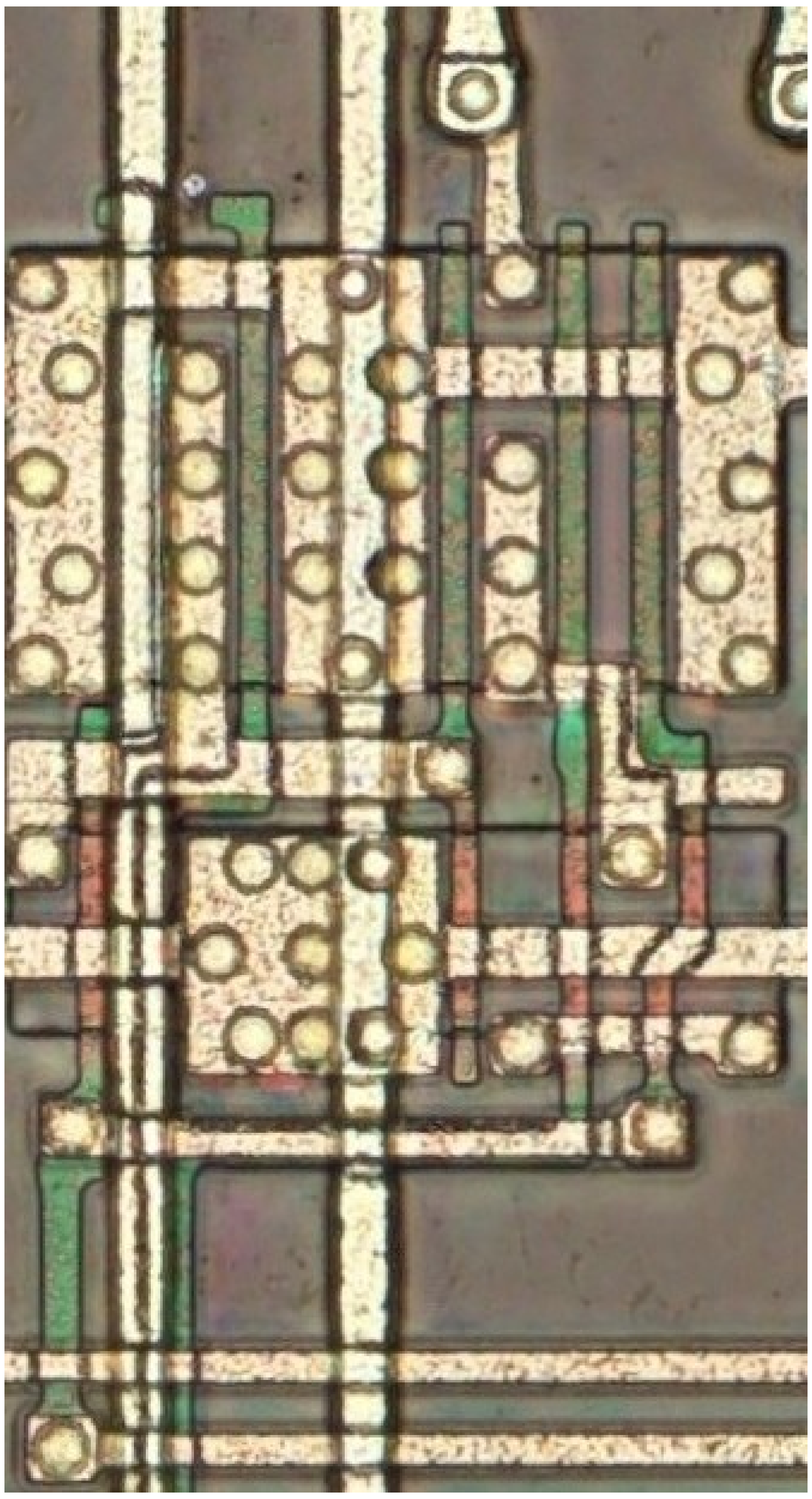
Same cell on
LA32
(MB87136A)

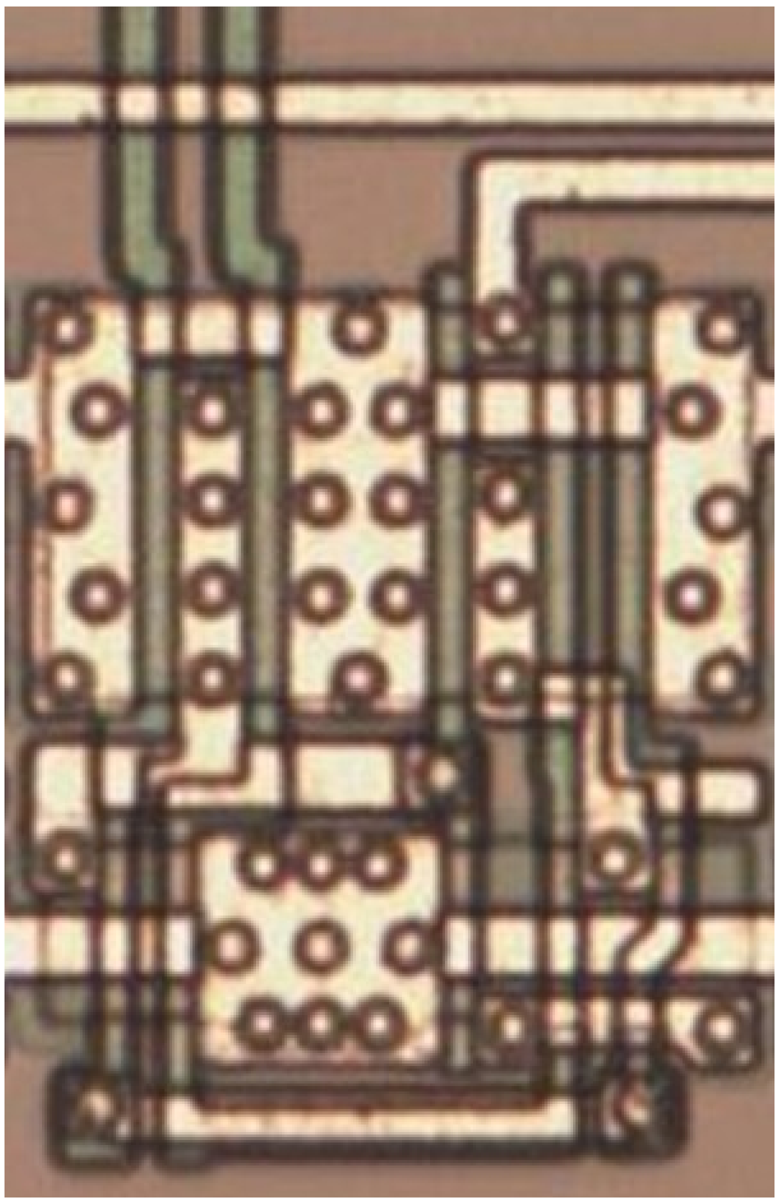


#000000ff Cell type 10



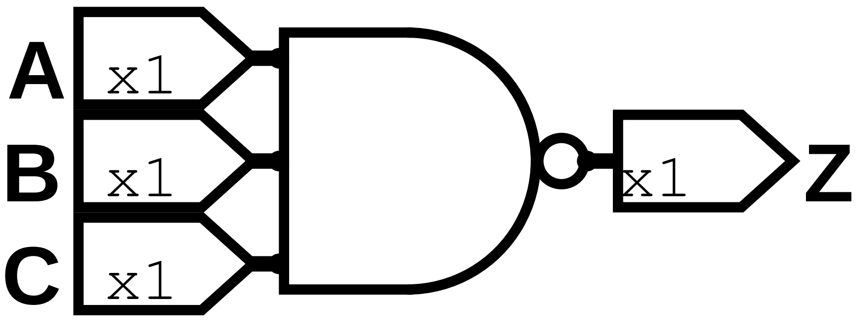




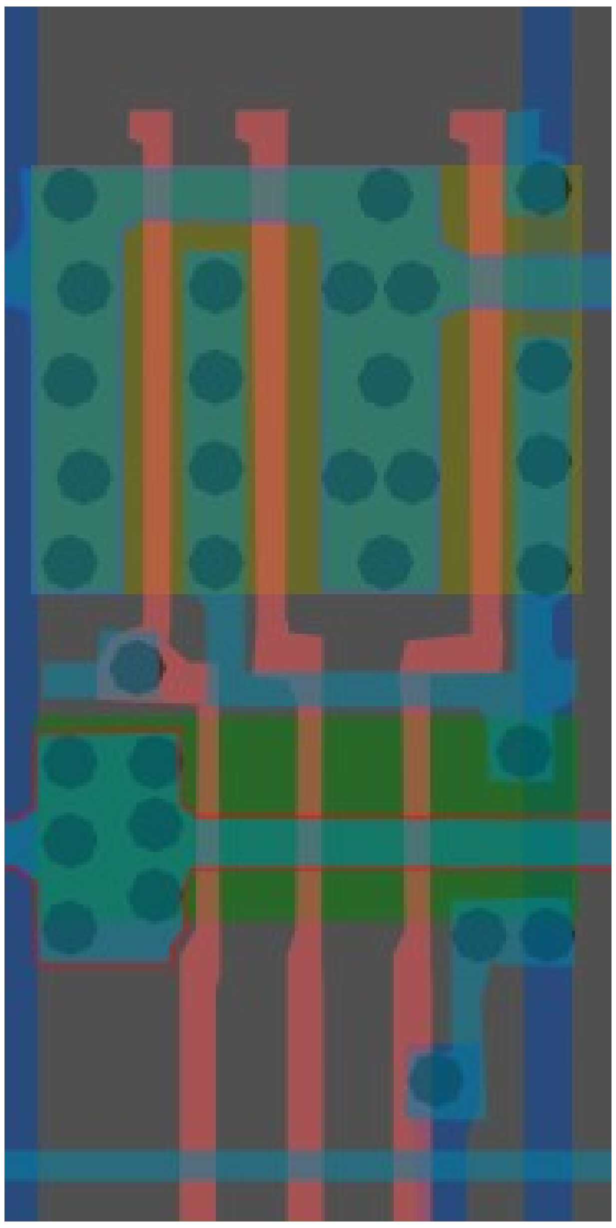


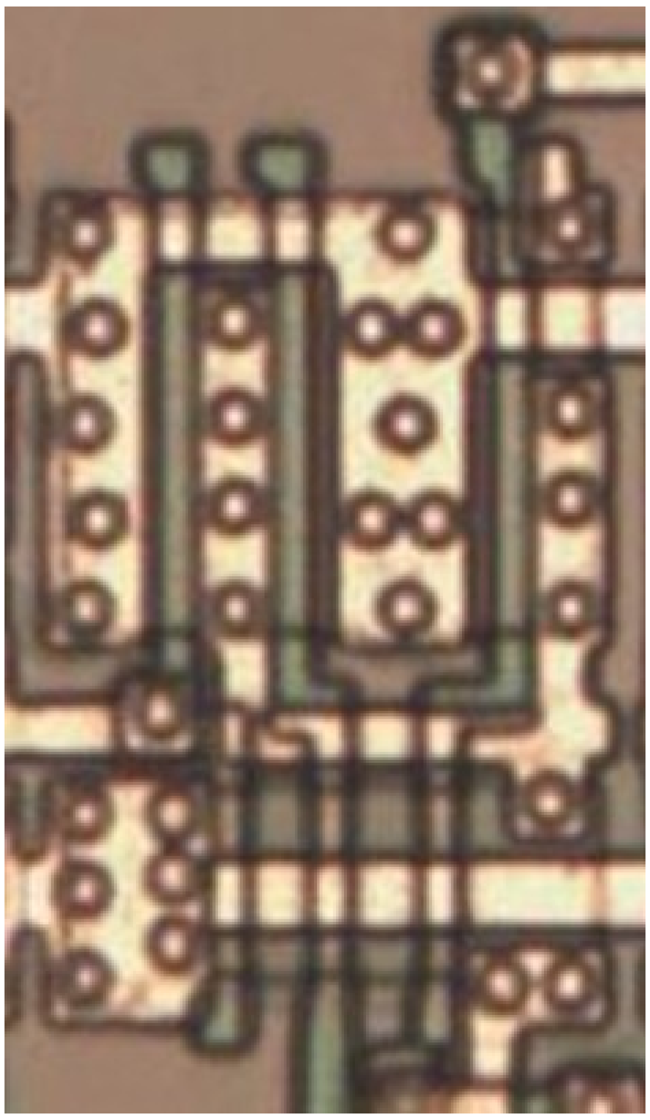
Same cell on
LA32
(MB87136A)

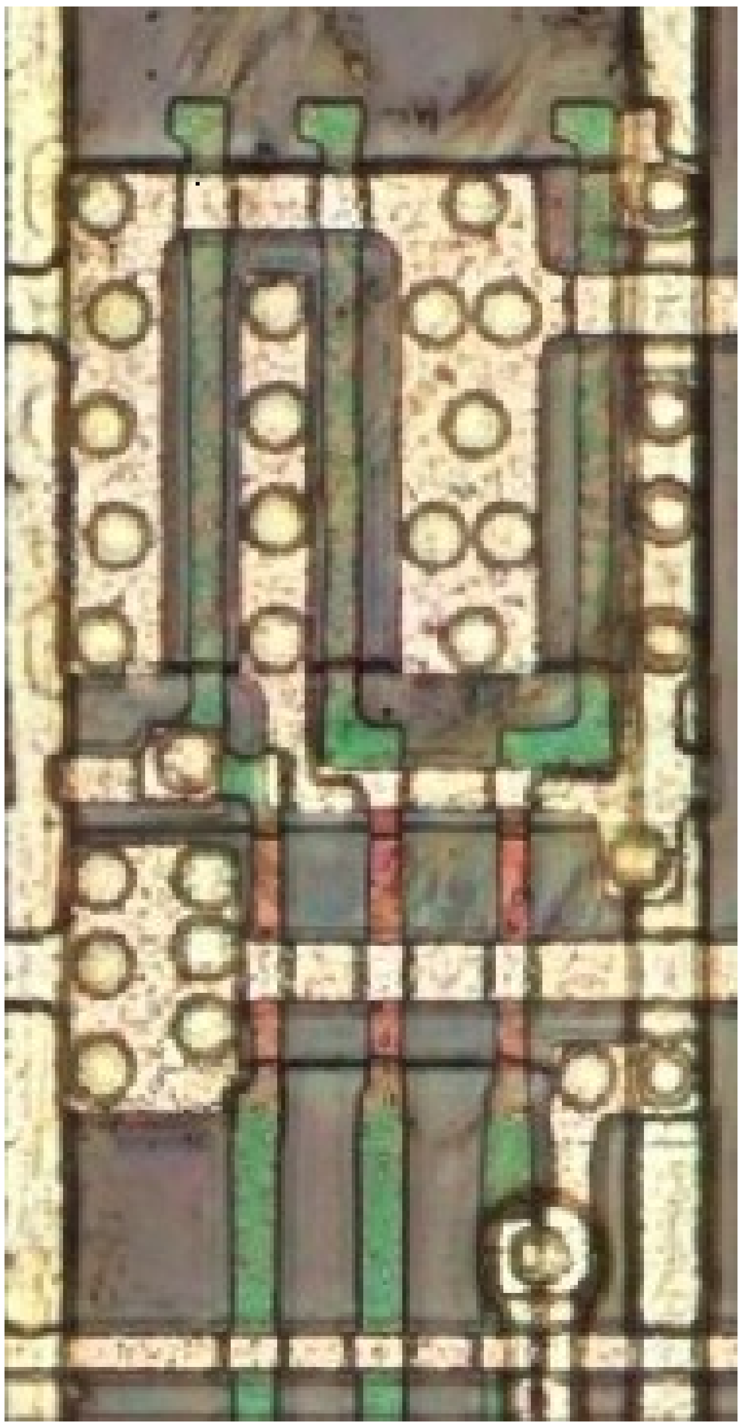
#f3ffffff 3-Input NAND Cell 11



A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

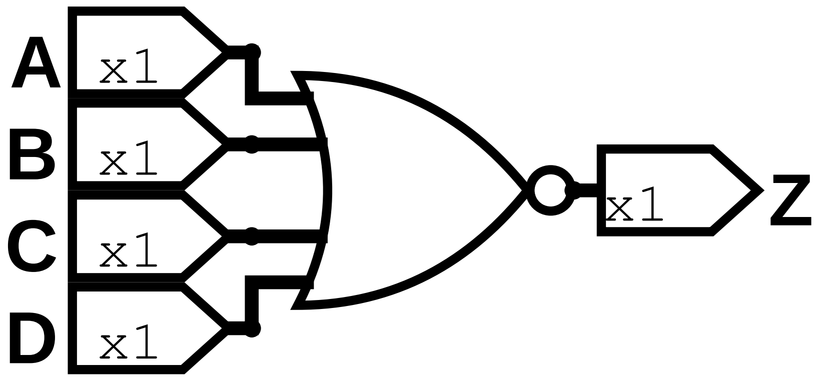




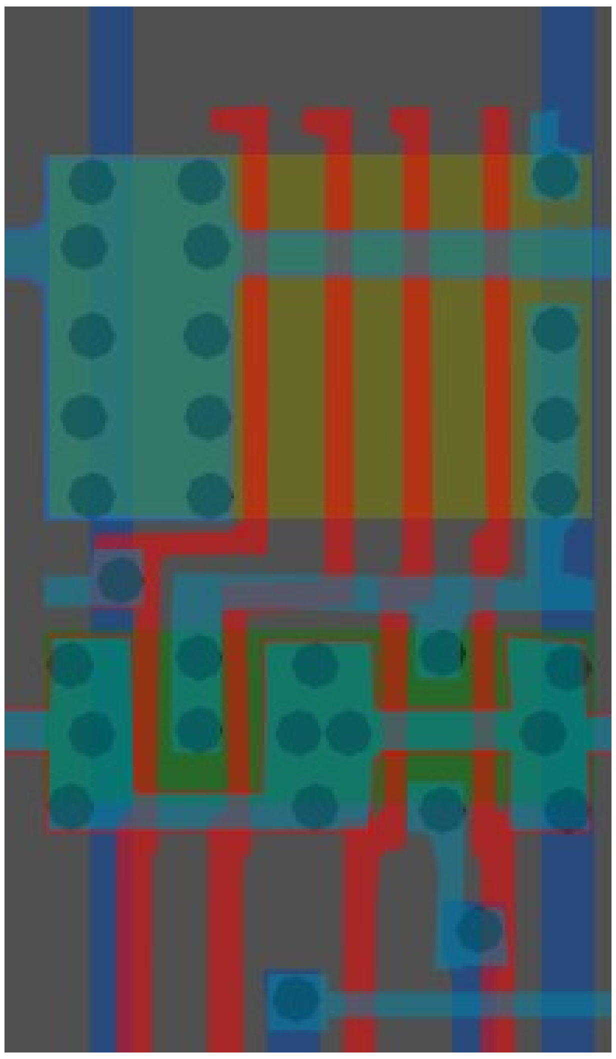


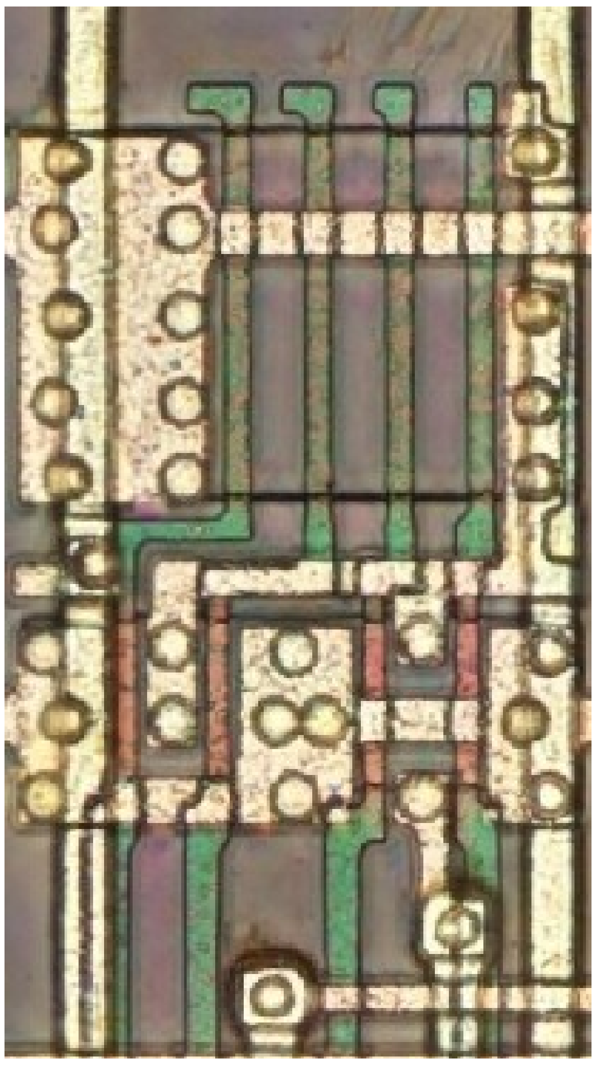
Same cell on
LA32
(MB87136A)

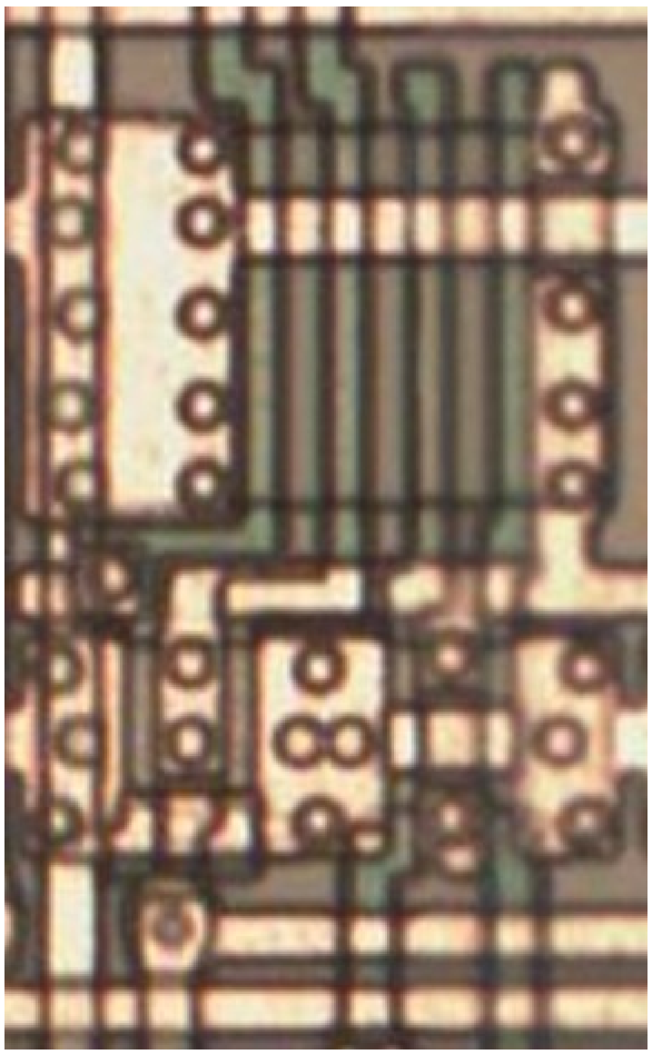
#007affff 4-Input NOR Cell 12



A	B	C	D	X
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0





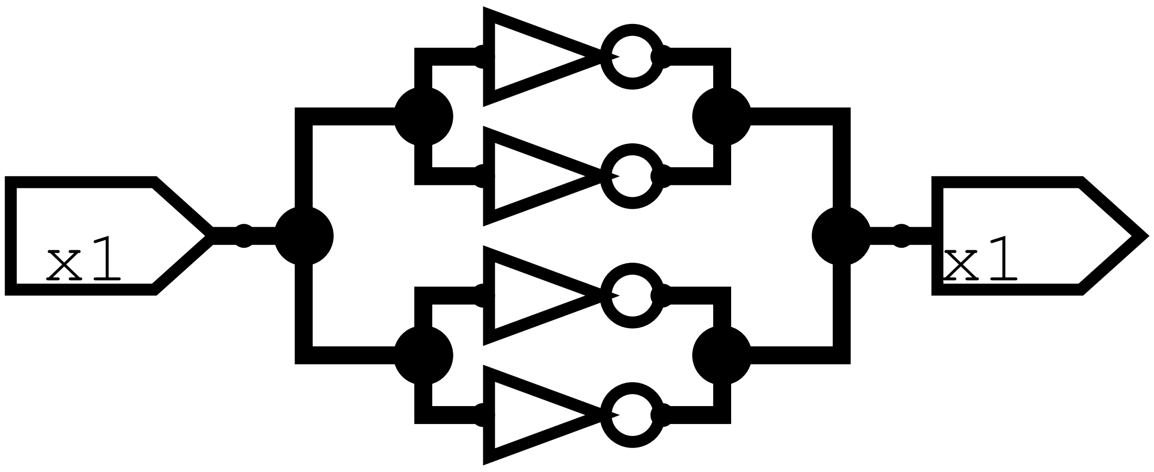


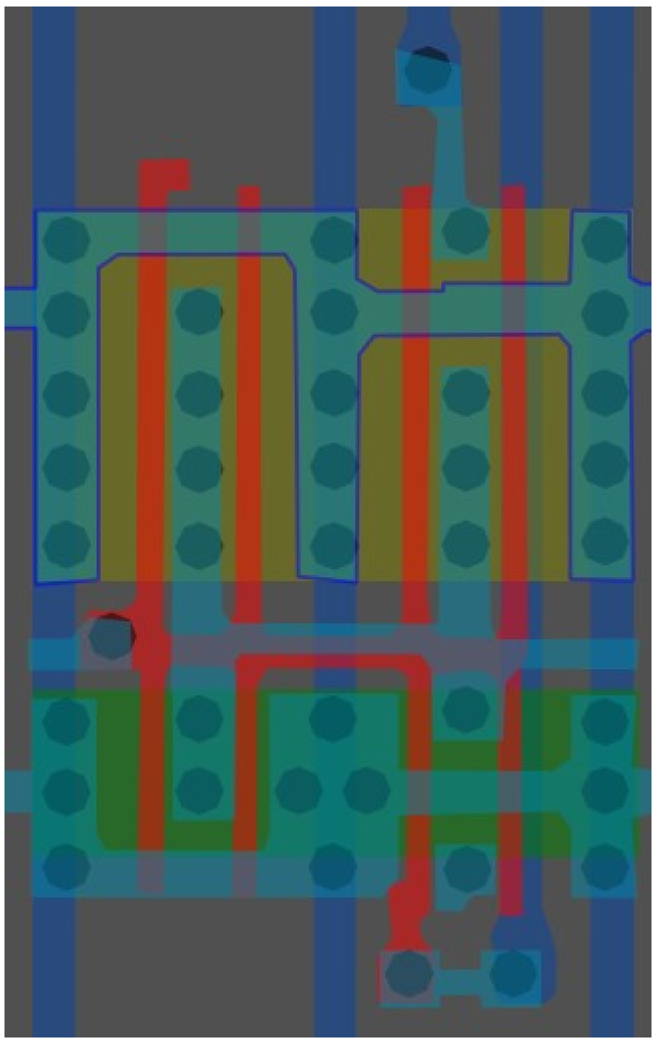
Same cell on LA32 (MB87136A)

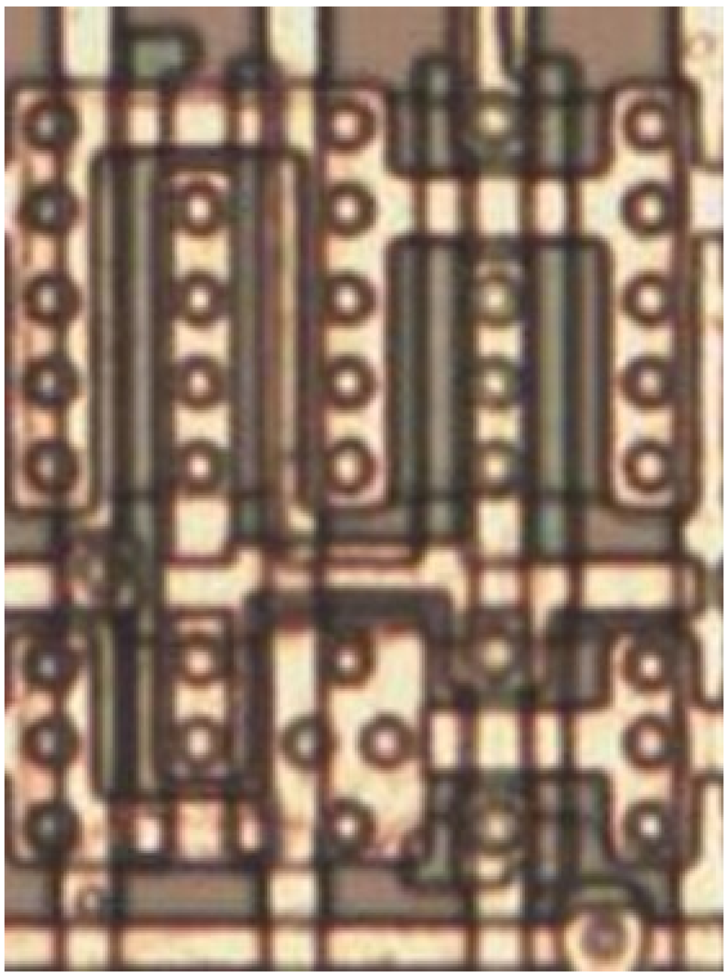
#b9ff43ff

Inverter/NOT x4

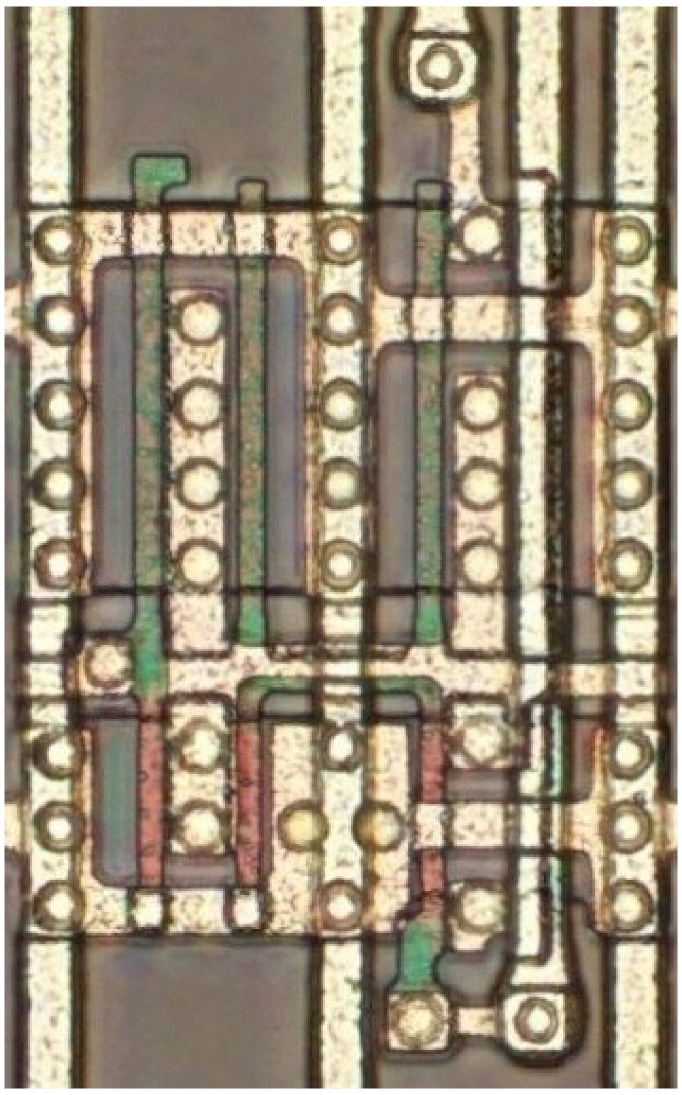
Cell 13a



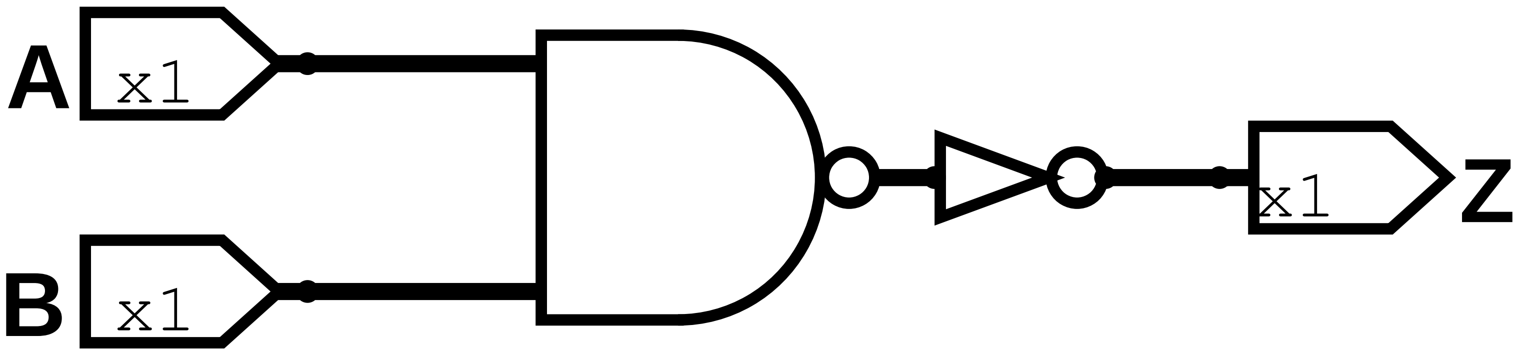




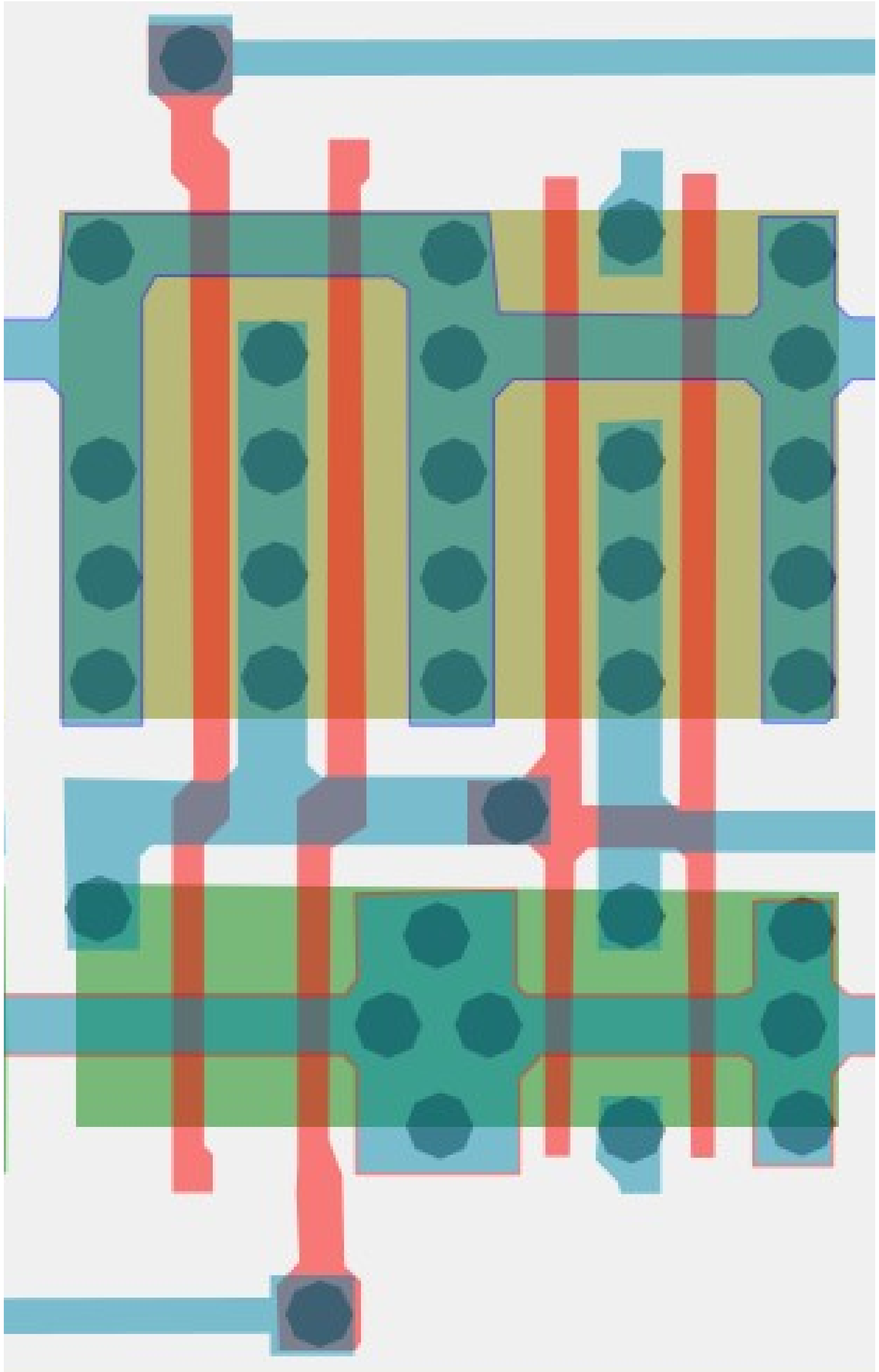
Same cell on
LA32
(MB87136A)

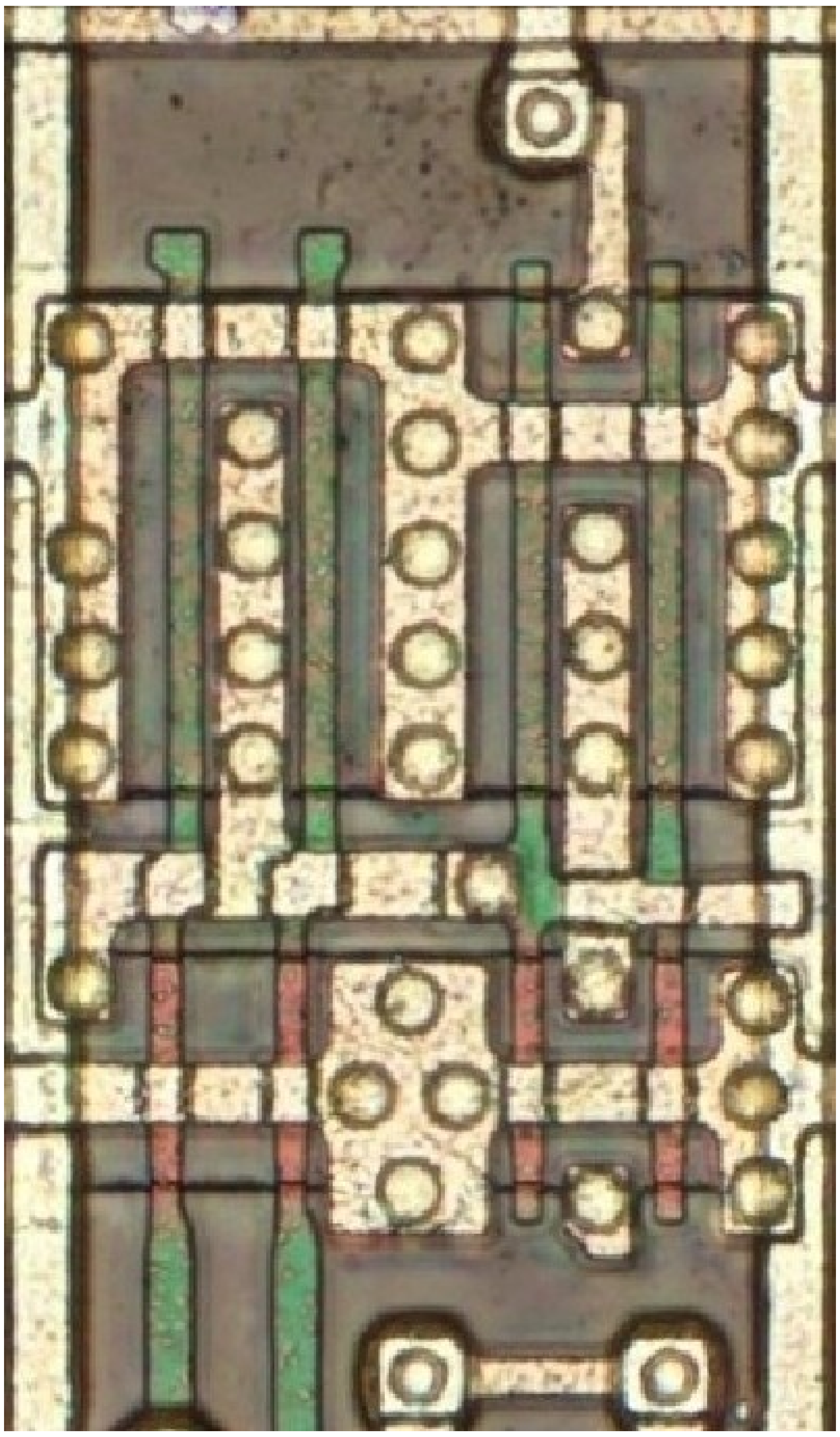


#bfa867ff 2-Input AND Cell 13b

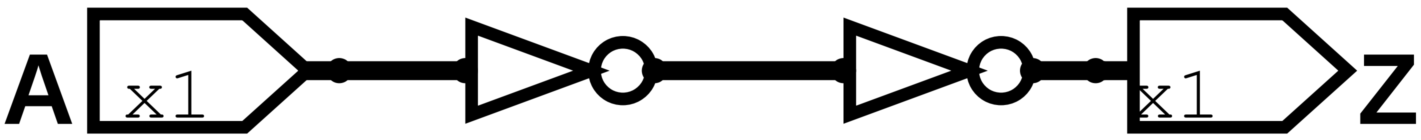


A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

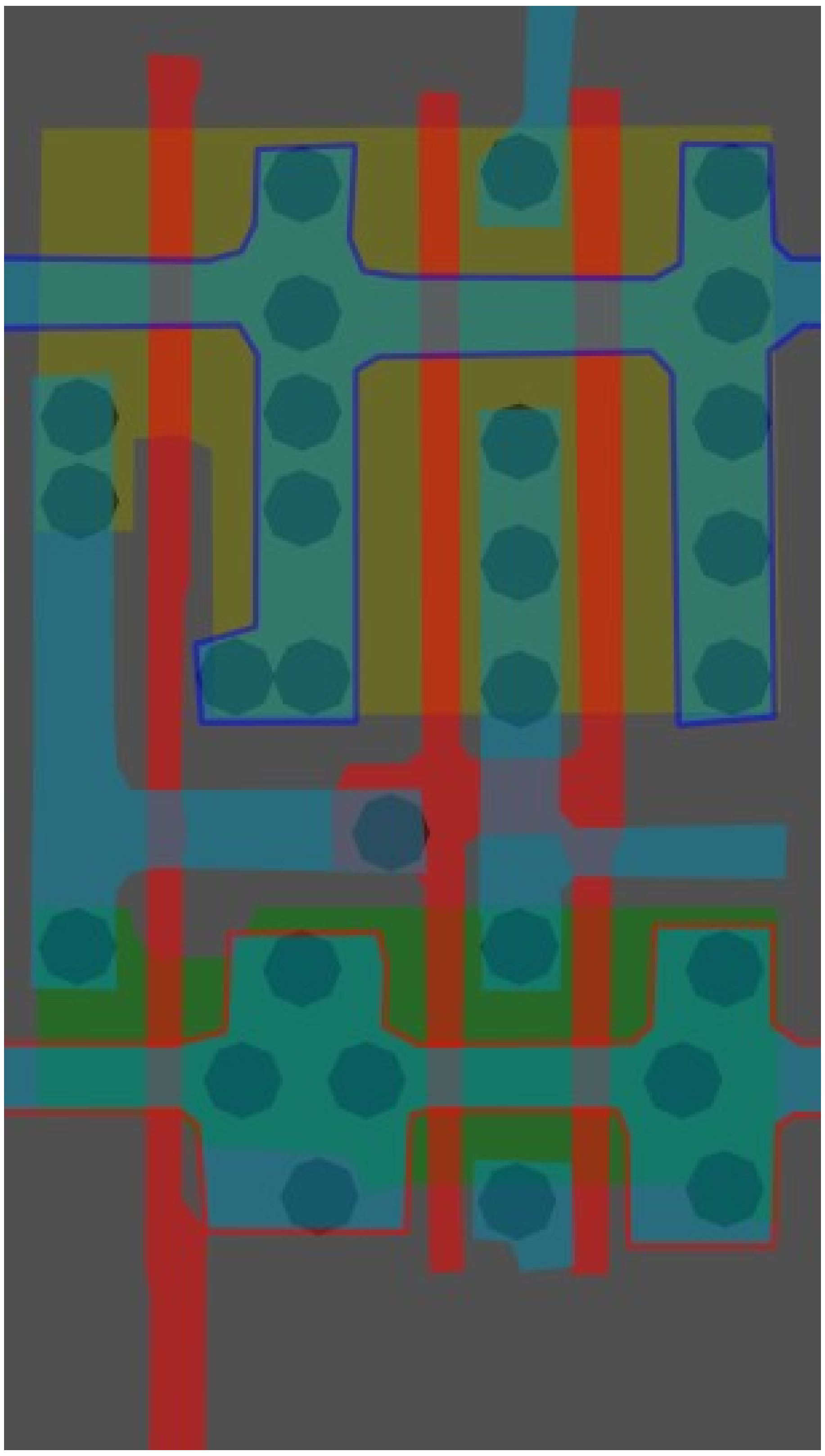


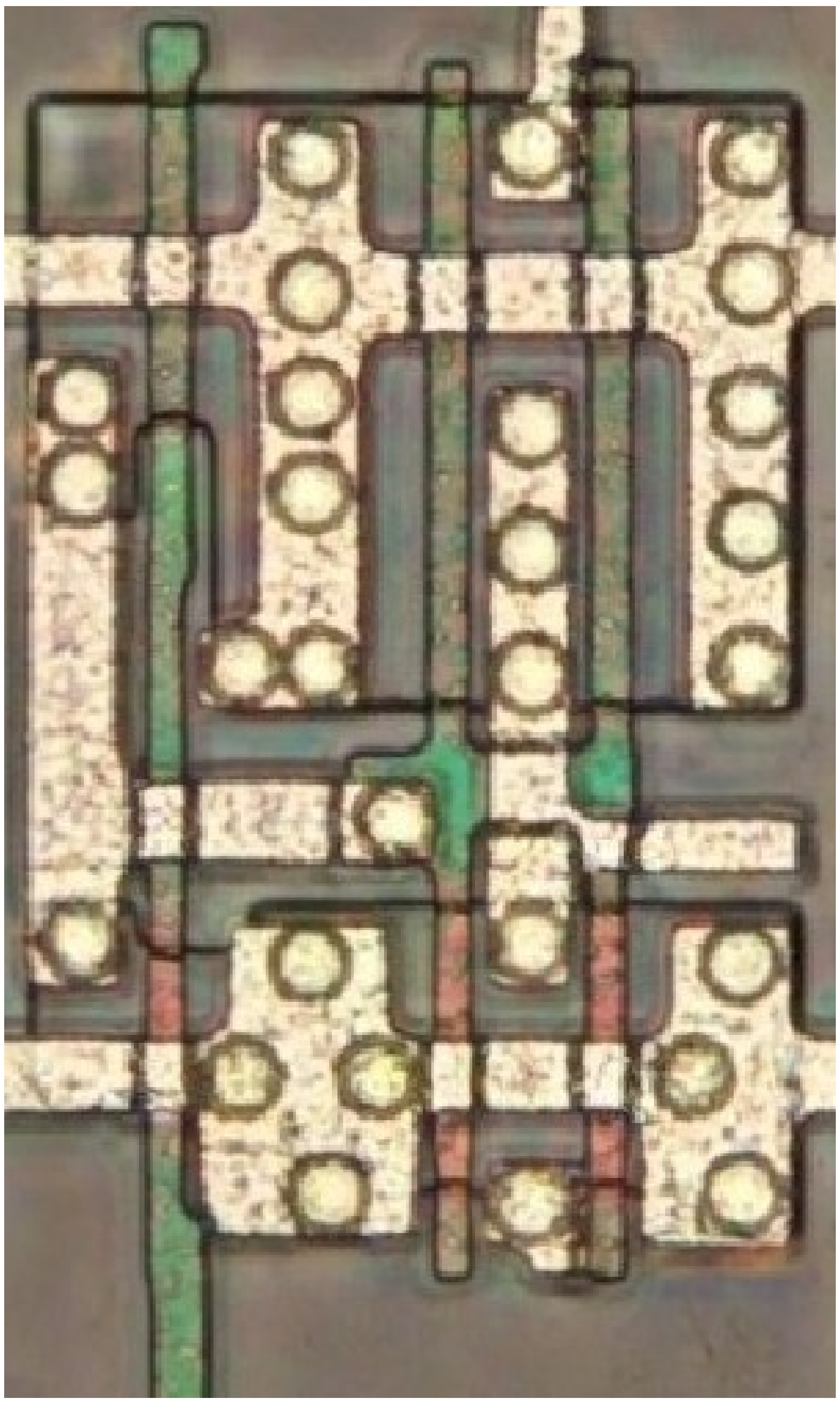


#a03ac0ff Buffer Cell 13c

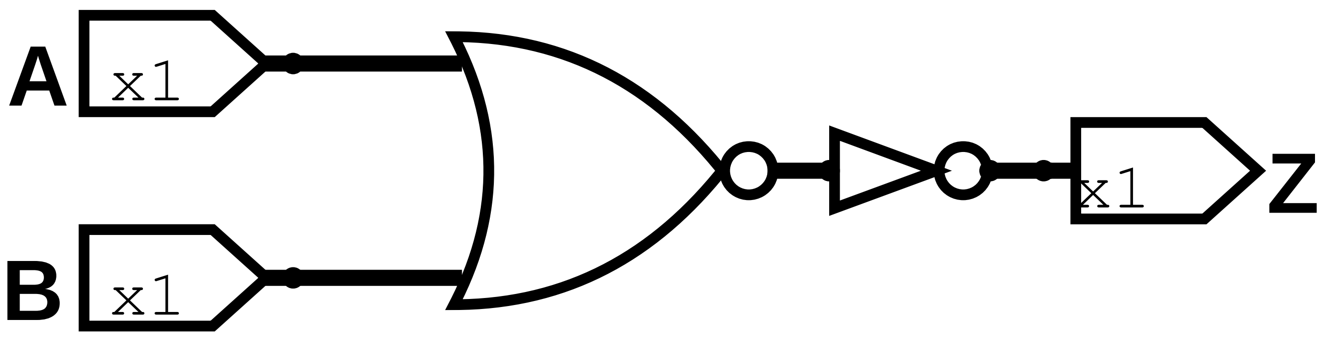


A	Z
0	0
1	1

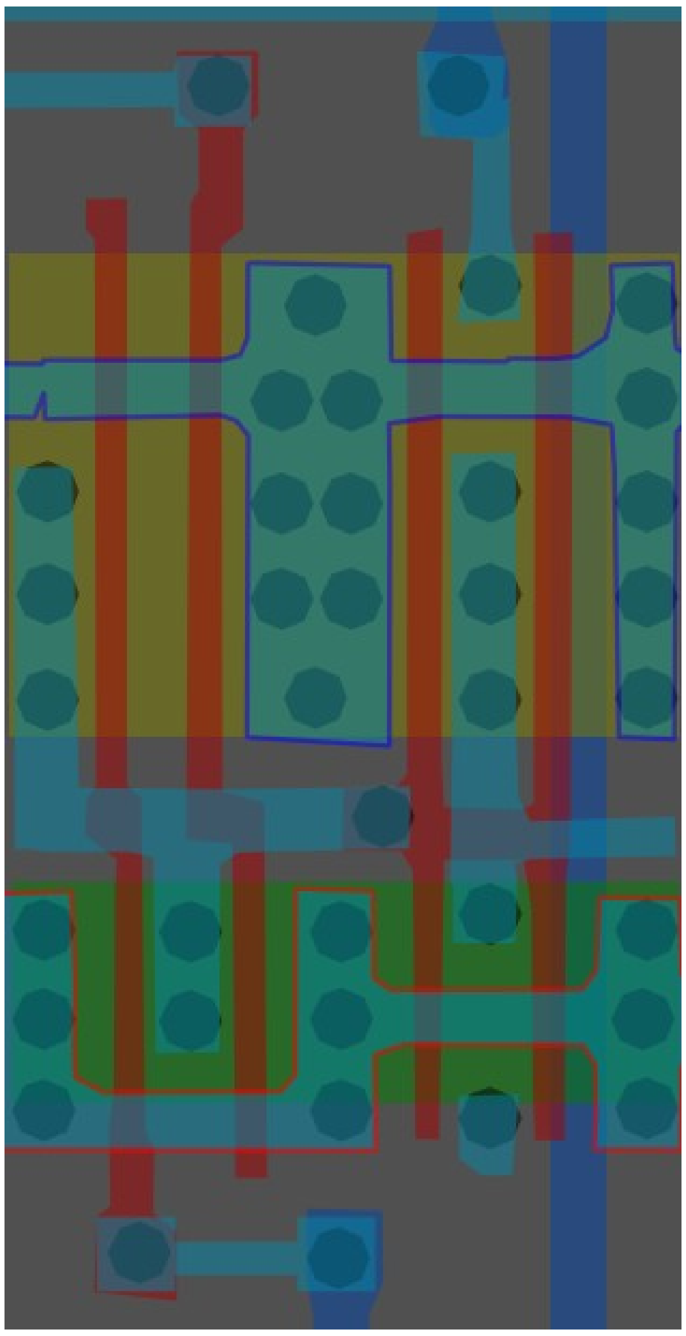


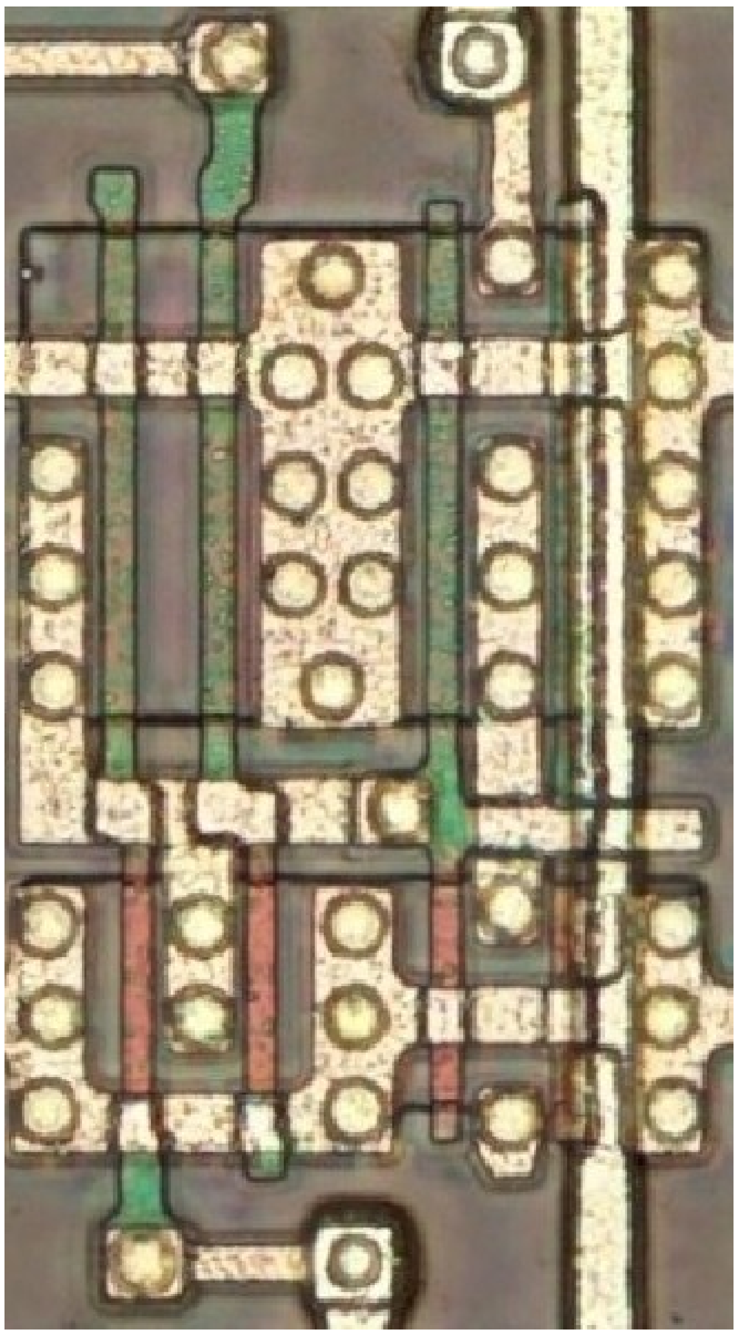


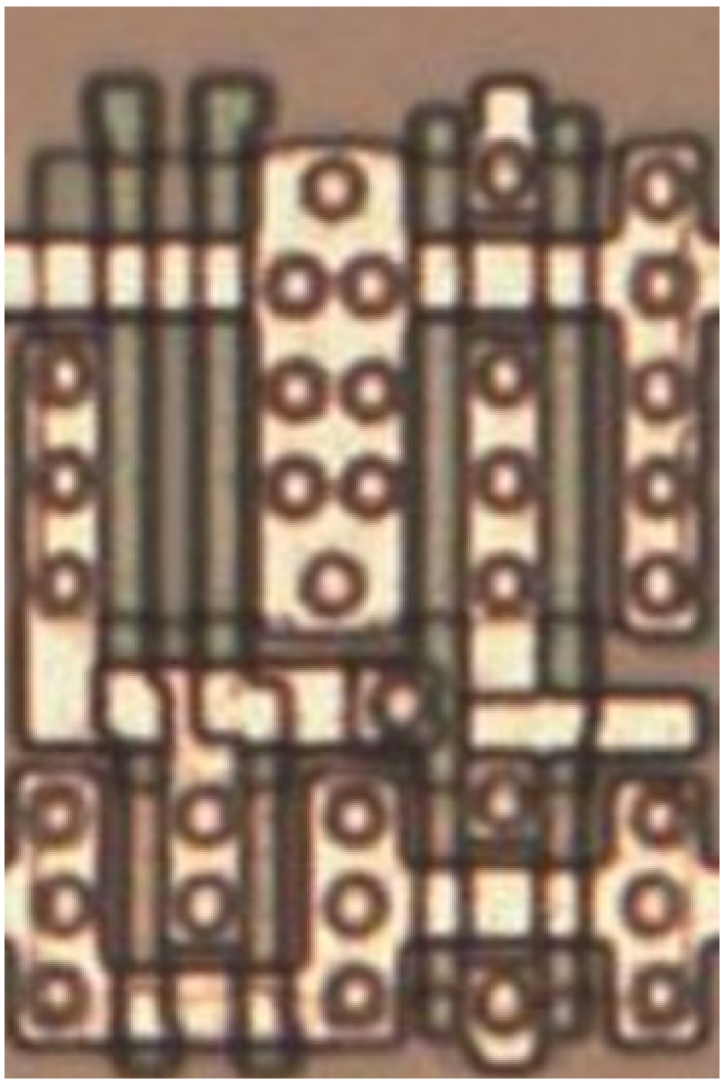
#ffbcacff | 2-Input OR x1 | Cell 13d



A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

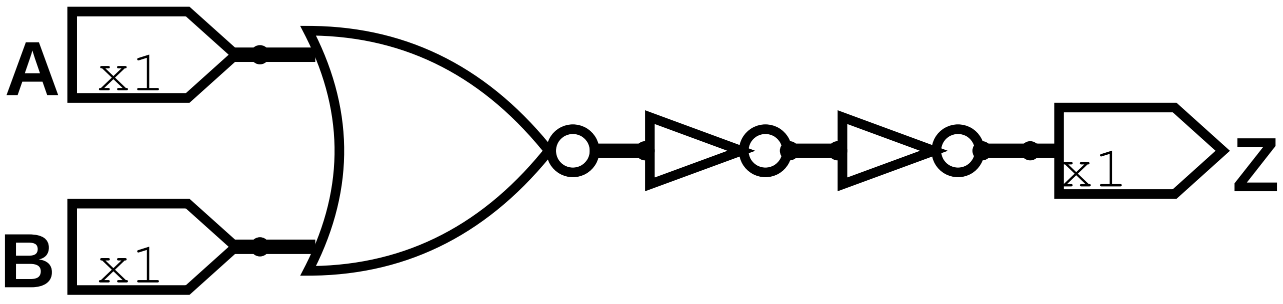




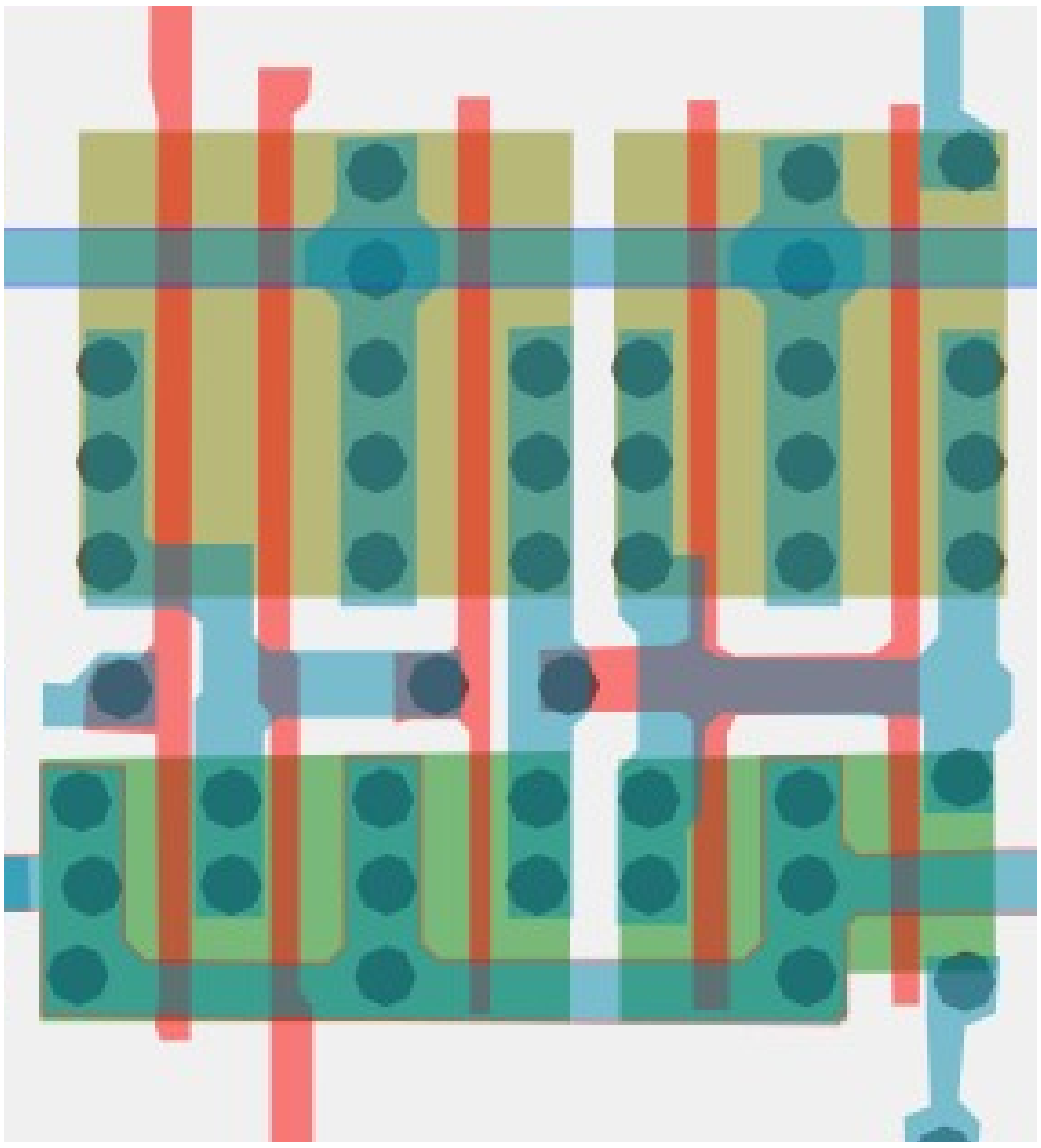


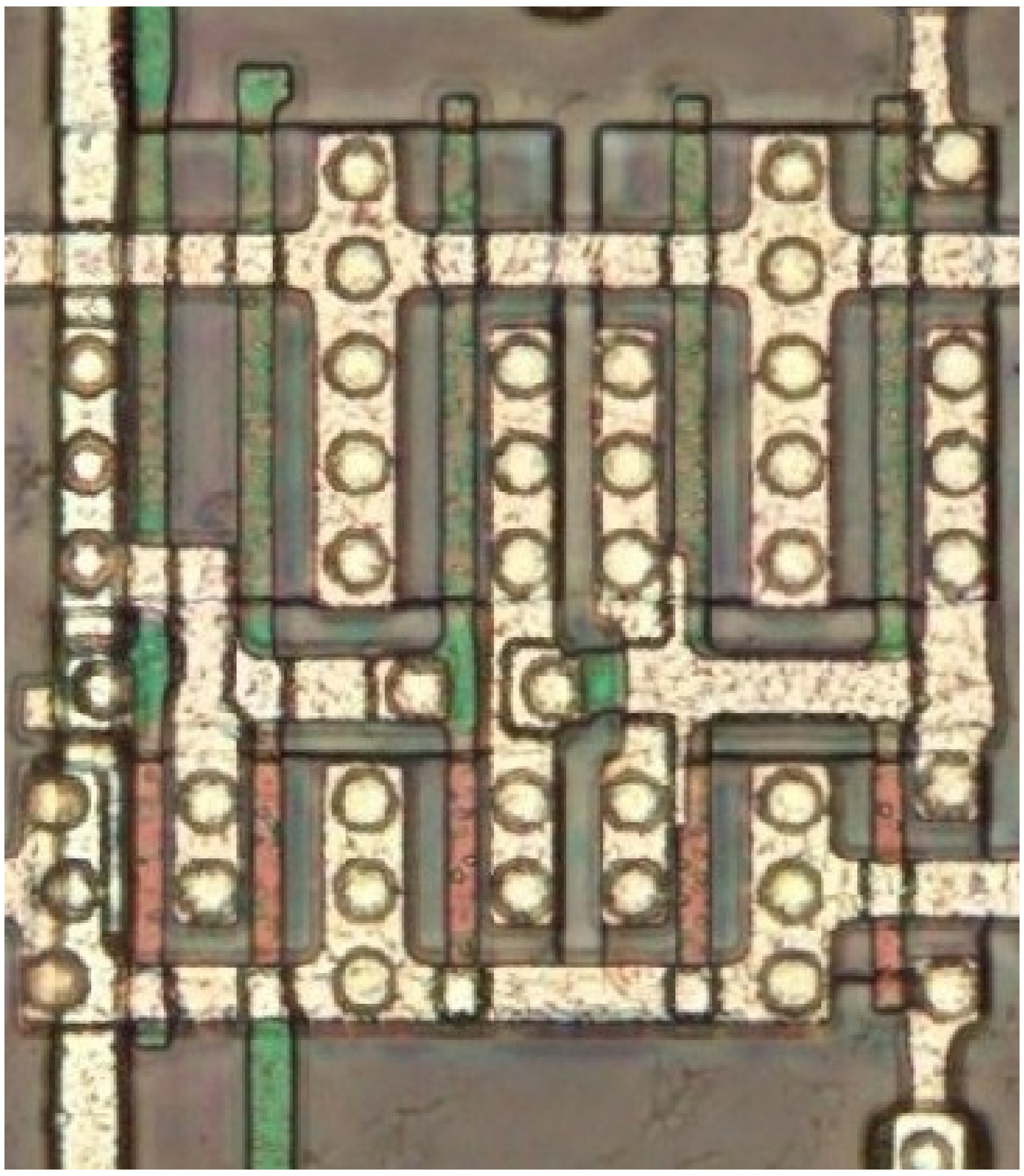
Same cell on LA32 (MB87136A)

#ff7b94ff | 2-Input NOR x2 | Cell 13e



A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

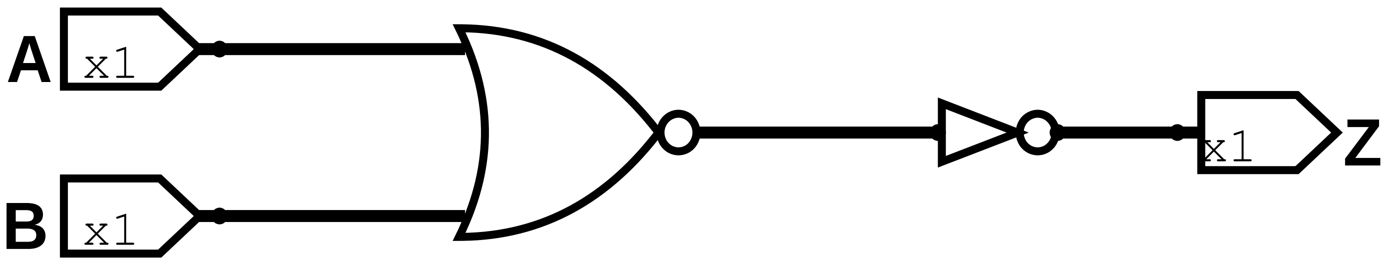




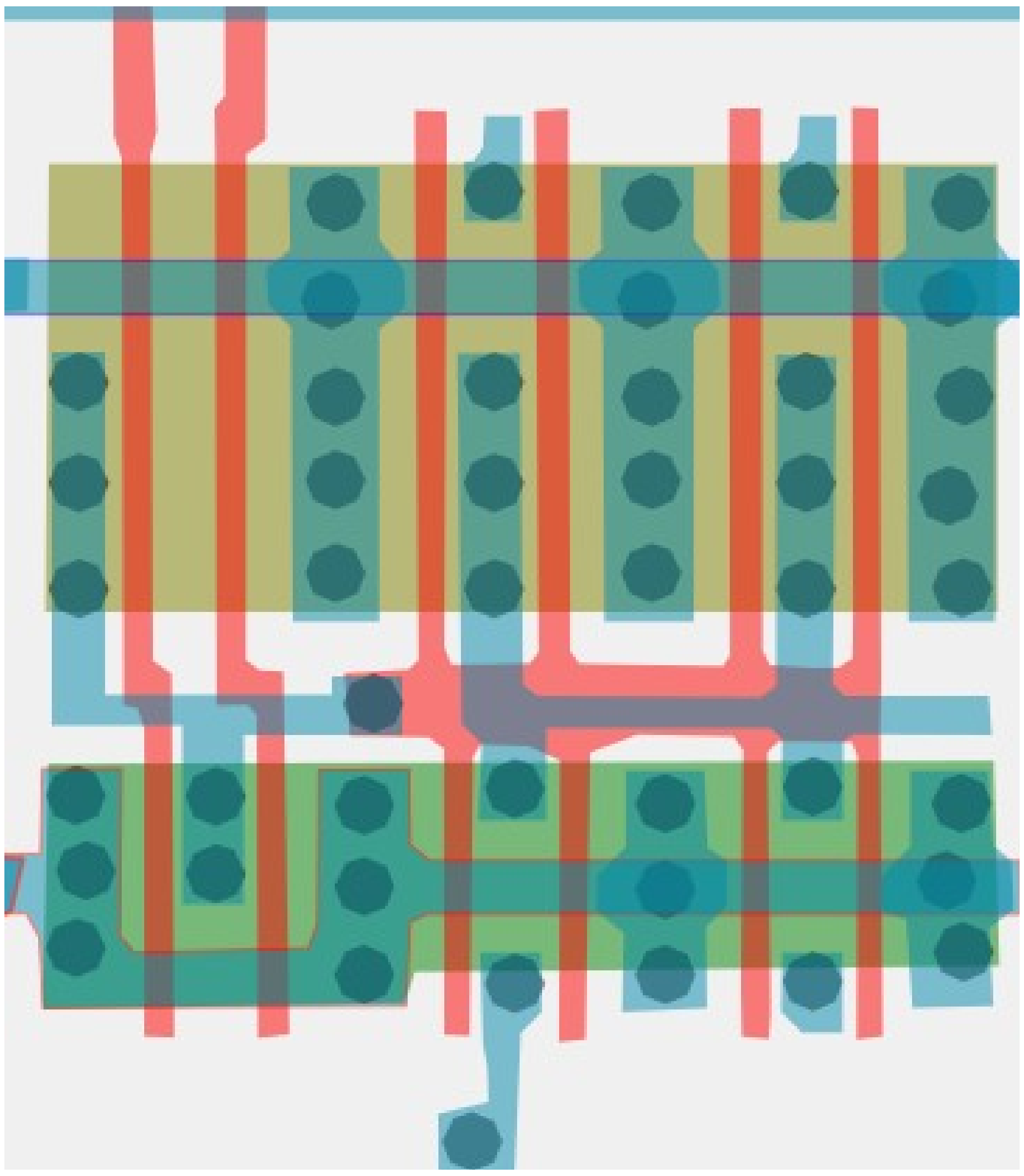
#fba968ff

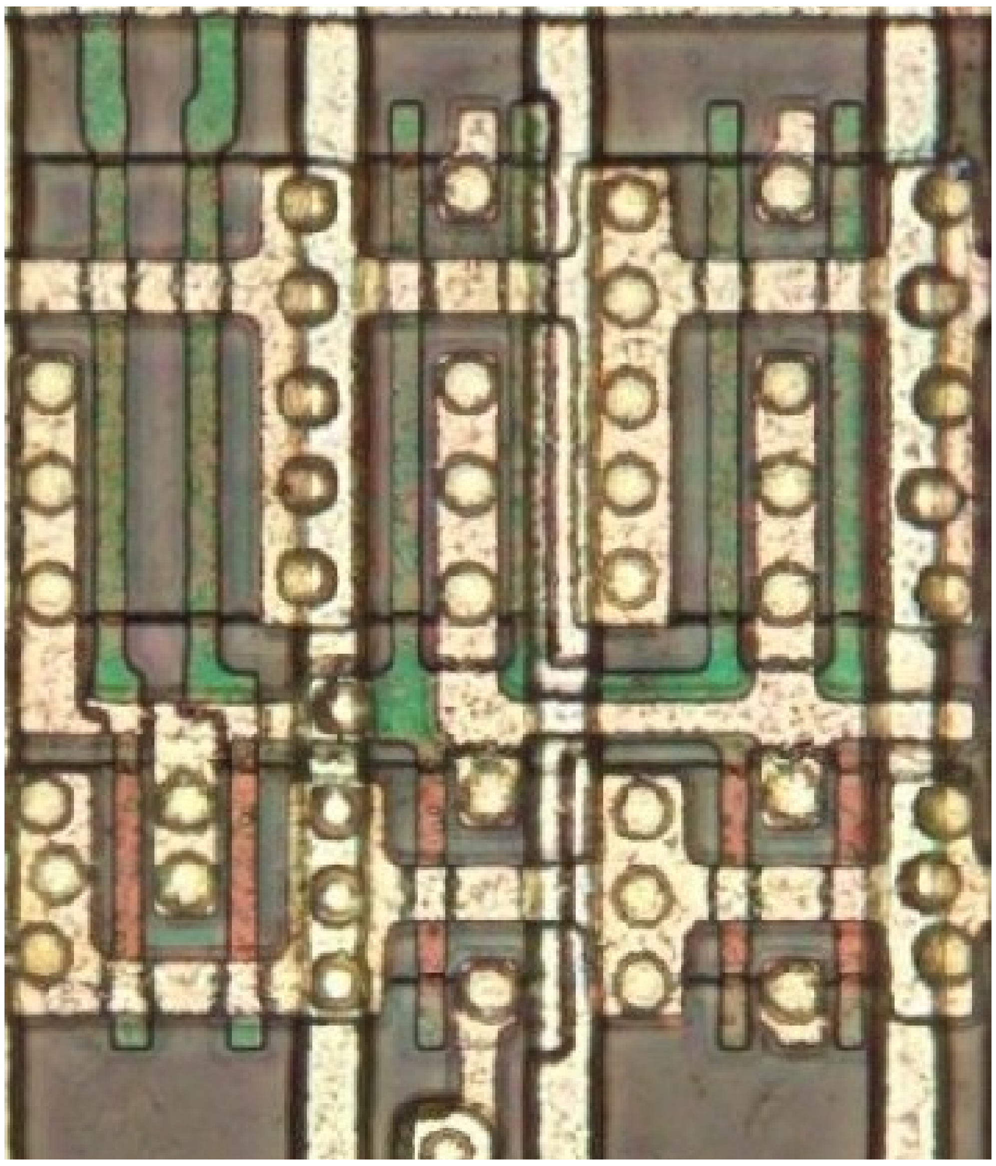
2-Input OR x2

Cell 13f

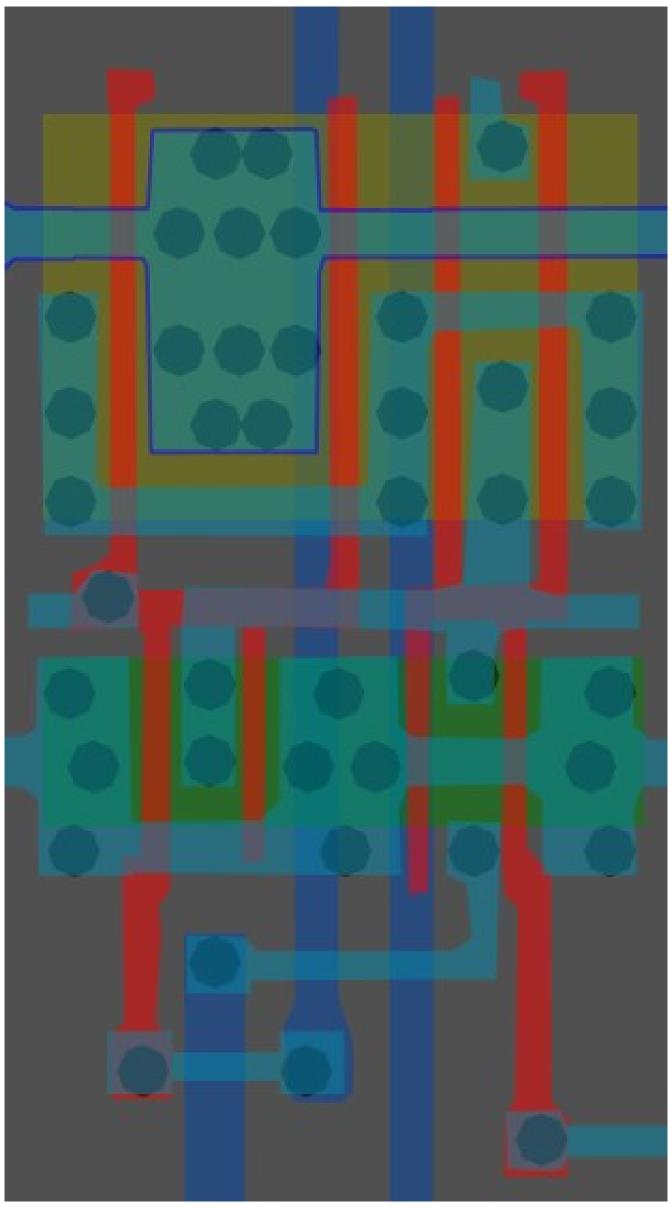


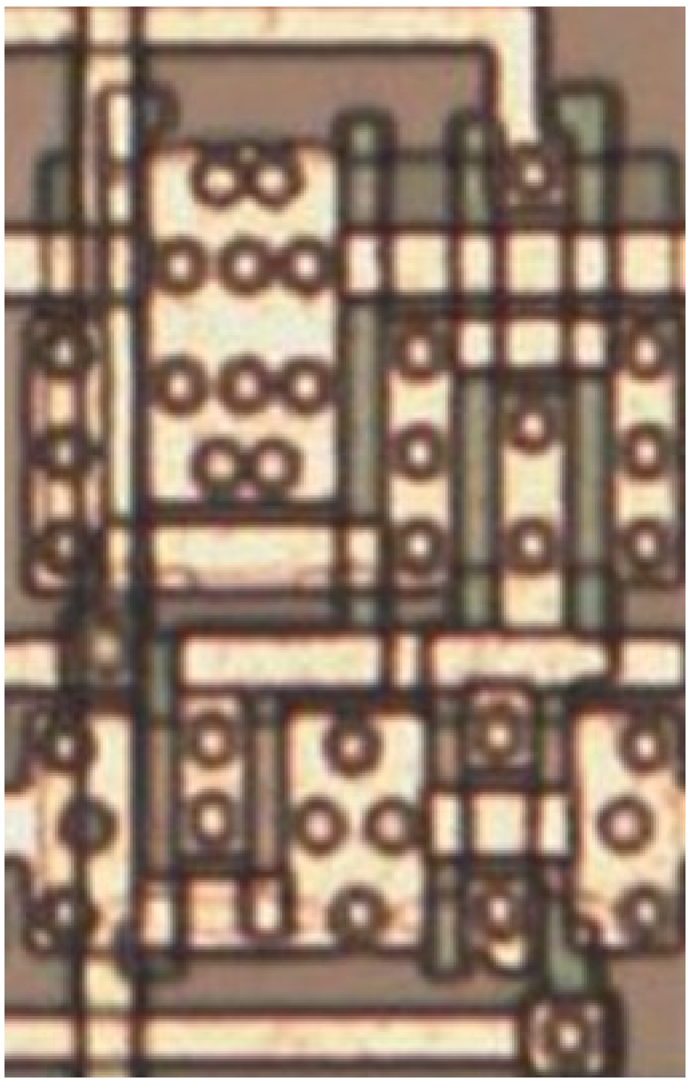
A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

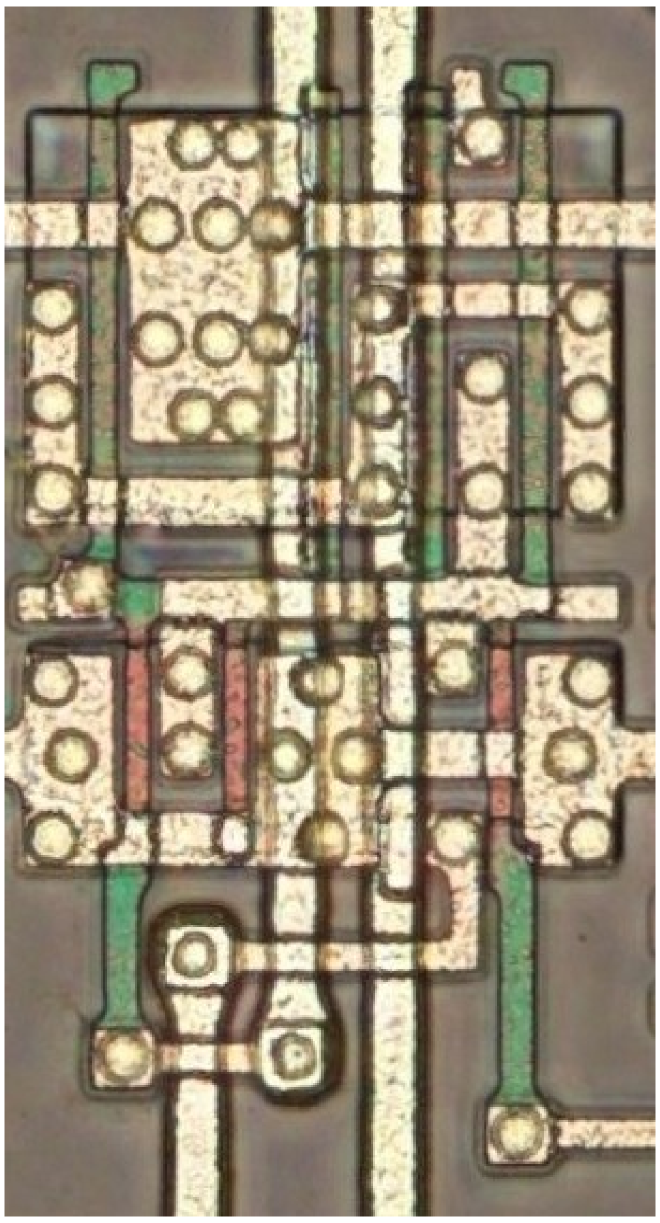




#f3ffa7ff Cell type 14

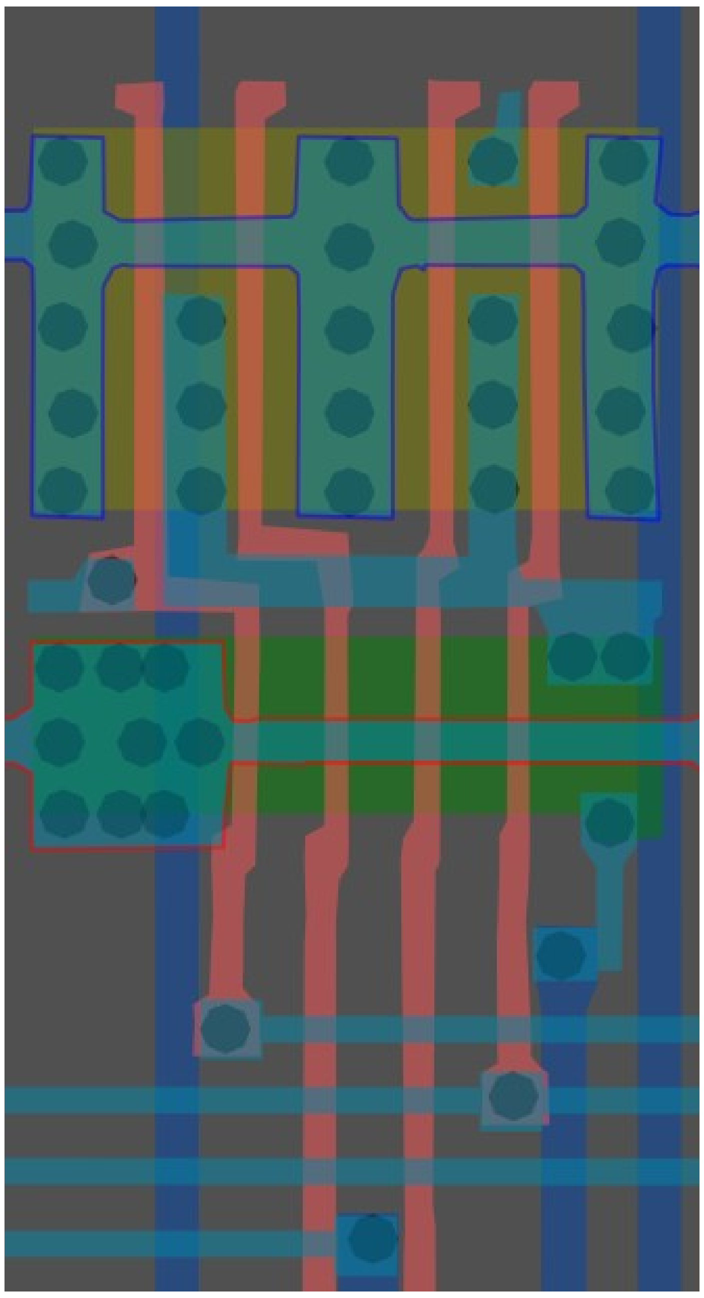


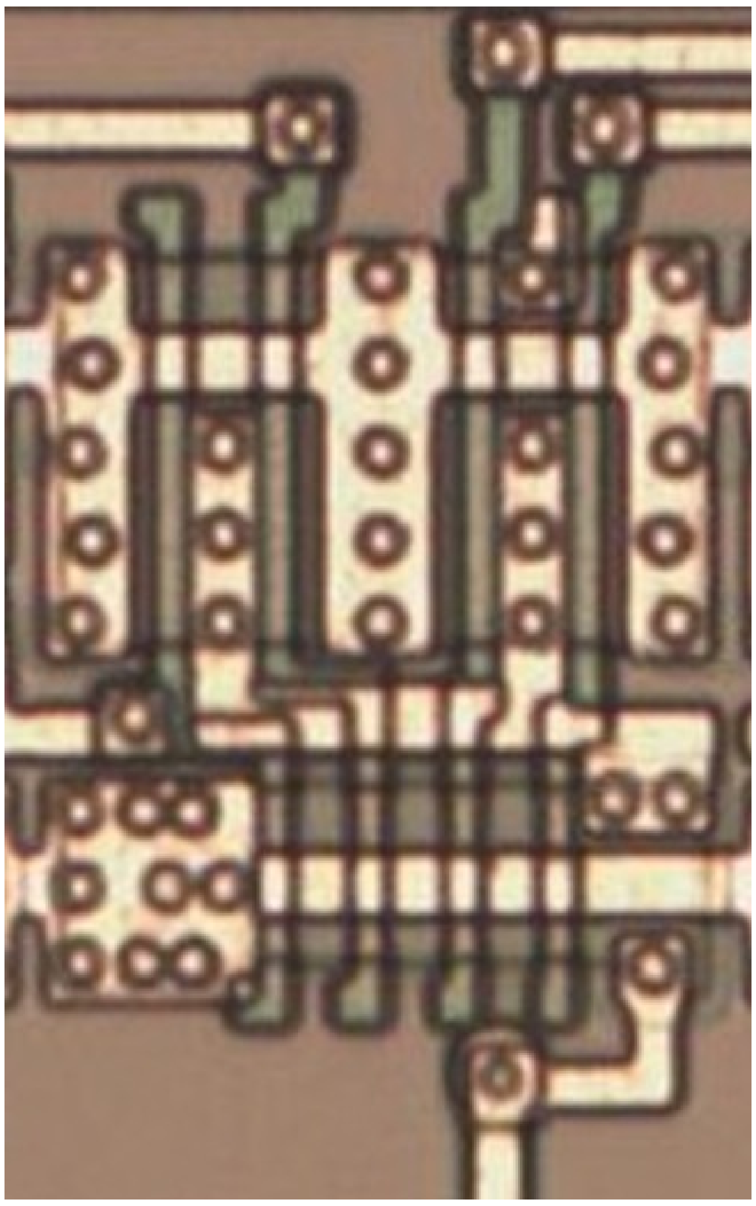




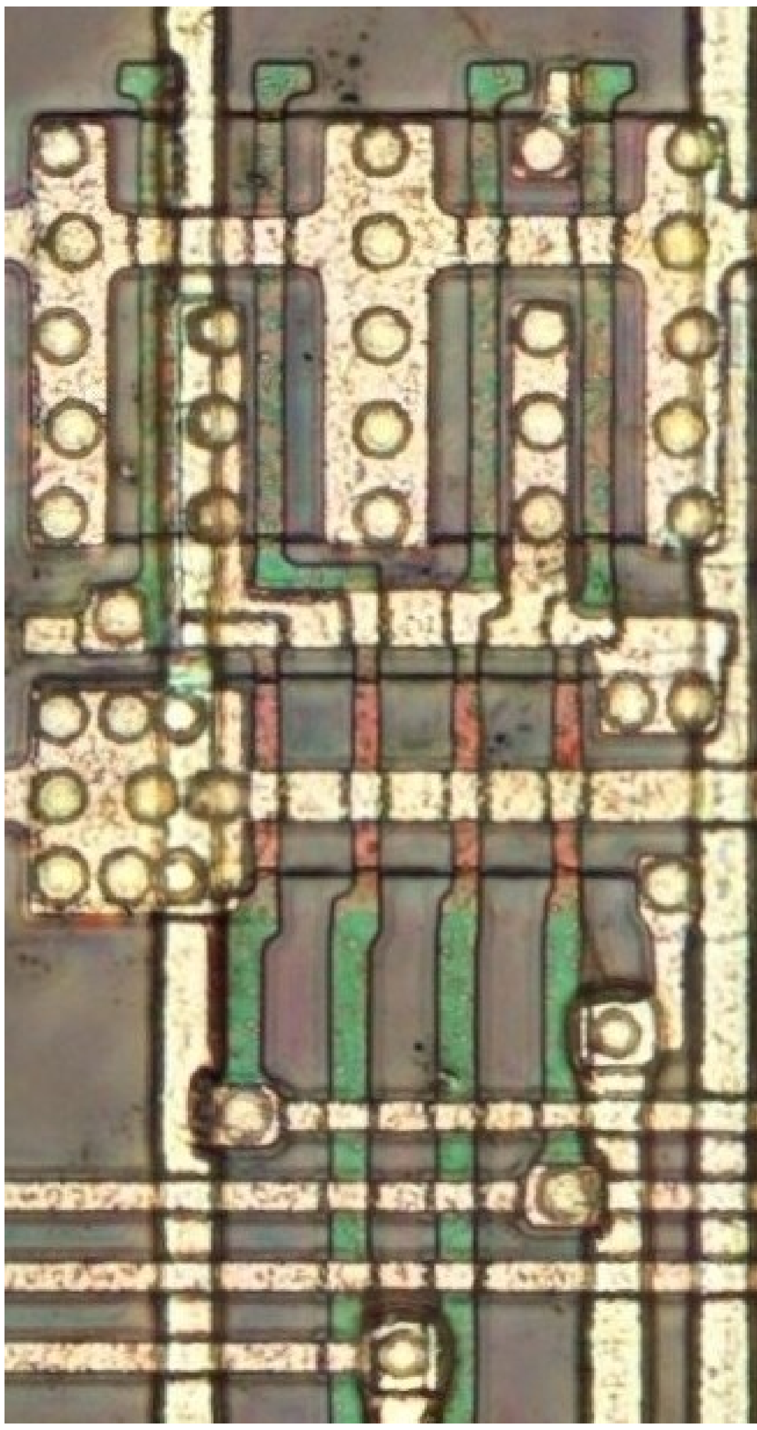
Same cell on
LA32
(MB87136A)

#00ffffff 4-Input NAND **Cell 15**



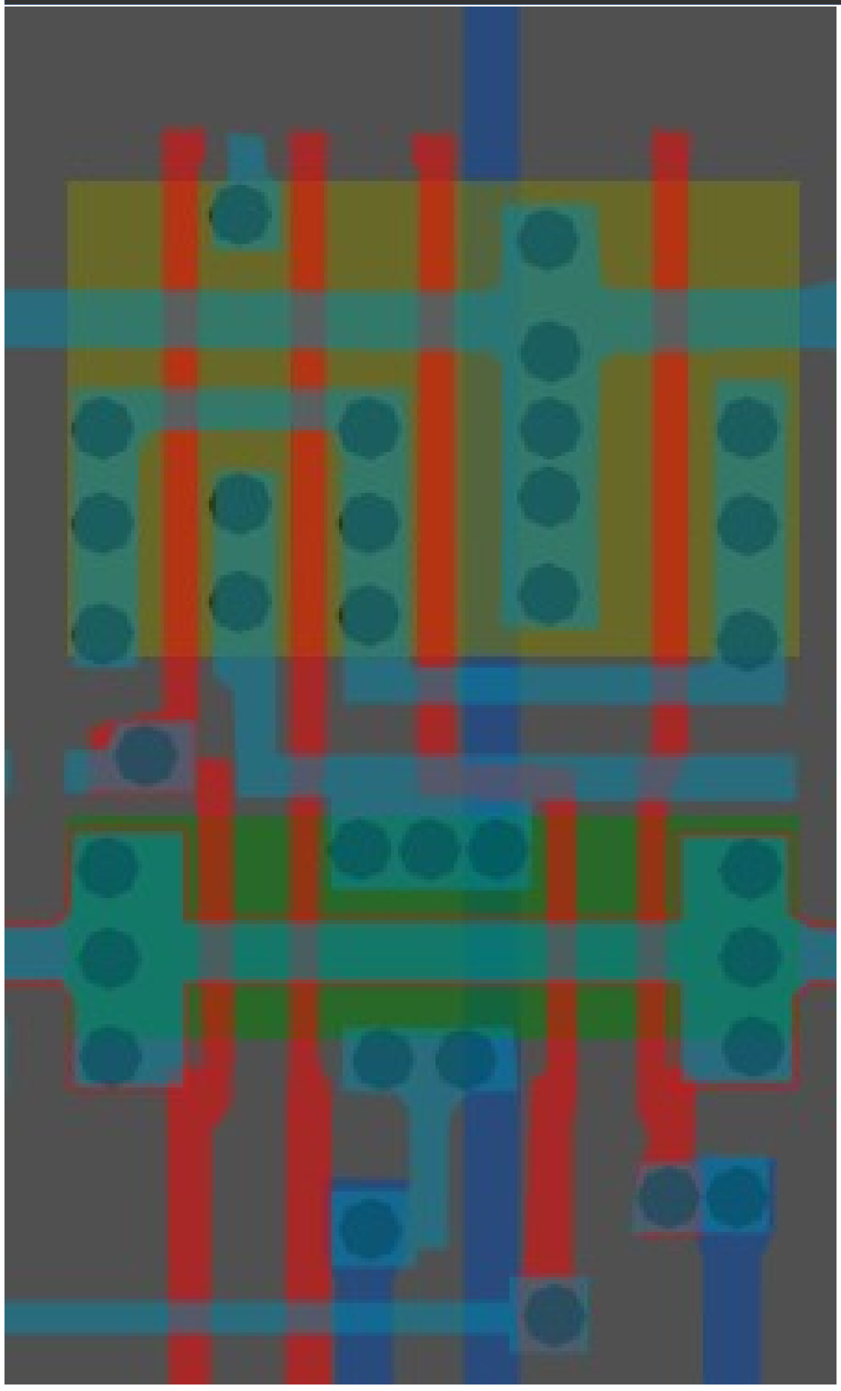


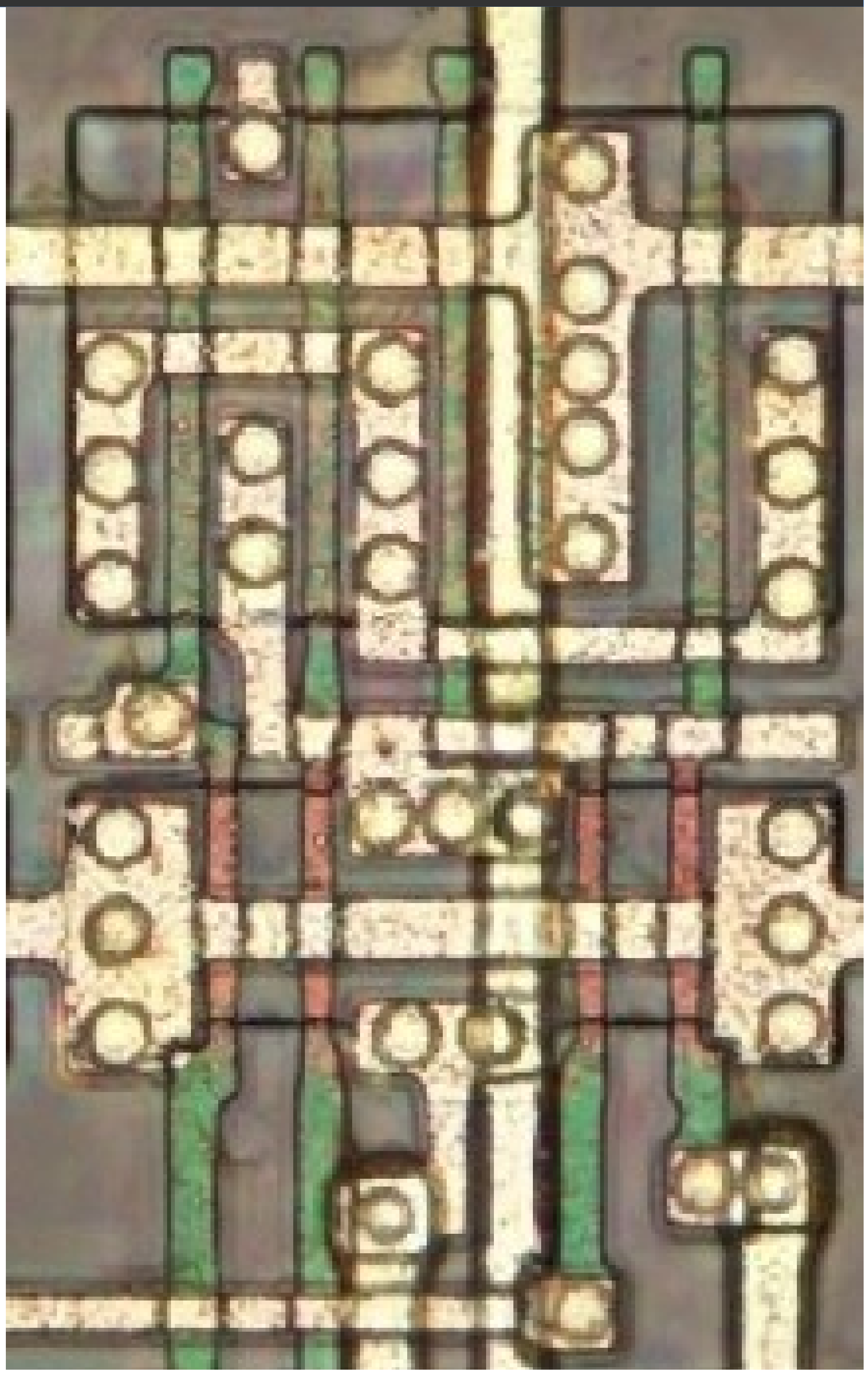
Same cell on
LA32
(MB87136A)



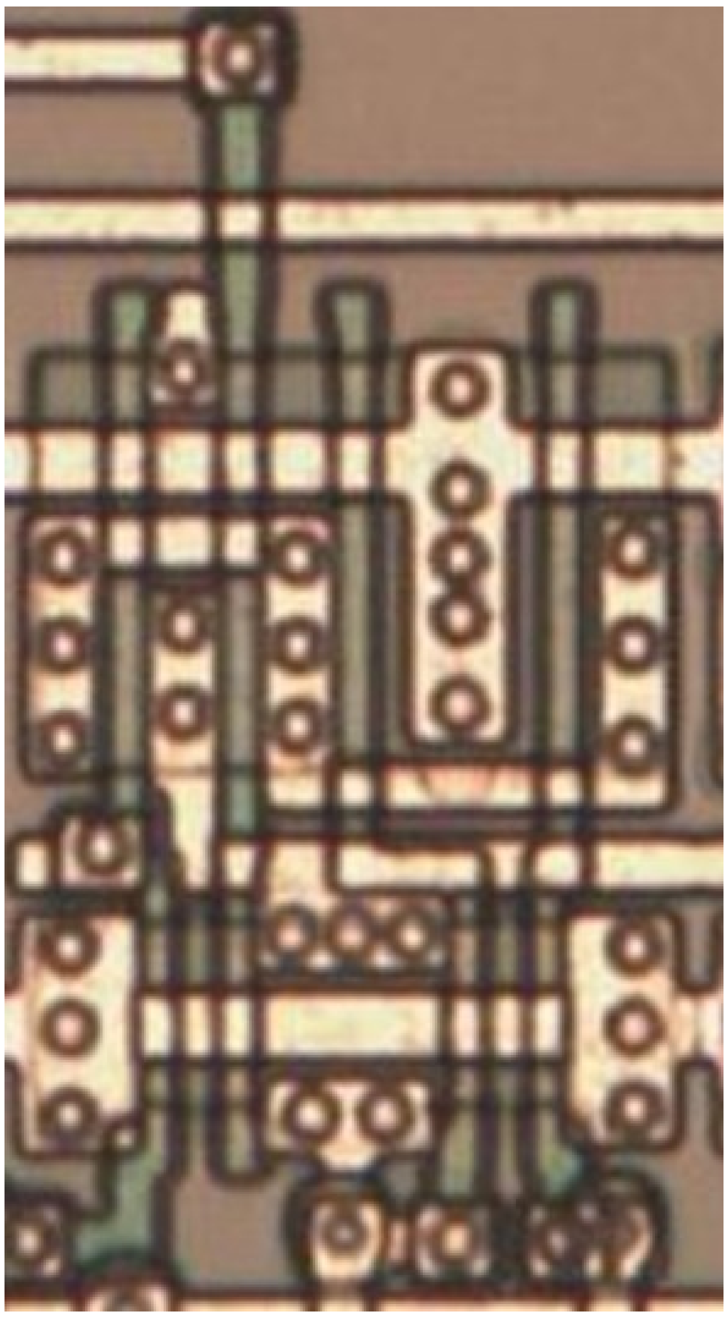
#ffb0ffff

Cell type 16



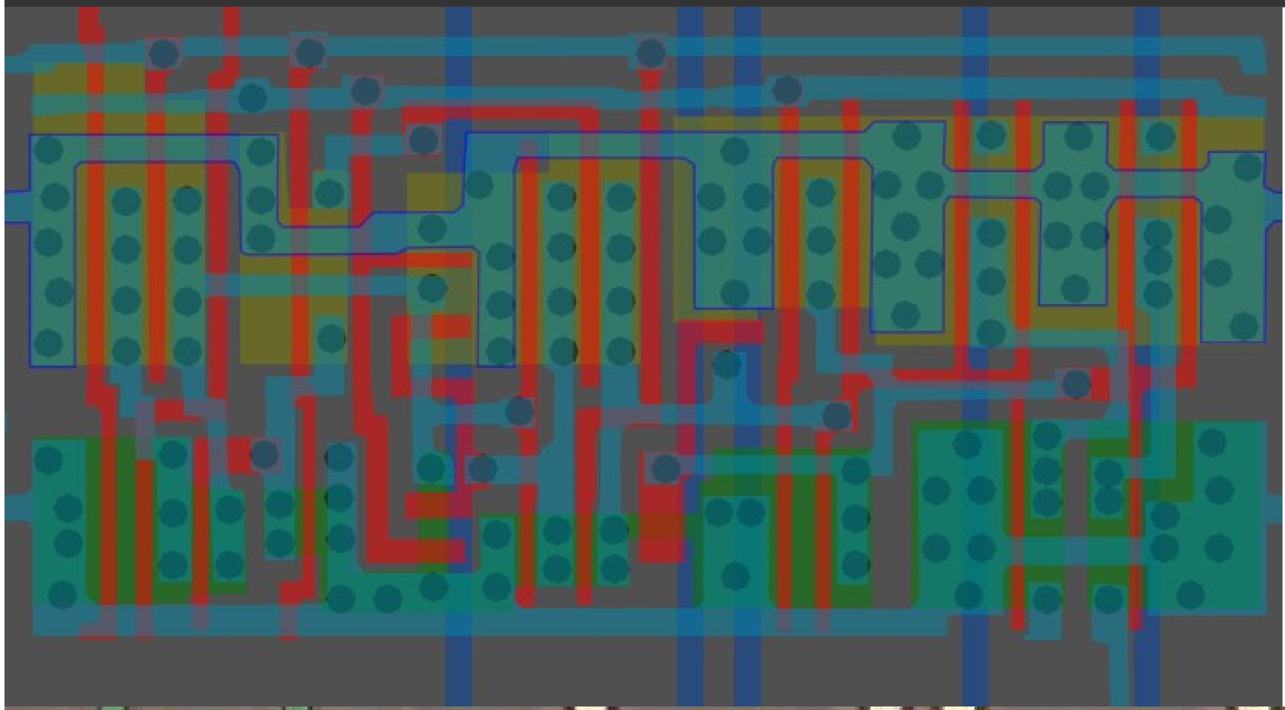


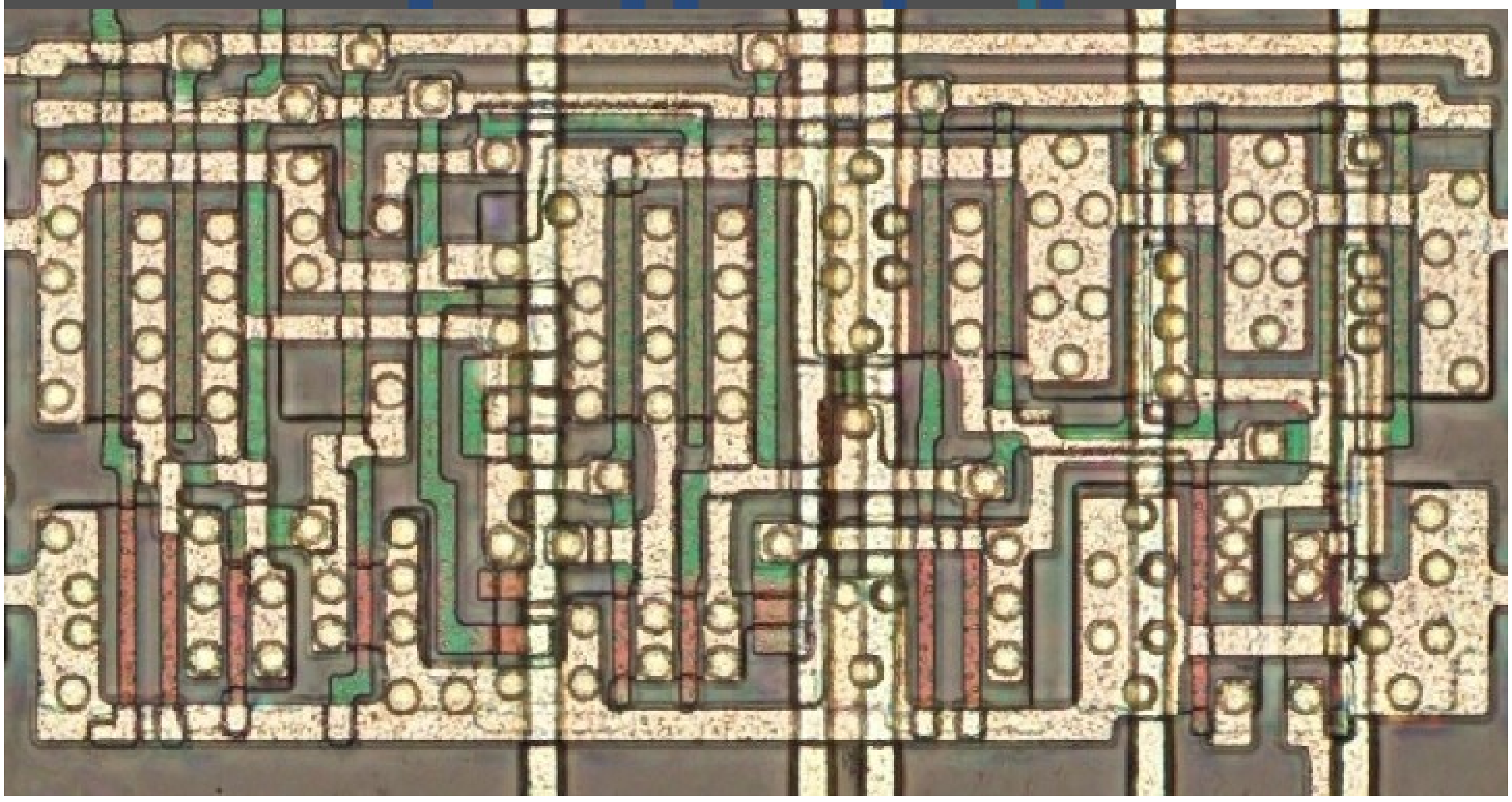
Same cell on
LA32
(MB87136A)

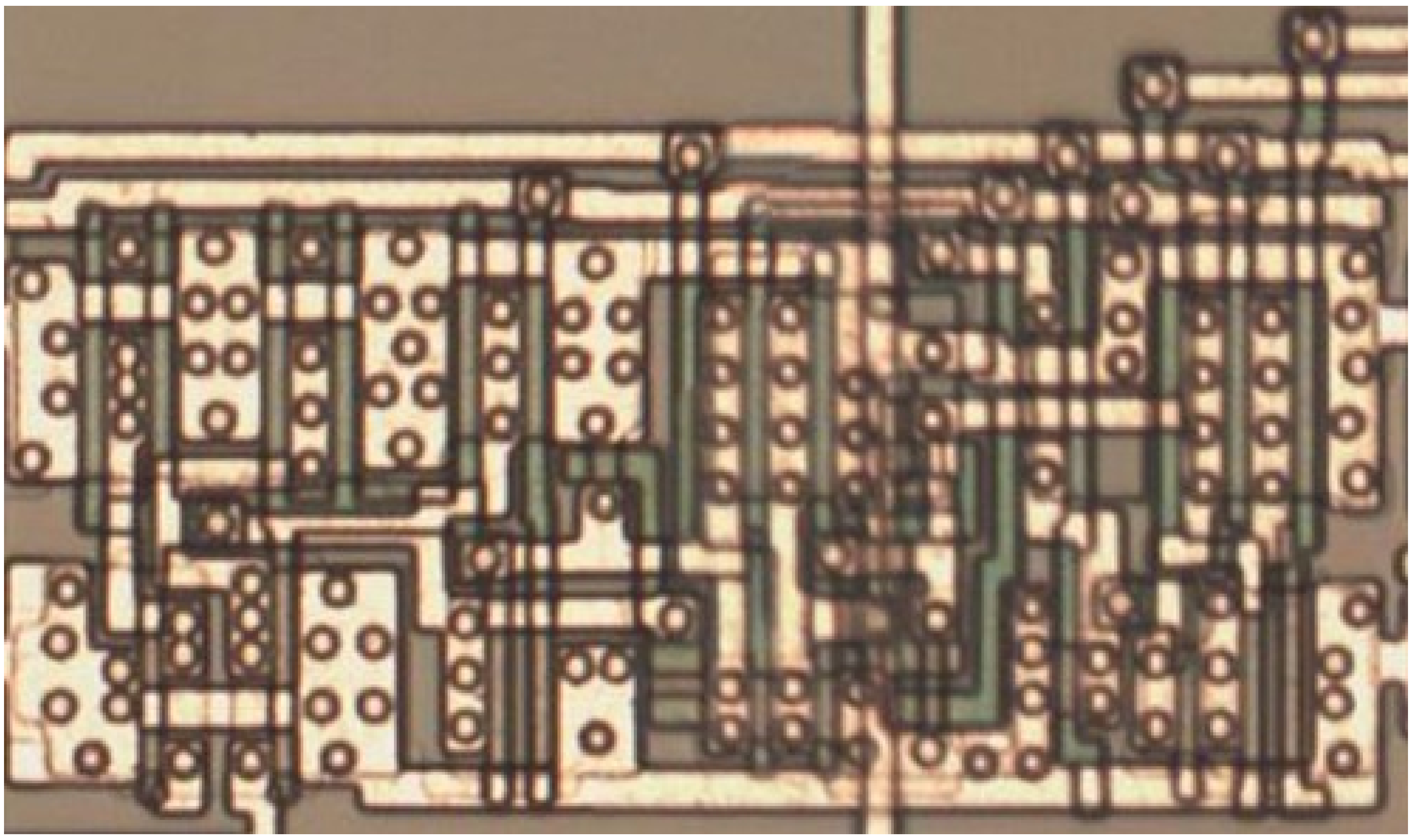


#0000ffff

Cell type 17



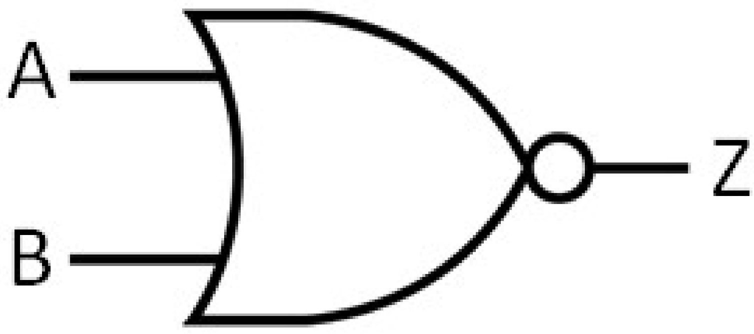


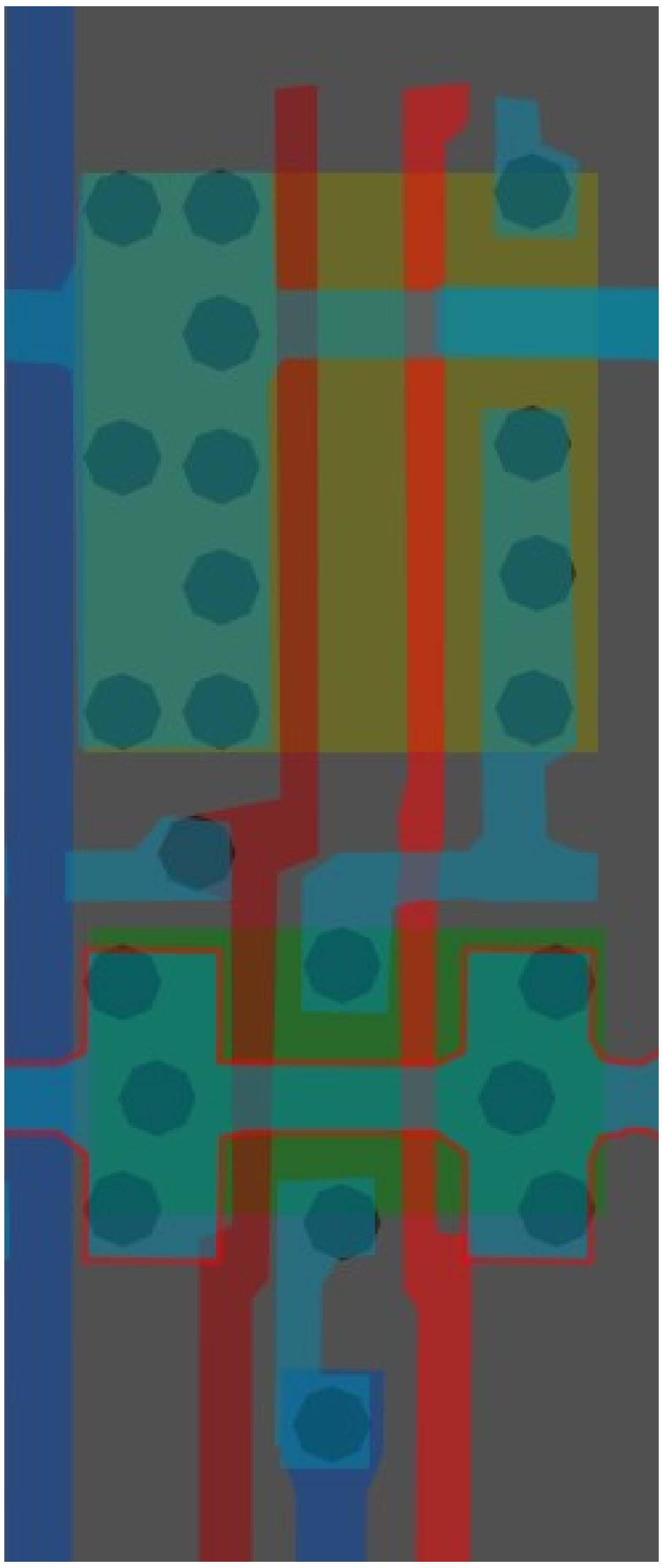


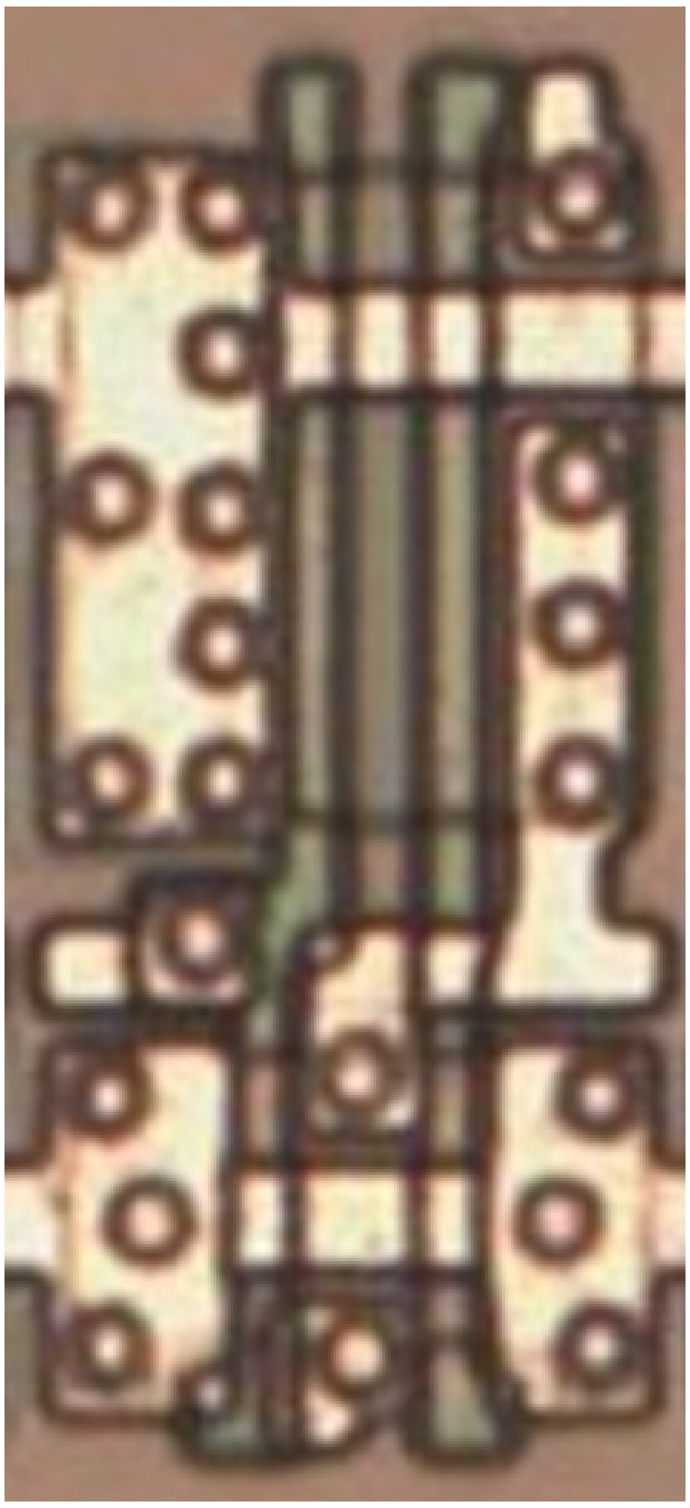
Same cell on LA32 (mirrored) (MB87136A)

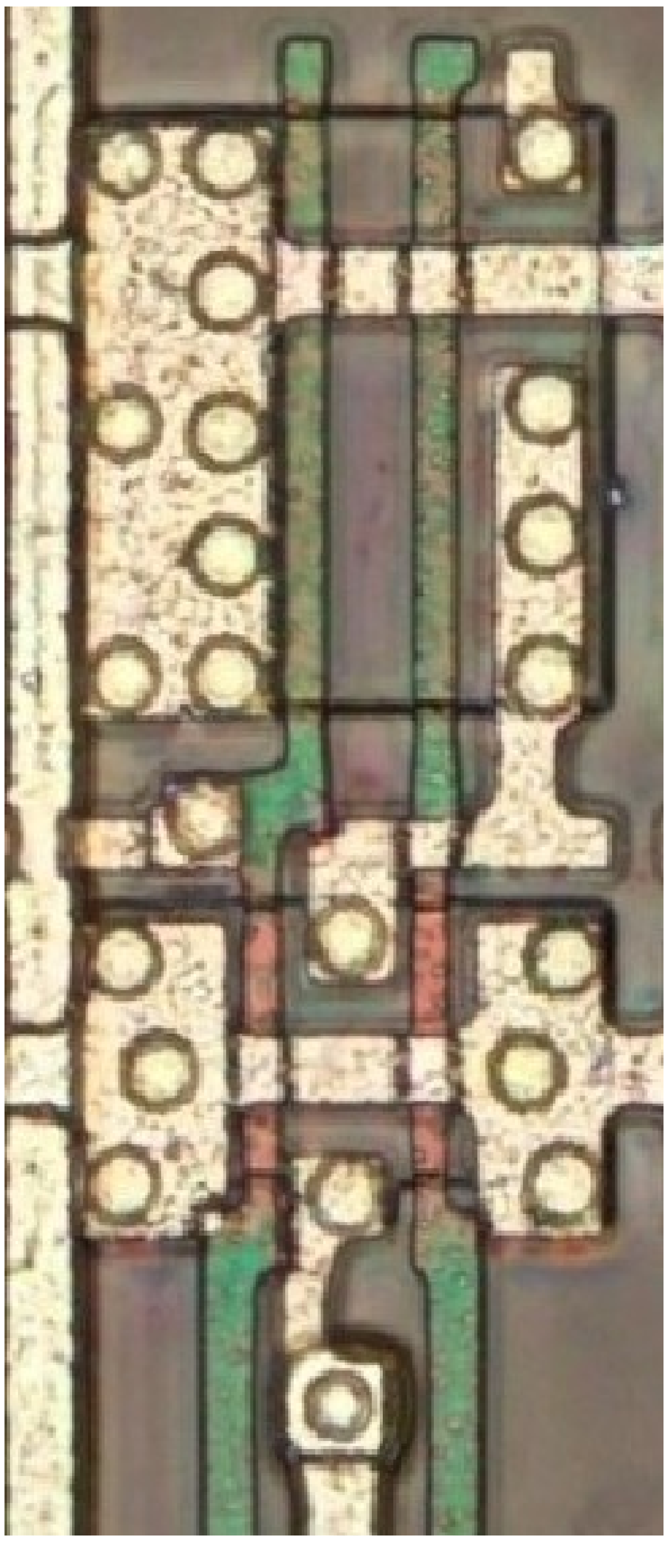
#005600ff 2-Input NOR Cell 18

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0



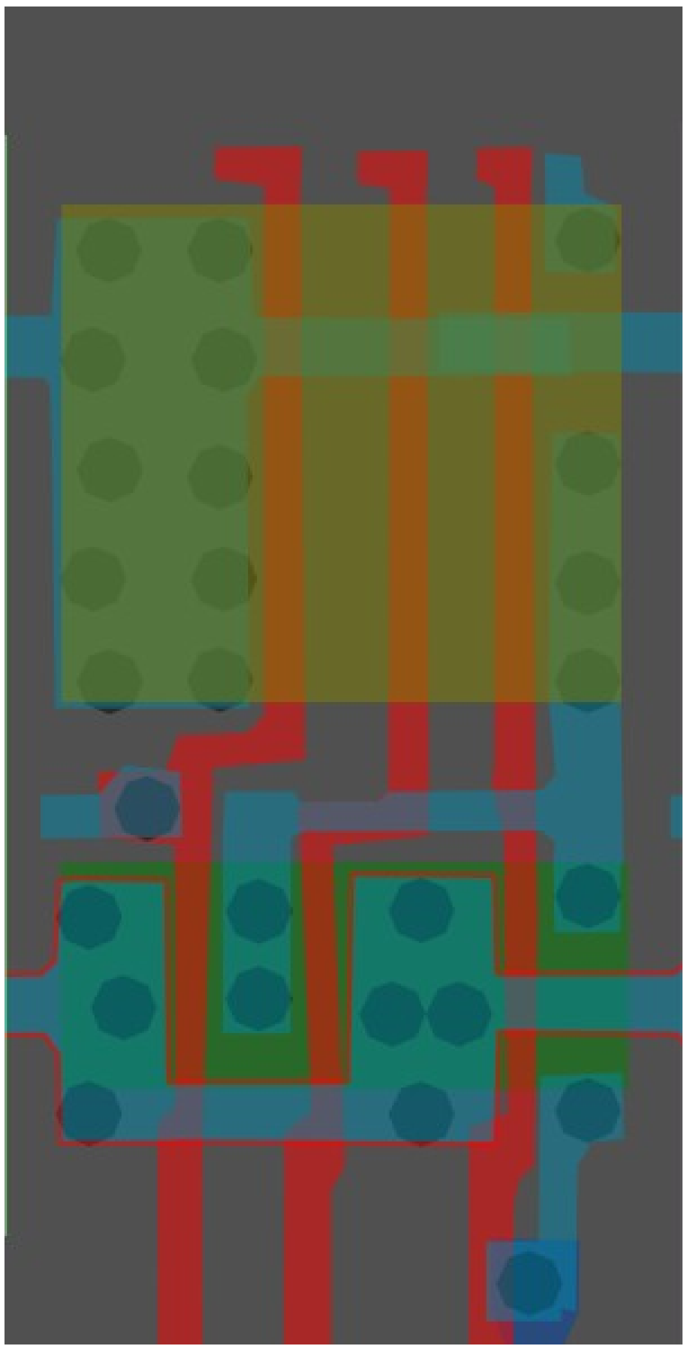


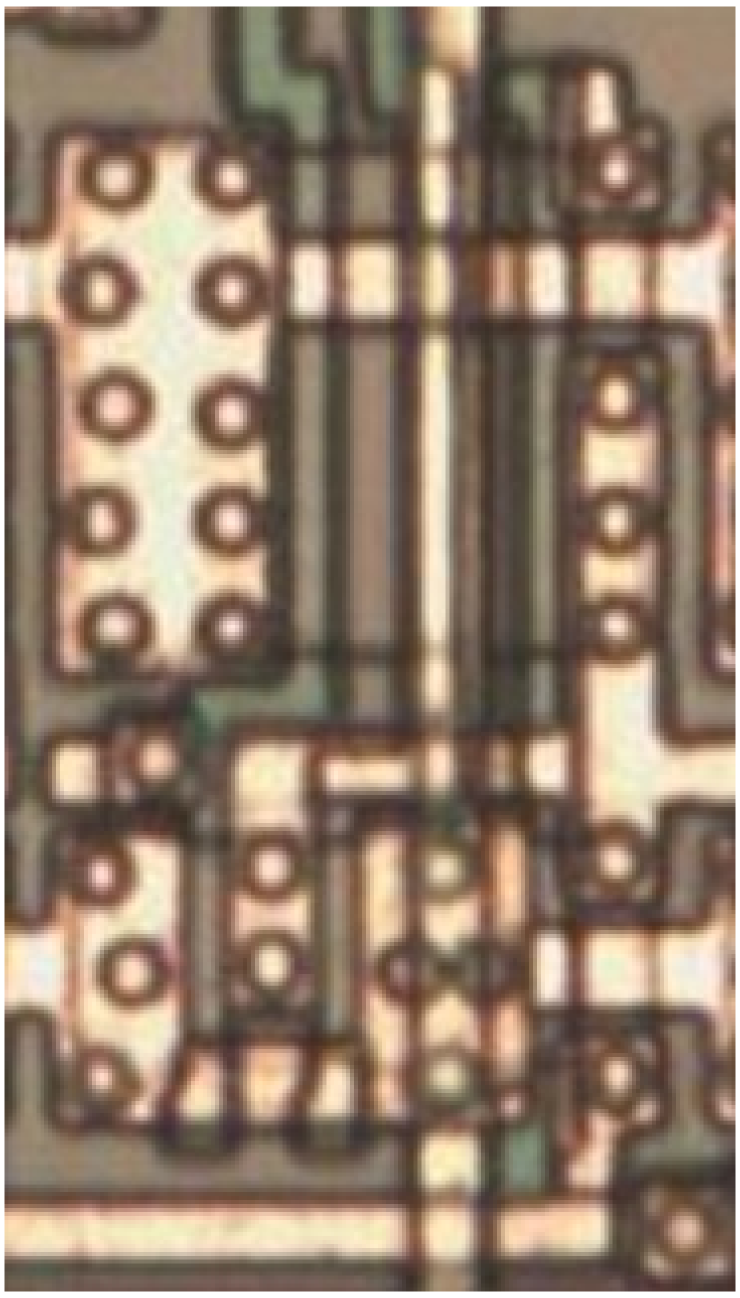




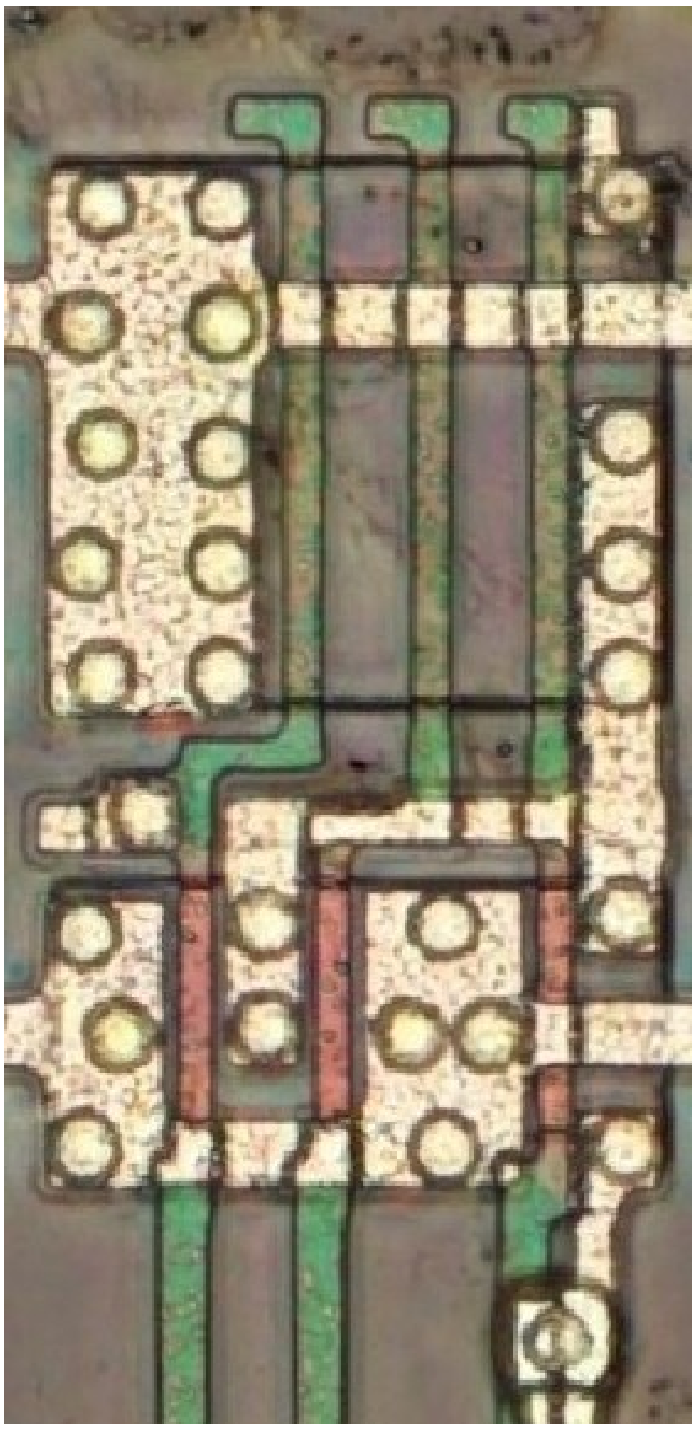
Same cell on
LA32
(MB87136A)

#970000ff 3-Input NOR Cell 19



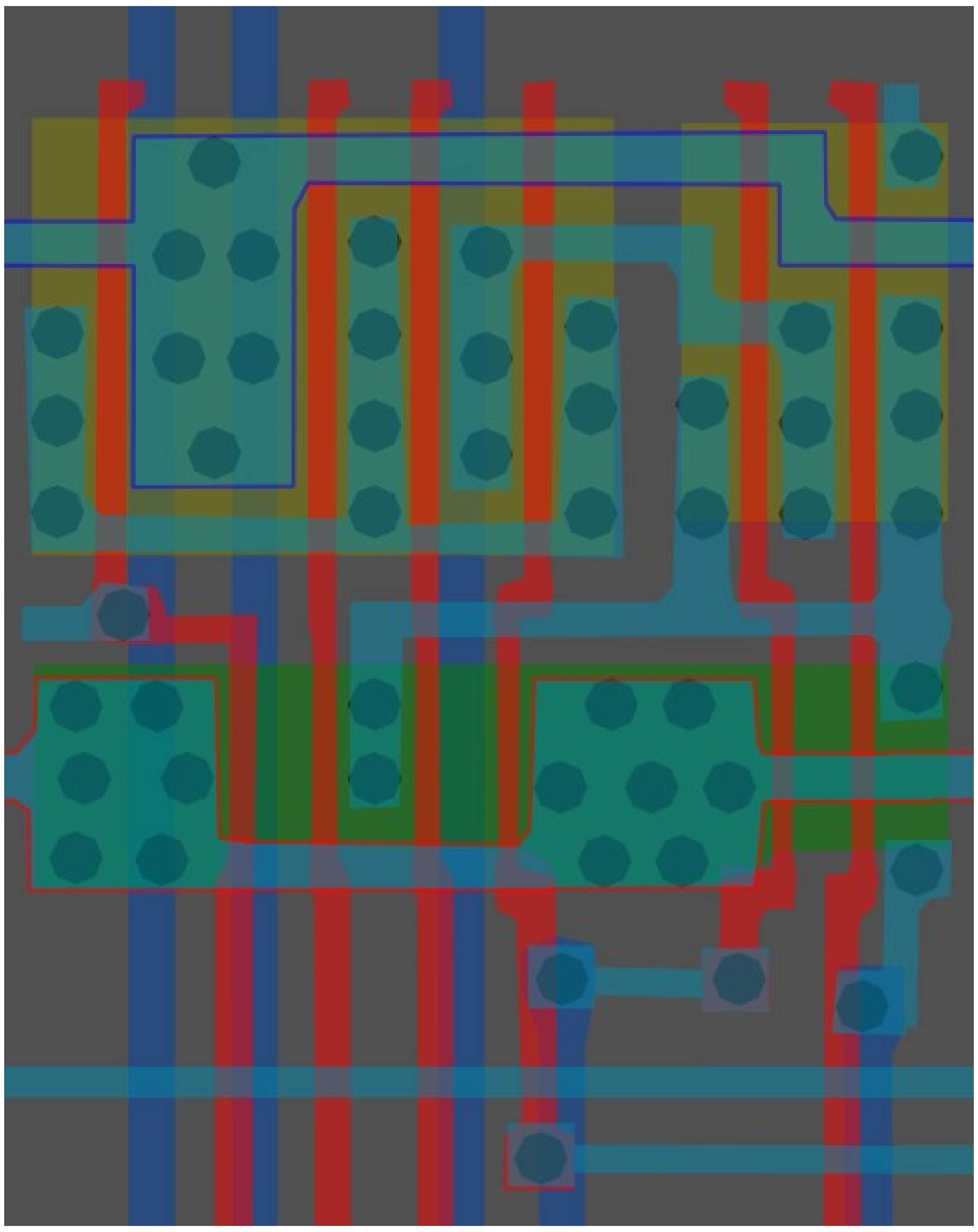


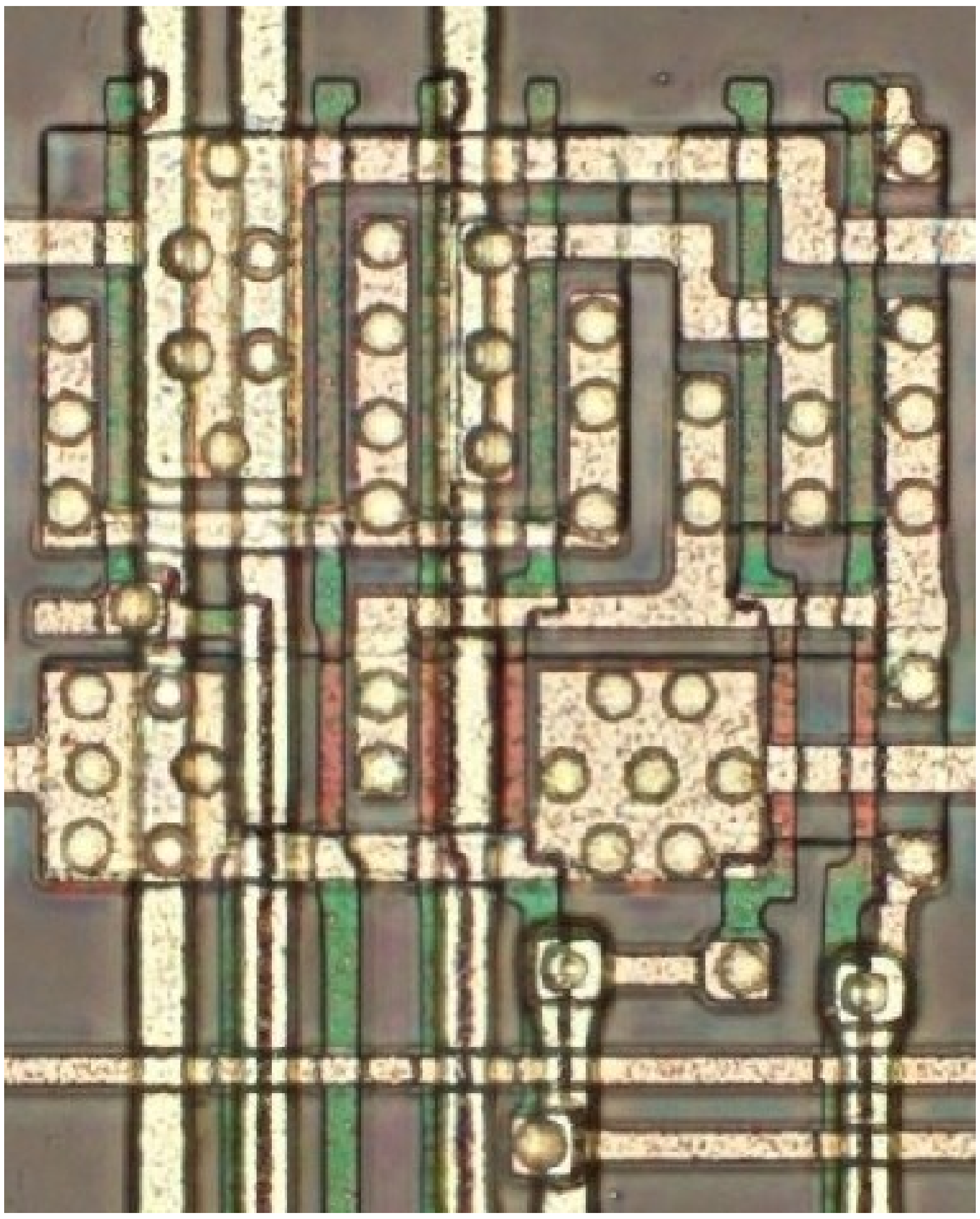
Same cell on
LA32
(MB87136A)



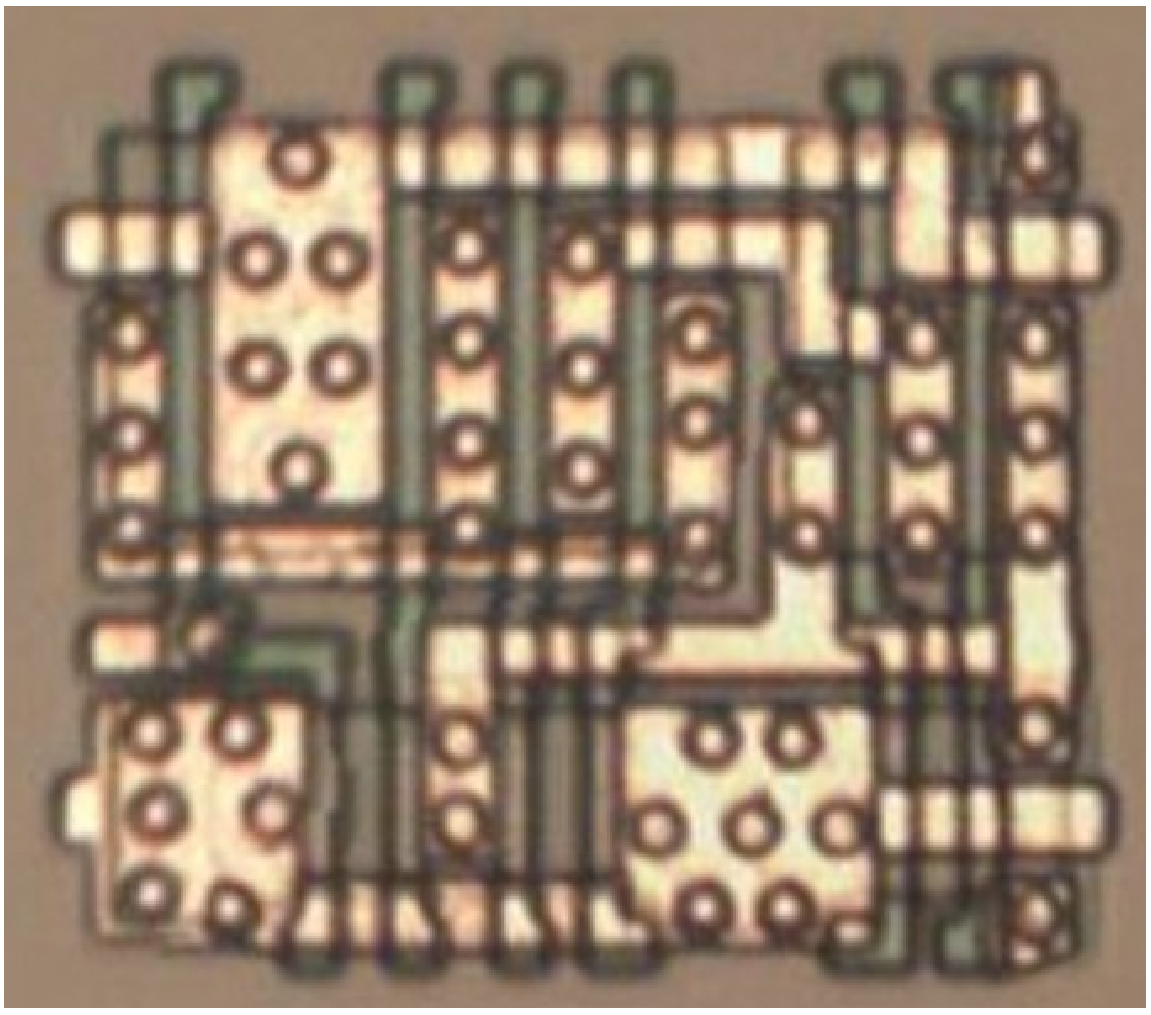
#ff0000ff

Cell type 20

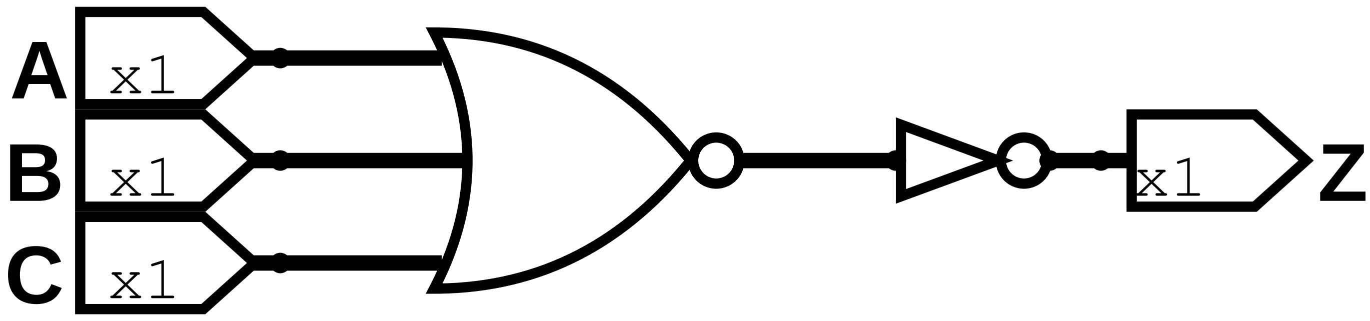




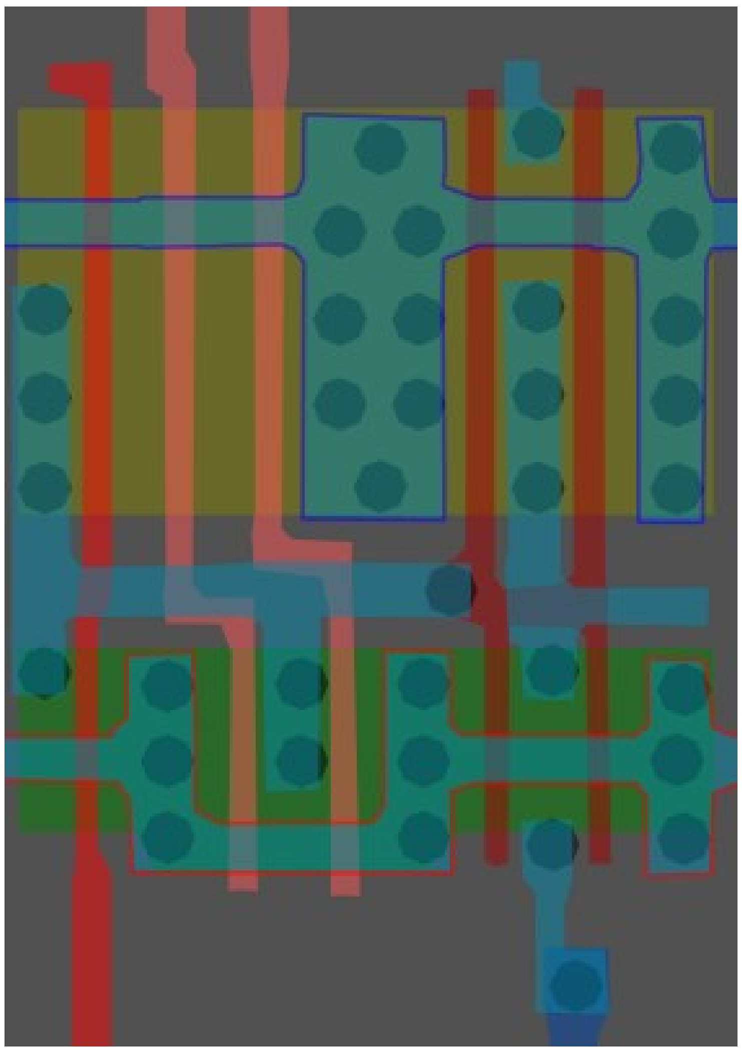
Same cell on LA32 (MB87136A)

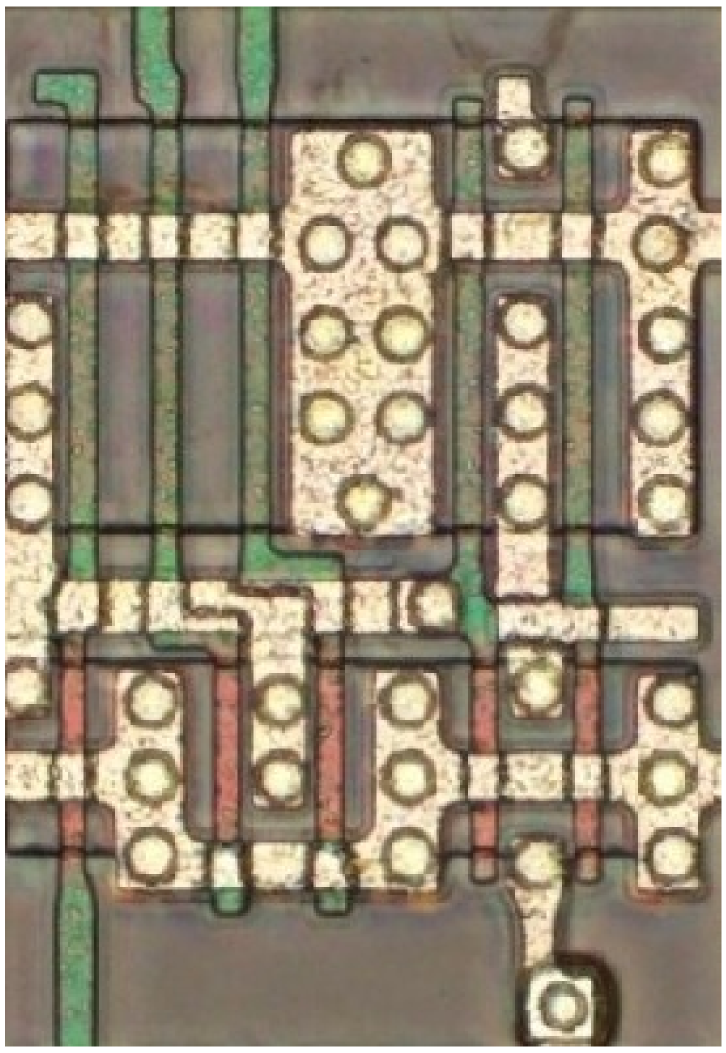


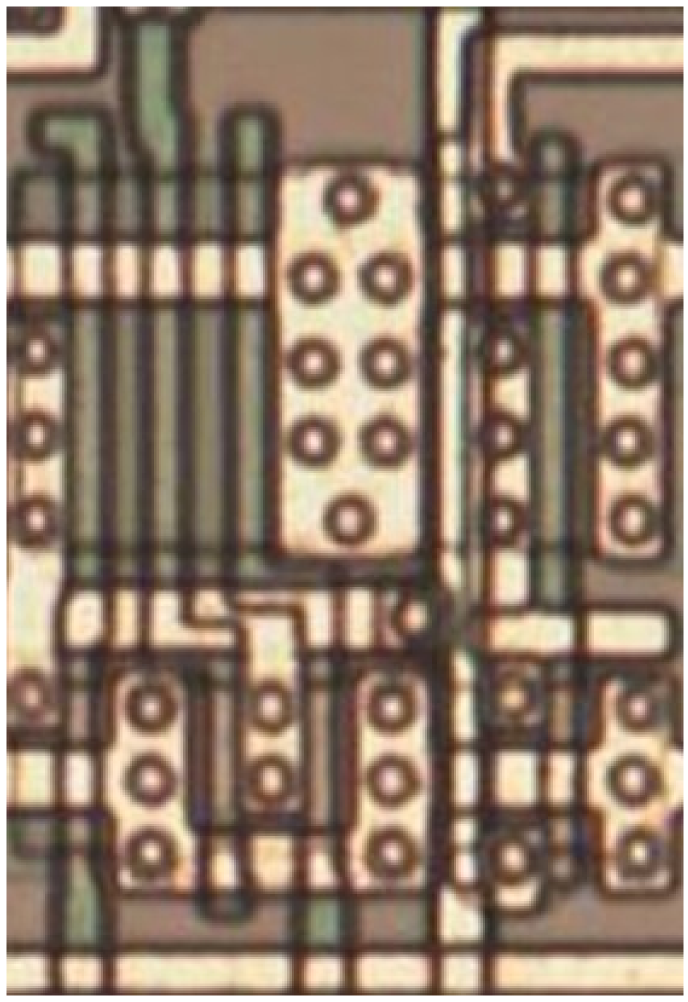
#218268ff 3-Input OR x1 Cell 21a



A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

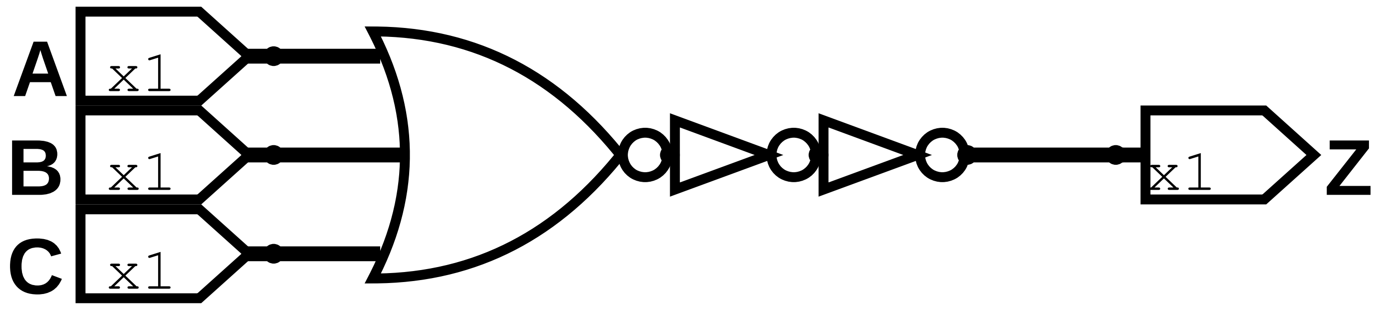


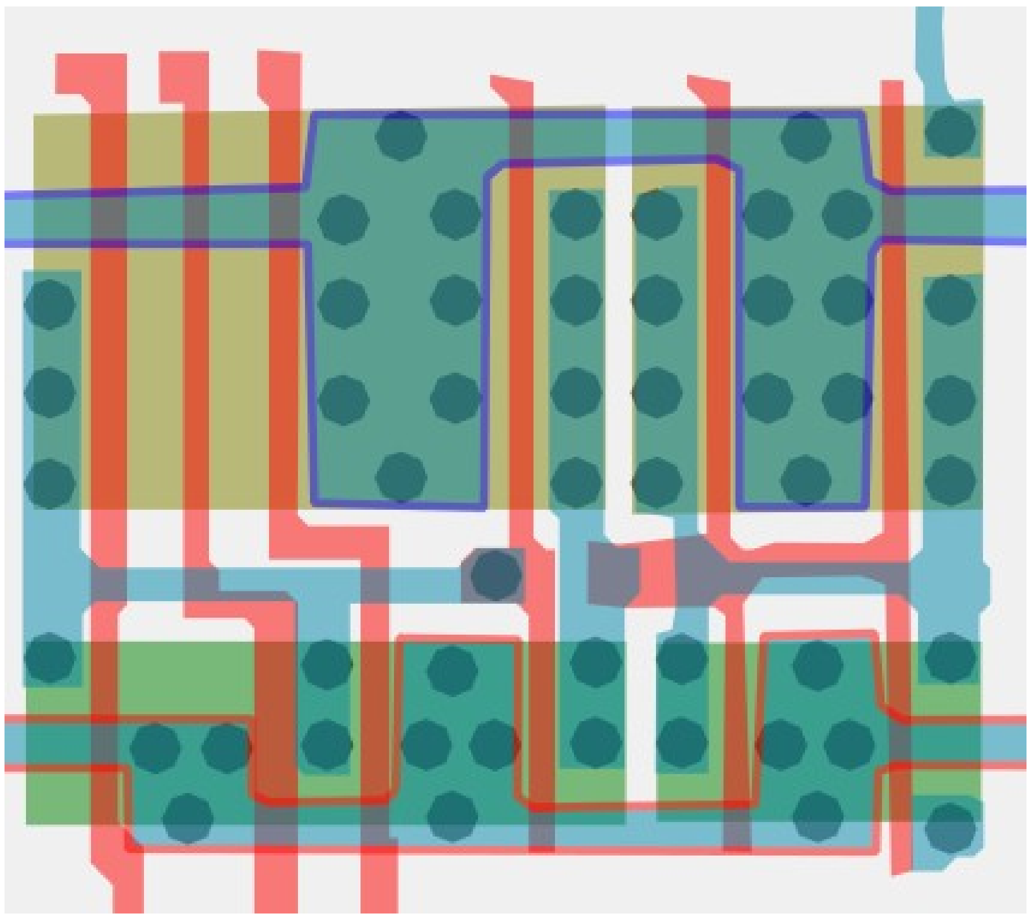


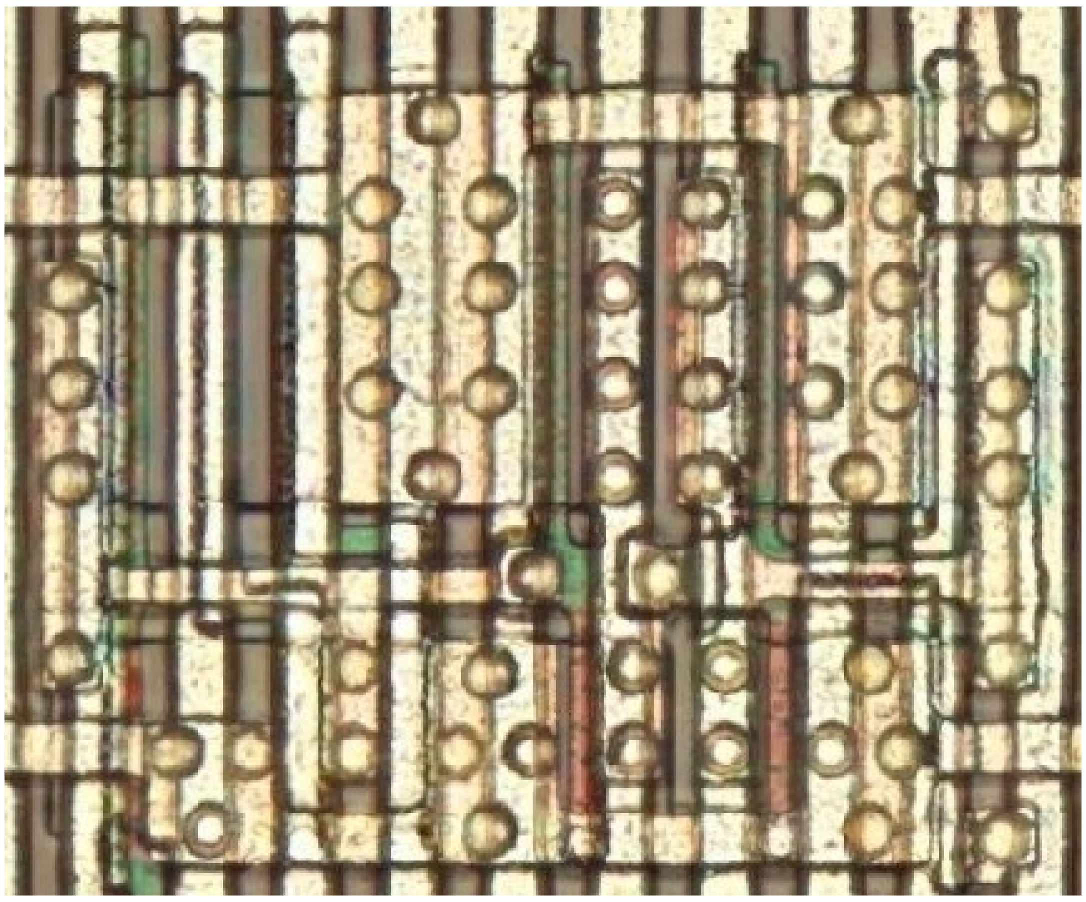


Same cell on LA32 (MB87136A)

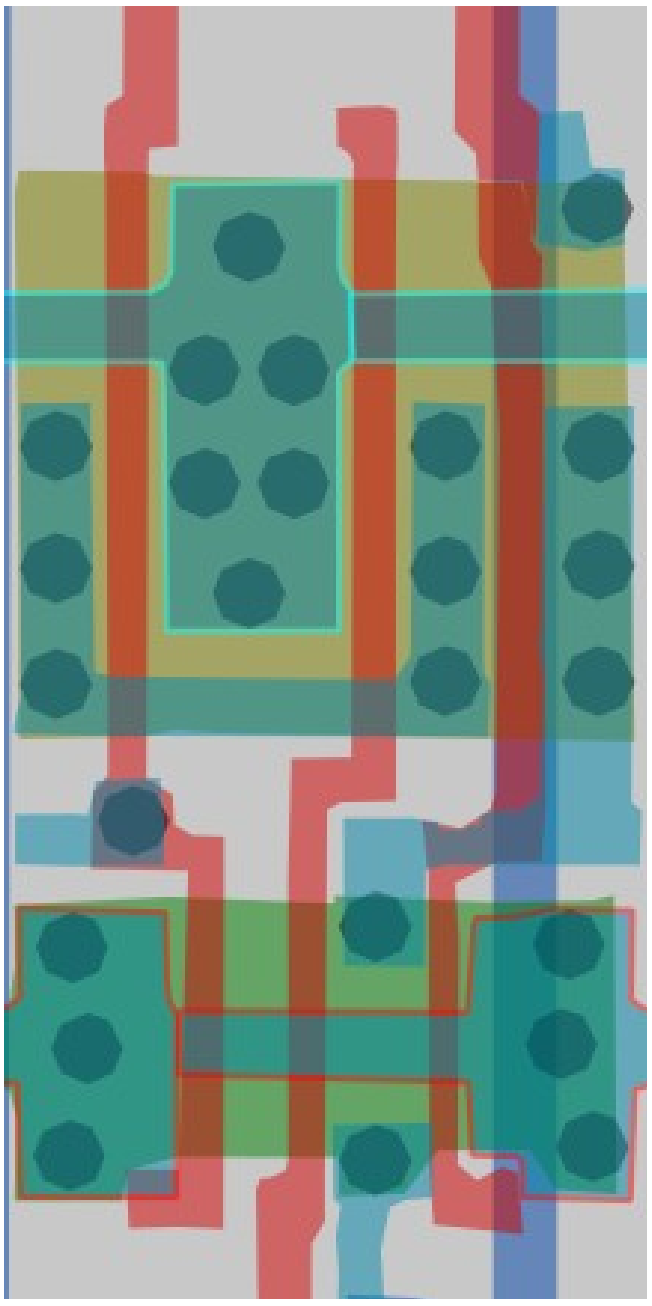
#115140ff 3-Input NOR Cell 21b

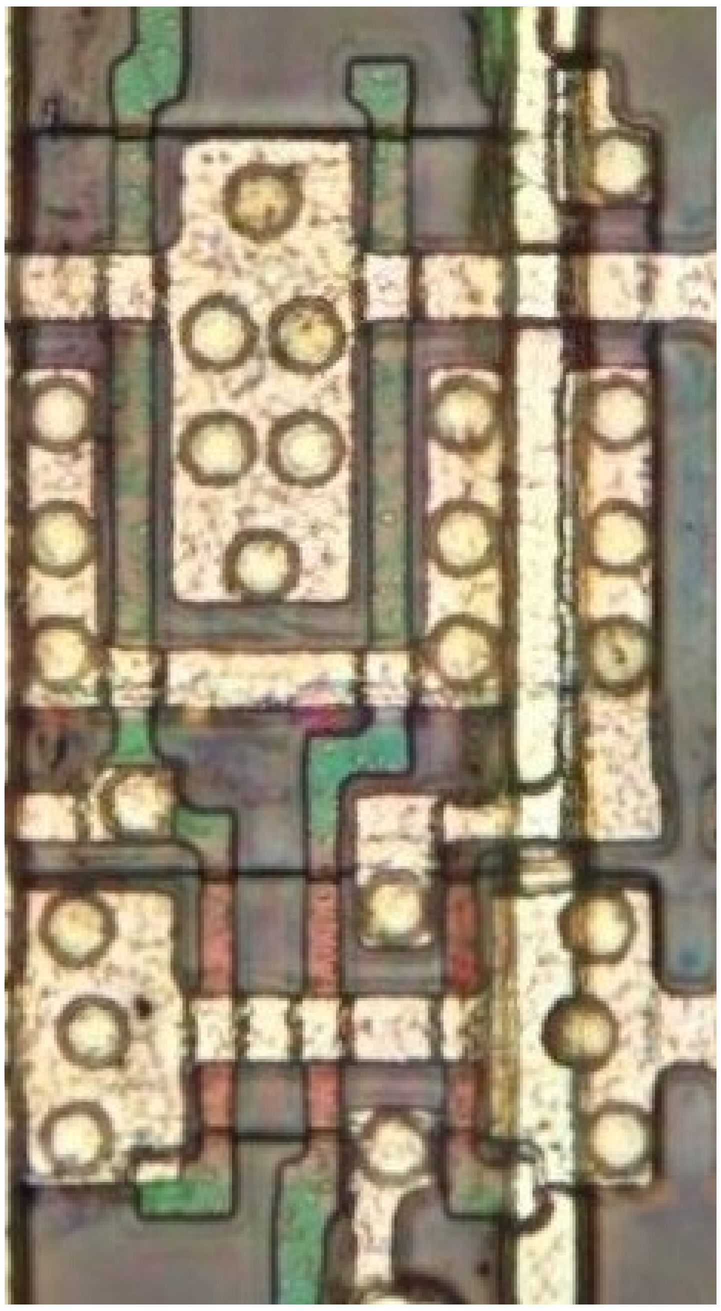






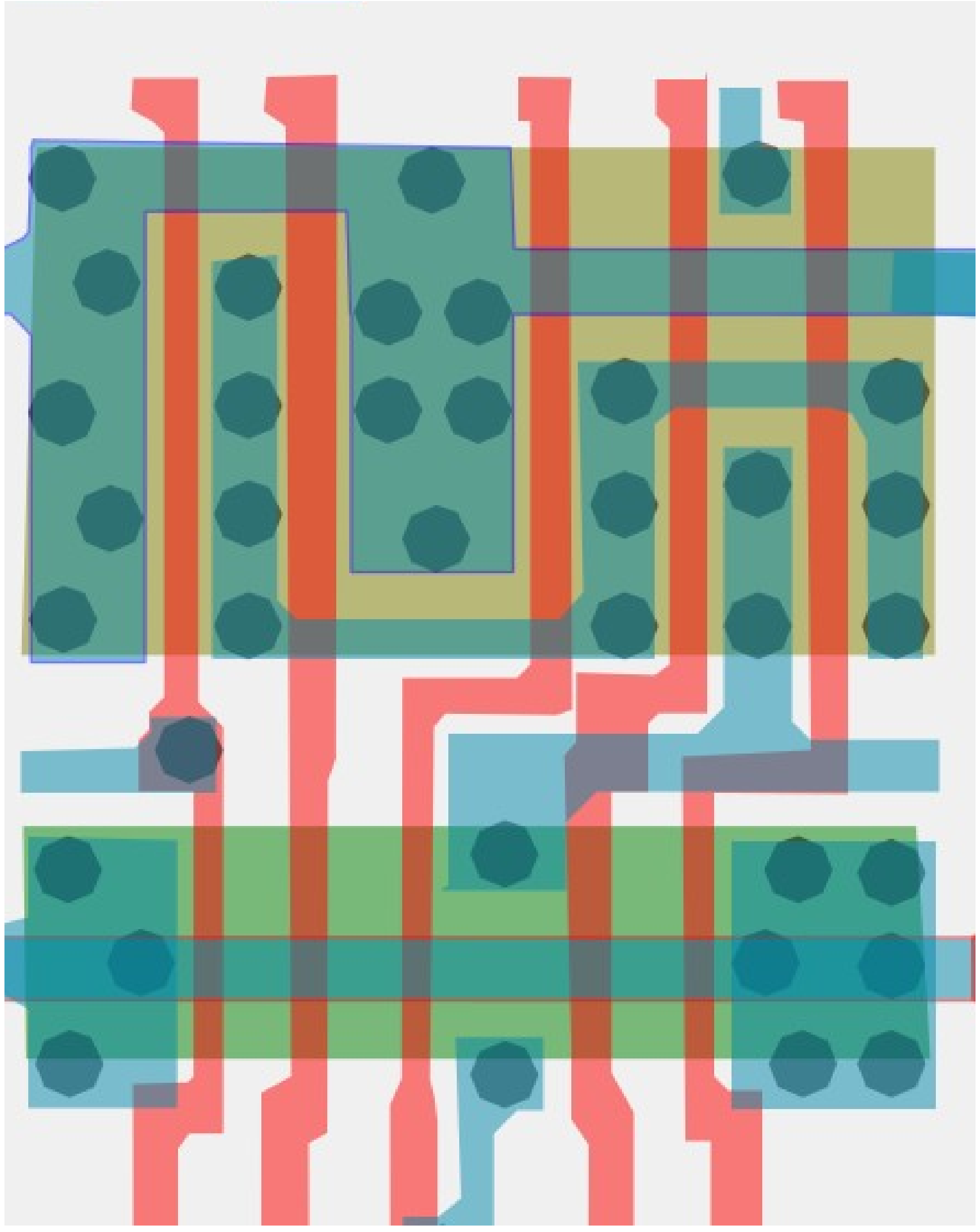
#d1796bff Cell type 24a

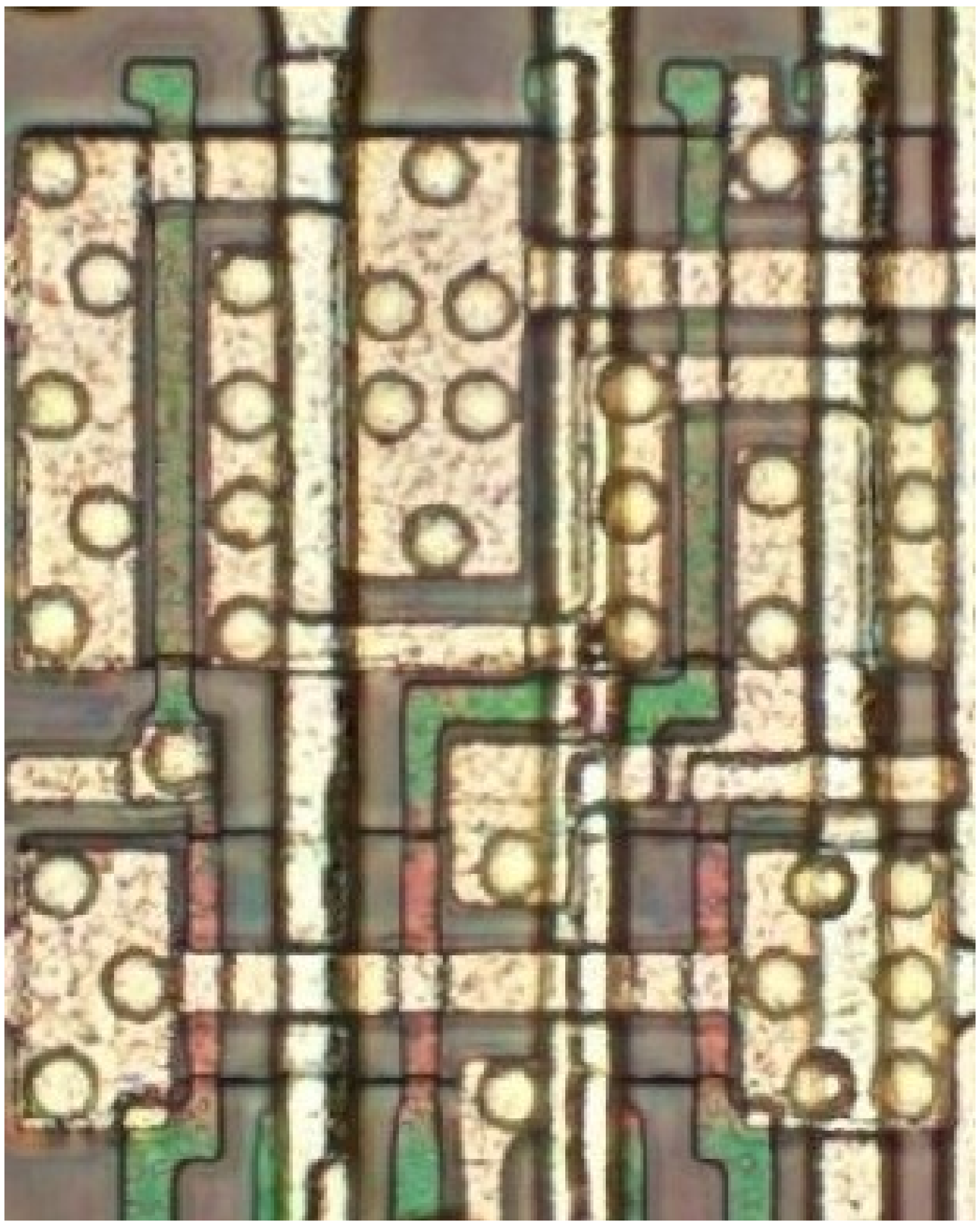




#7f8157ff

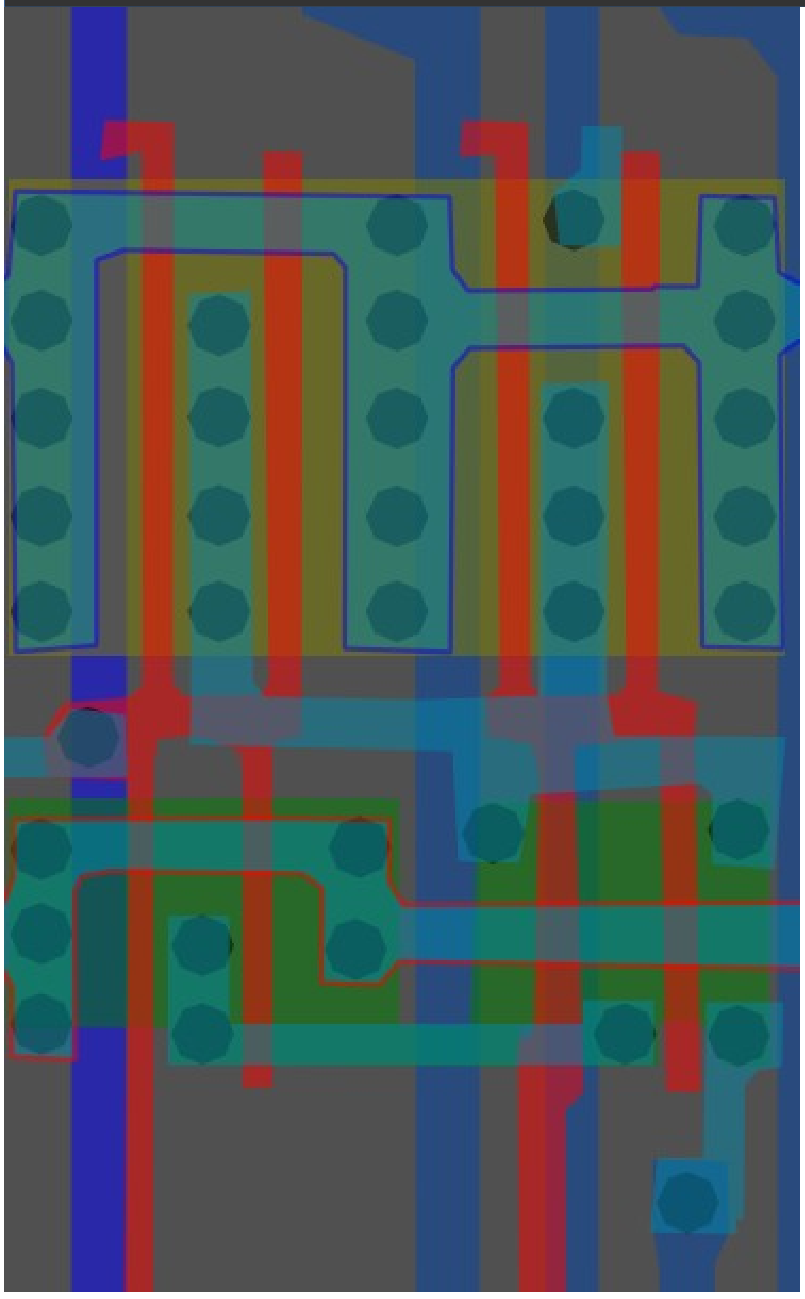
Cell type 24b

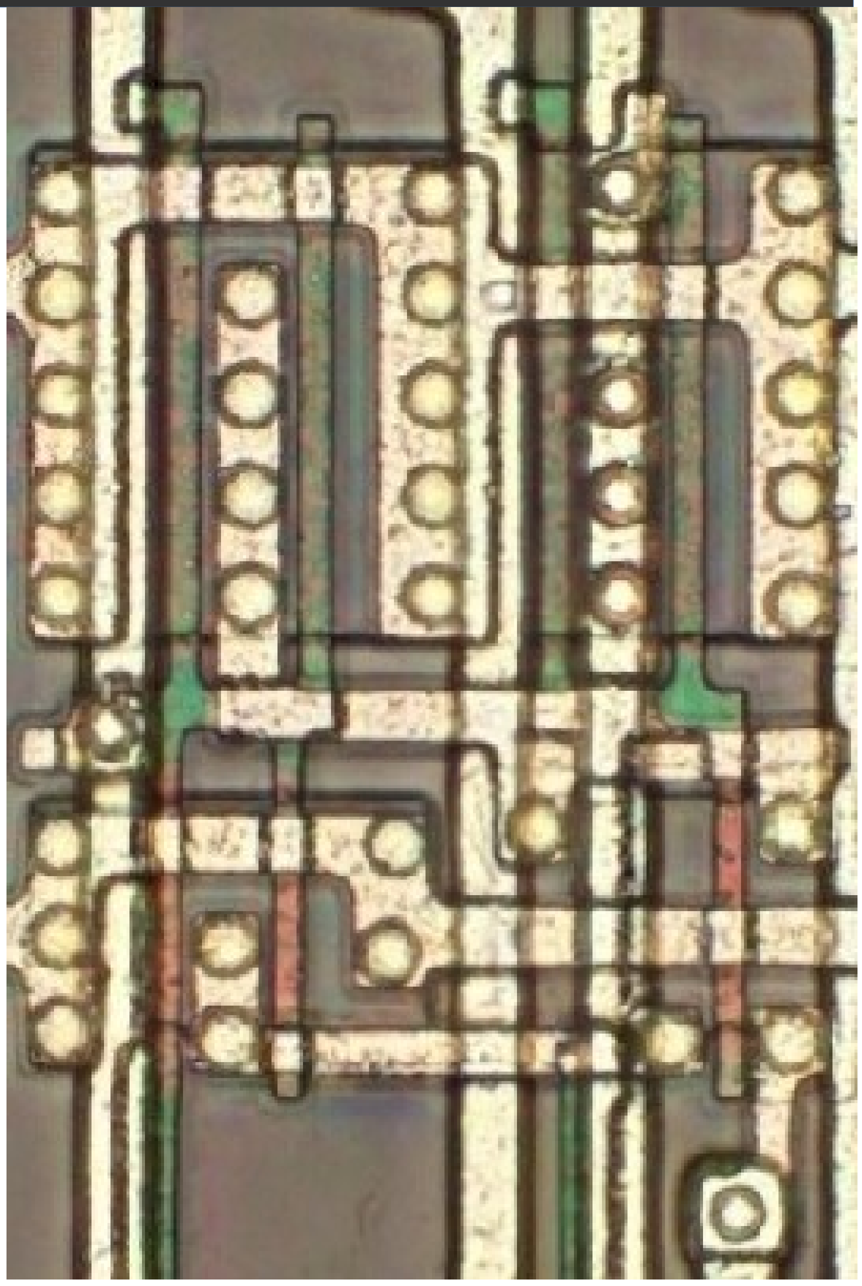




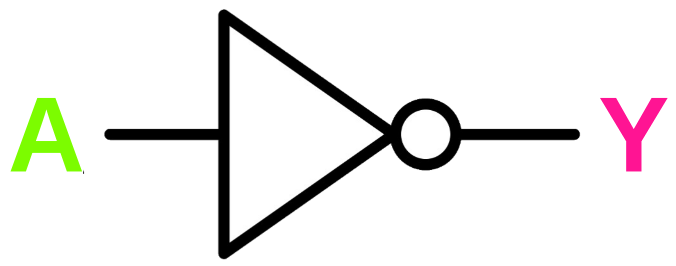
#ff9972ff

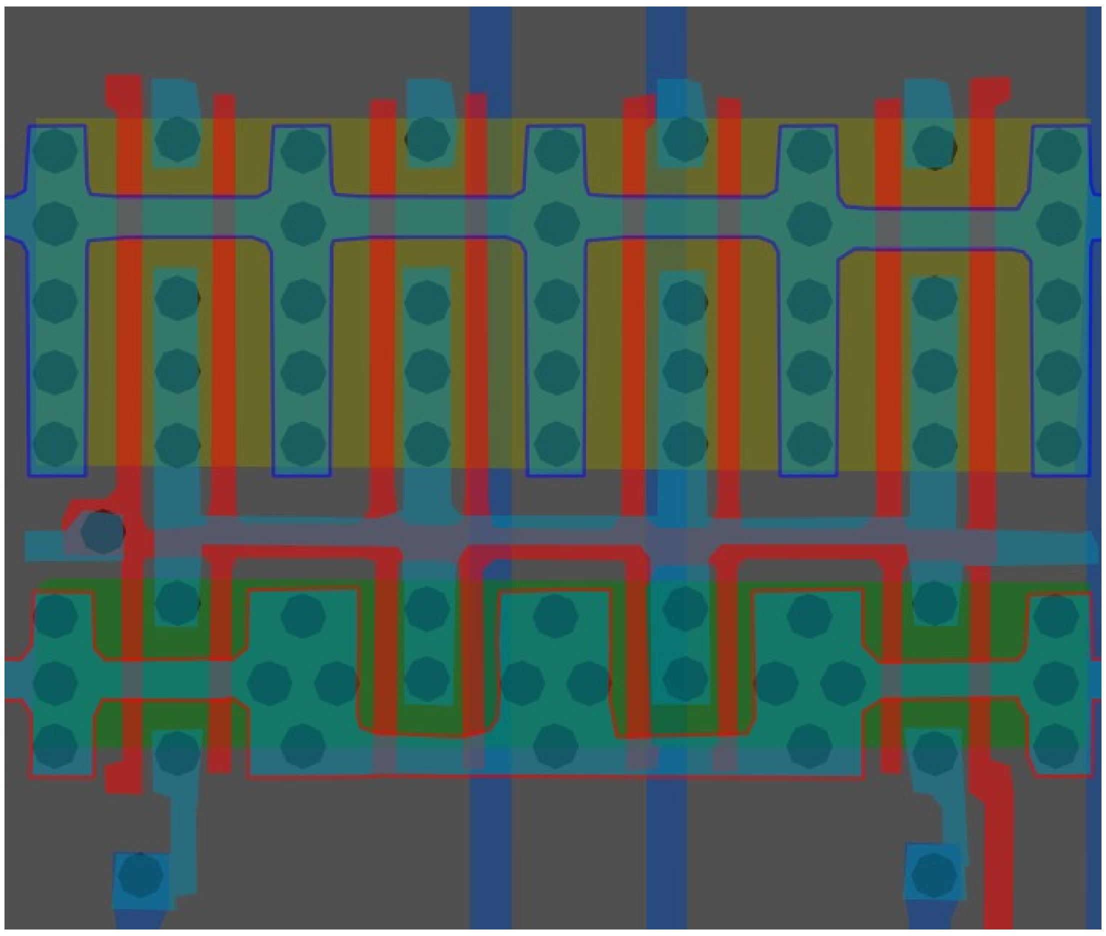
Cell type 25

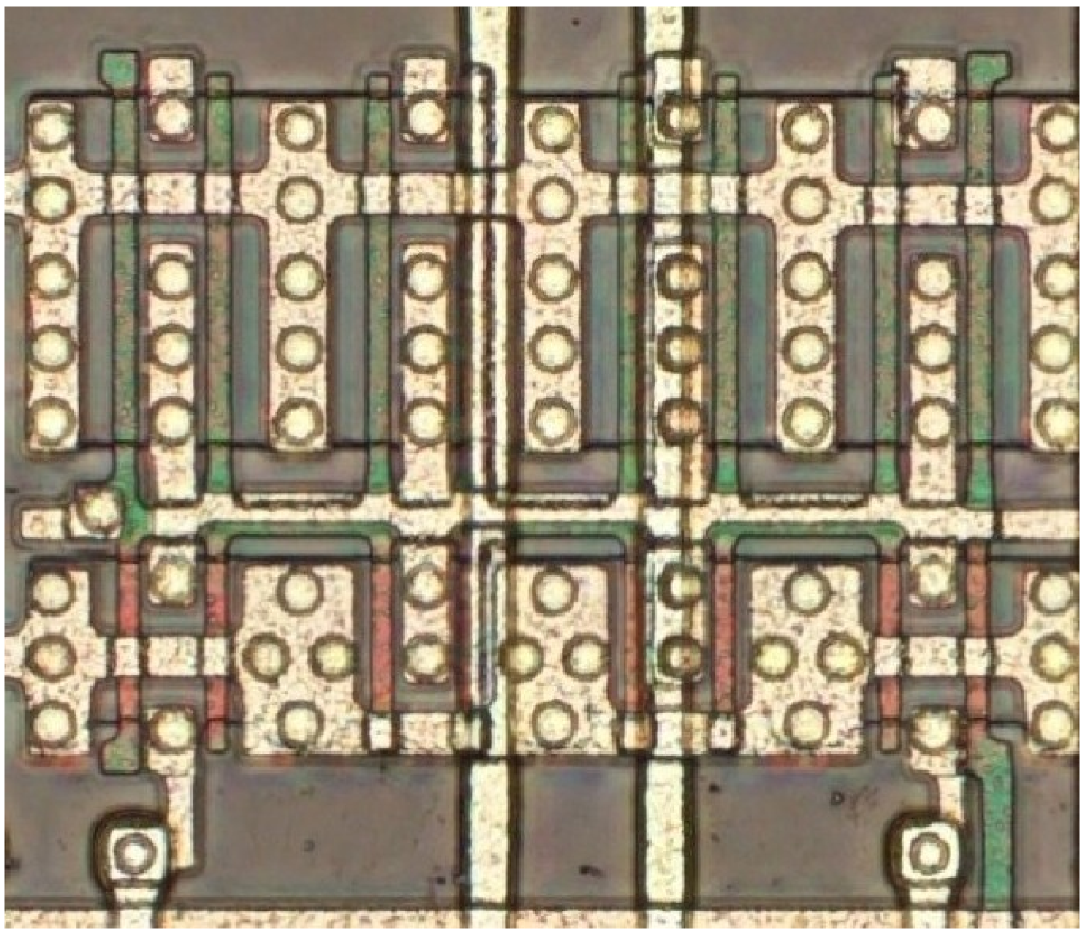


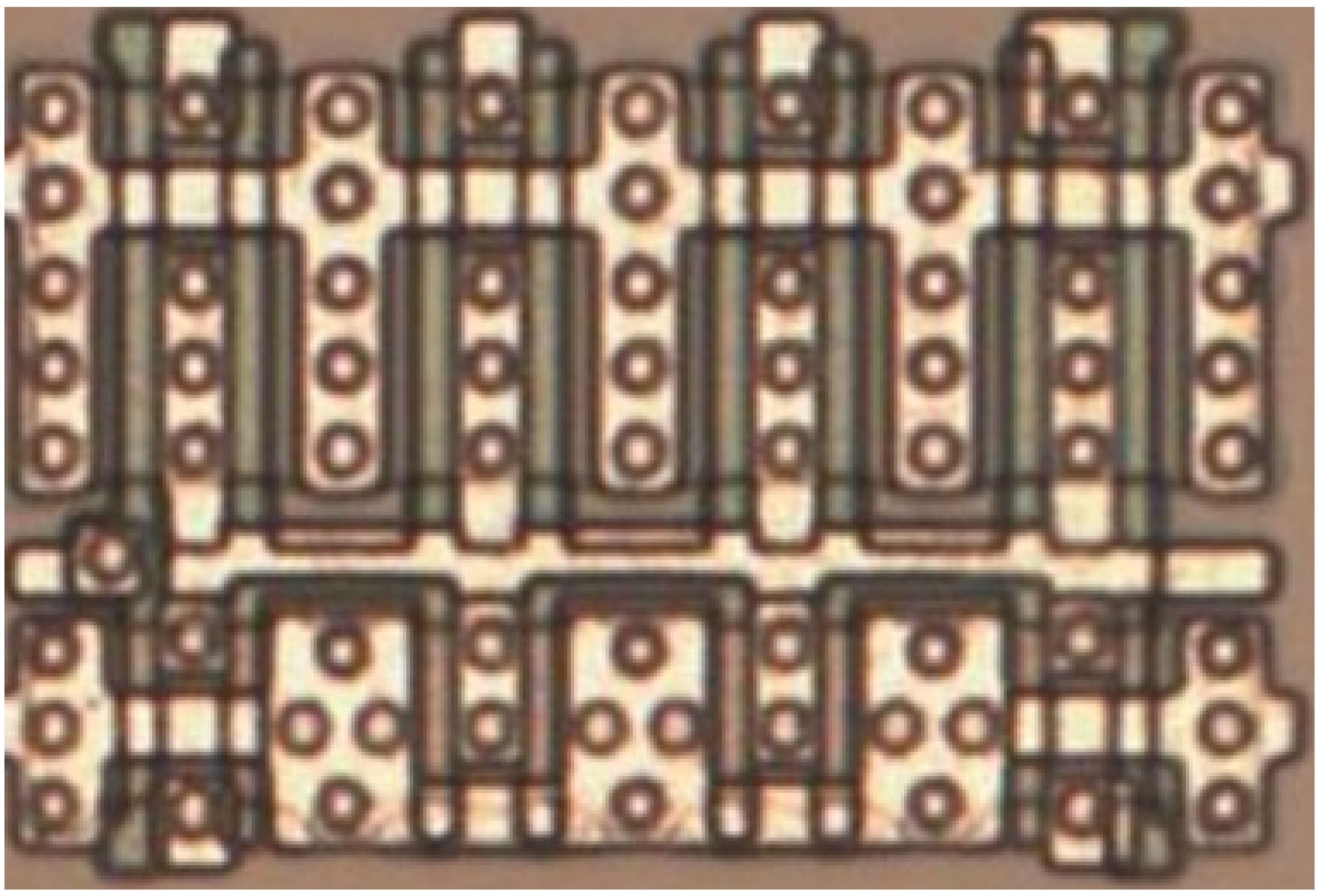


#ffb94cff Inverter/NOT x8 Cell 26



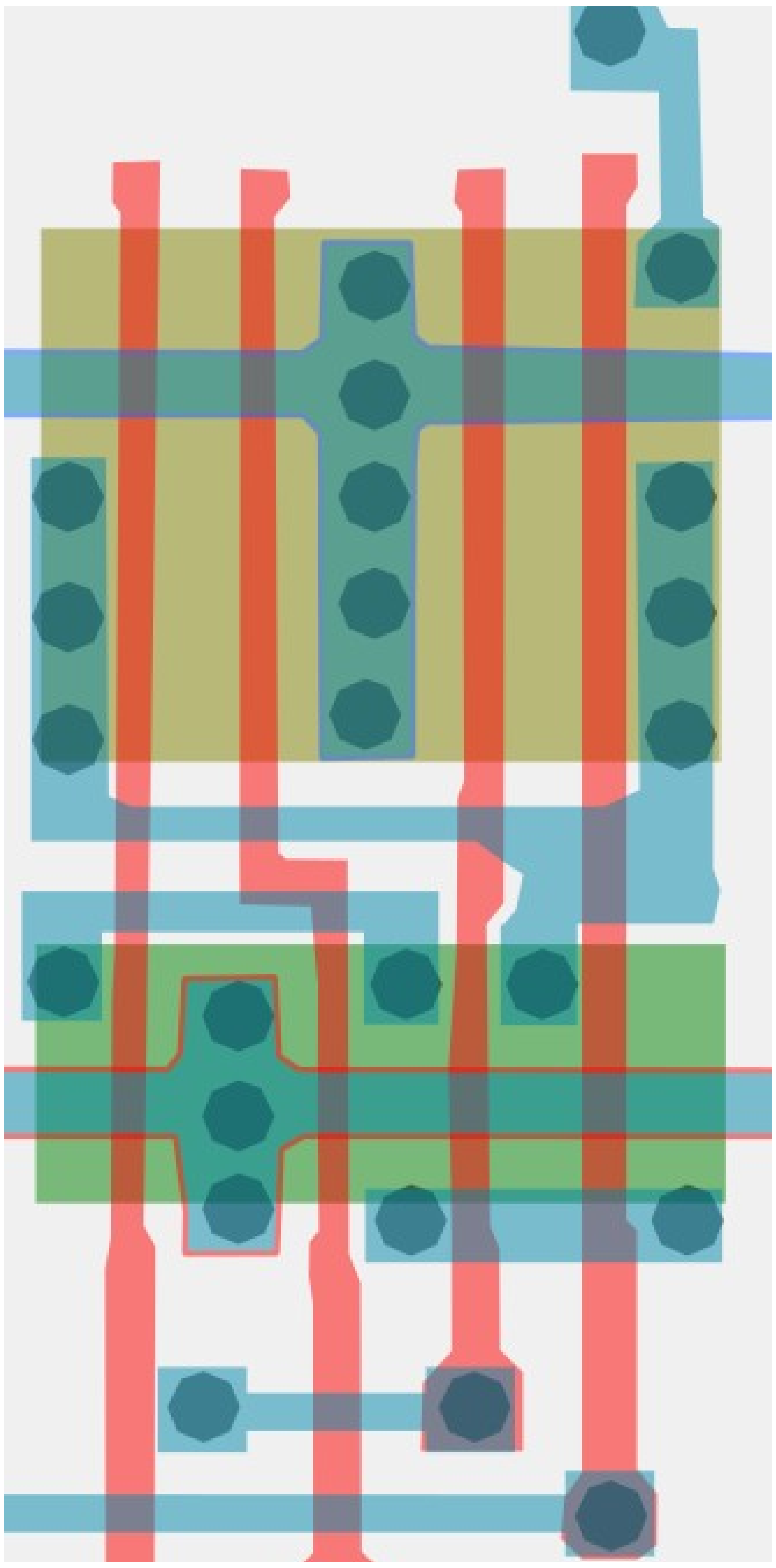


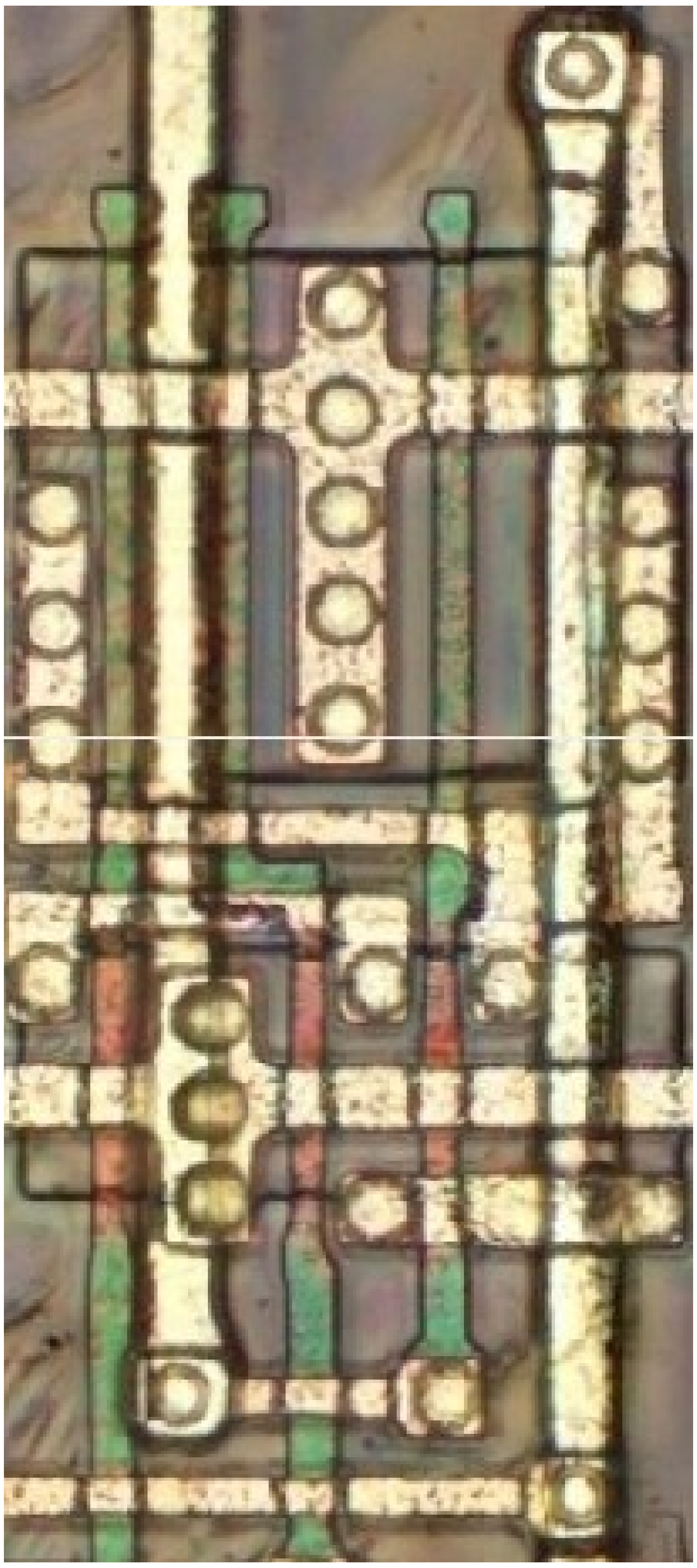




Same cell on
LA32
(MB87136A)

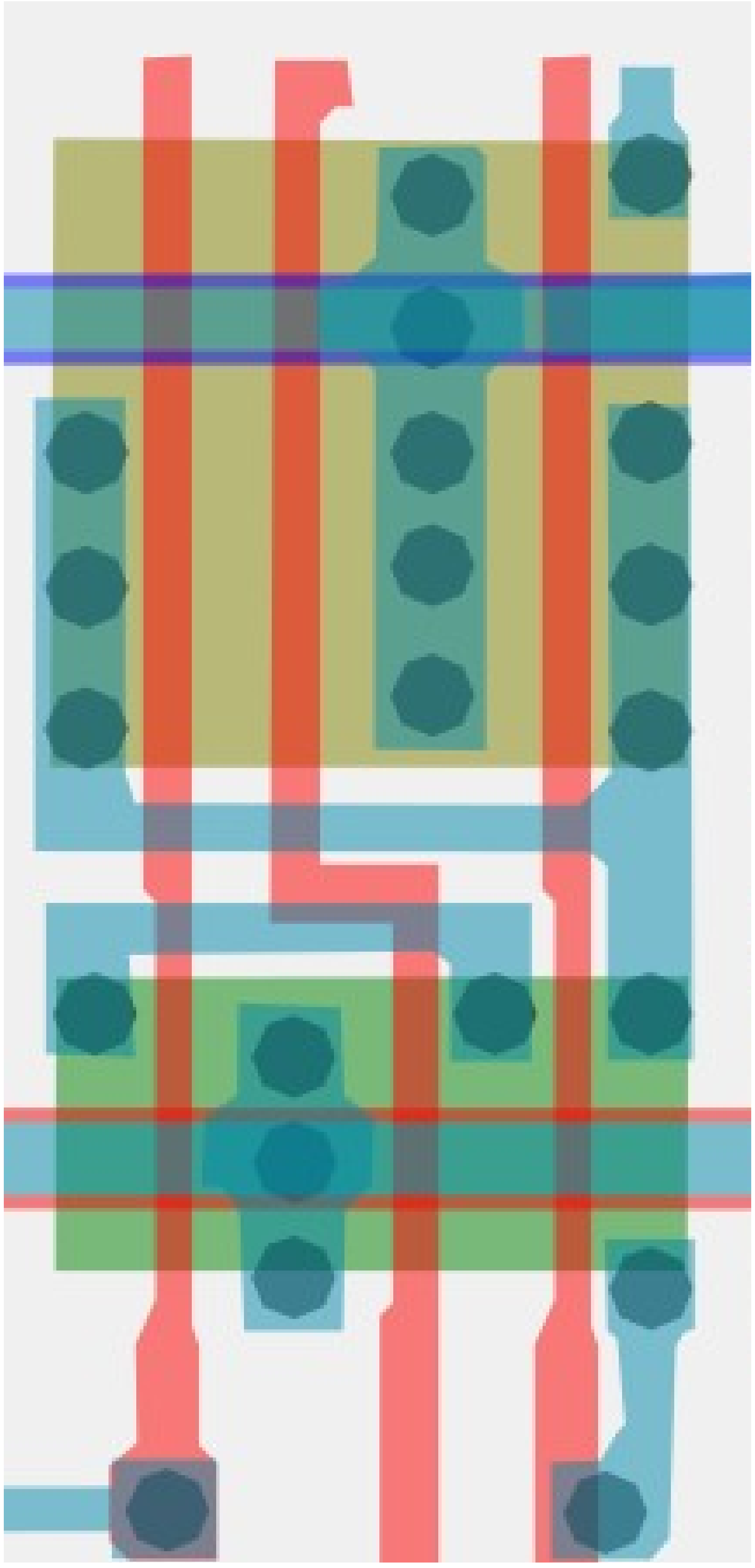
#bec1c3ff Cell type 27a

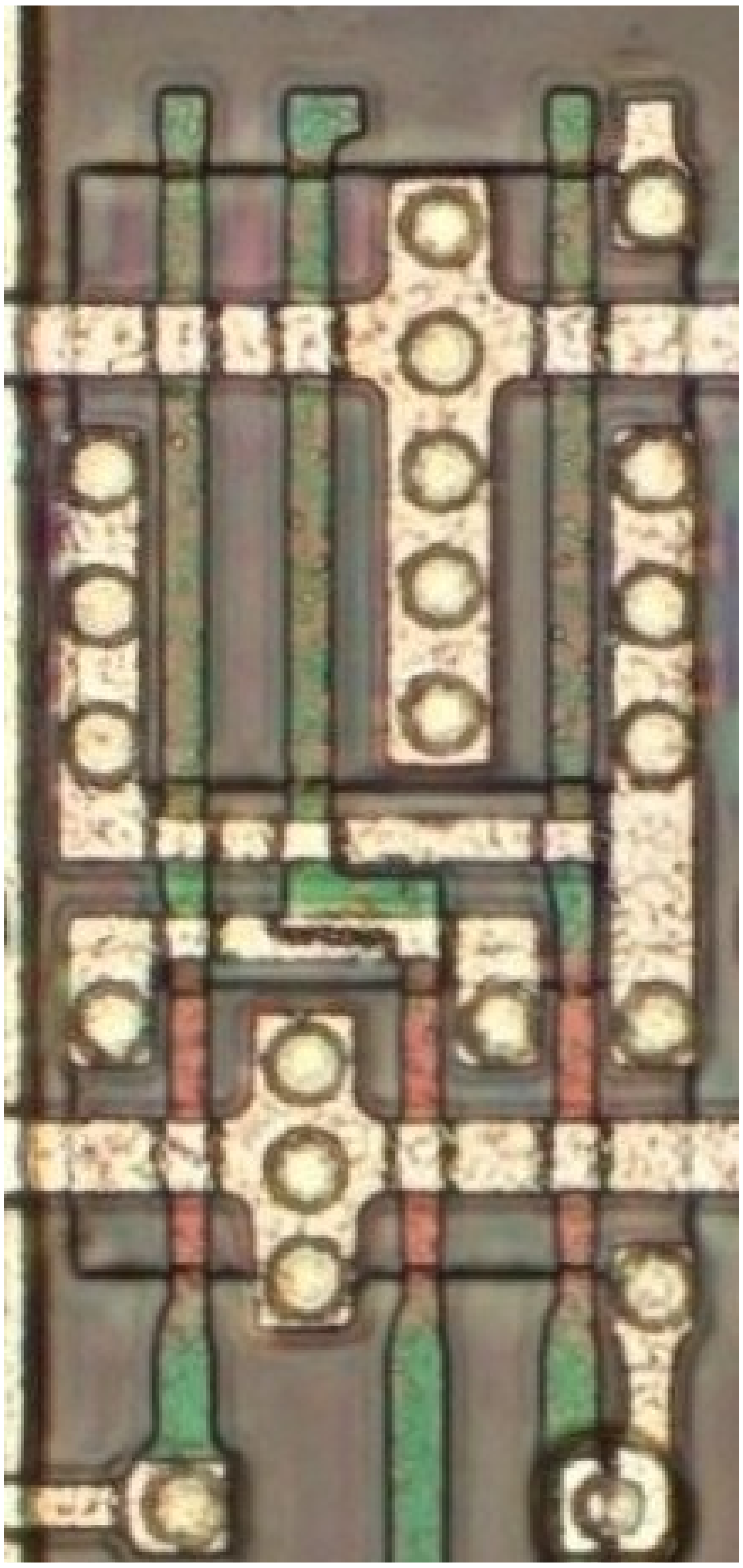




#787878ff

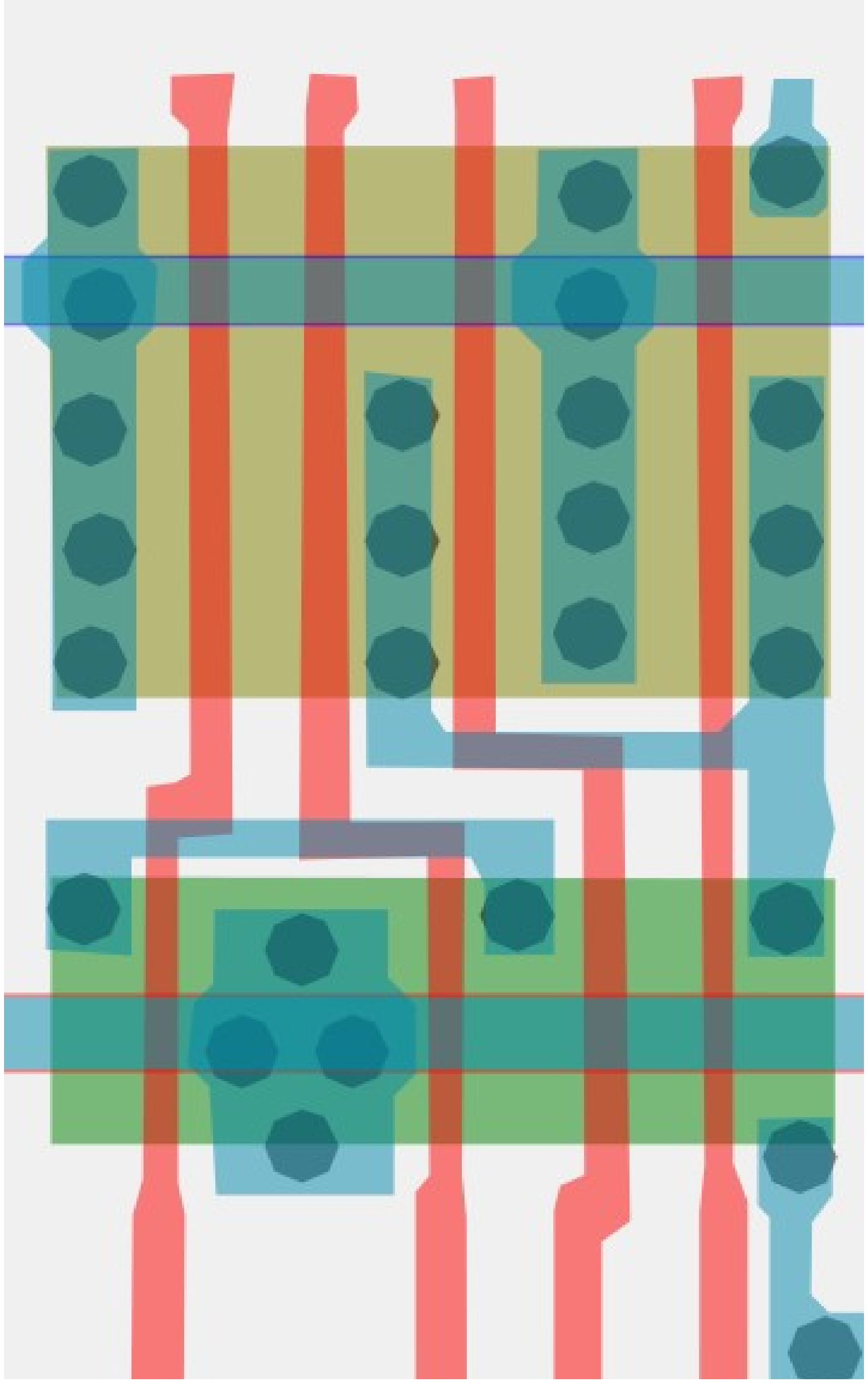
Cell type 27b

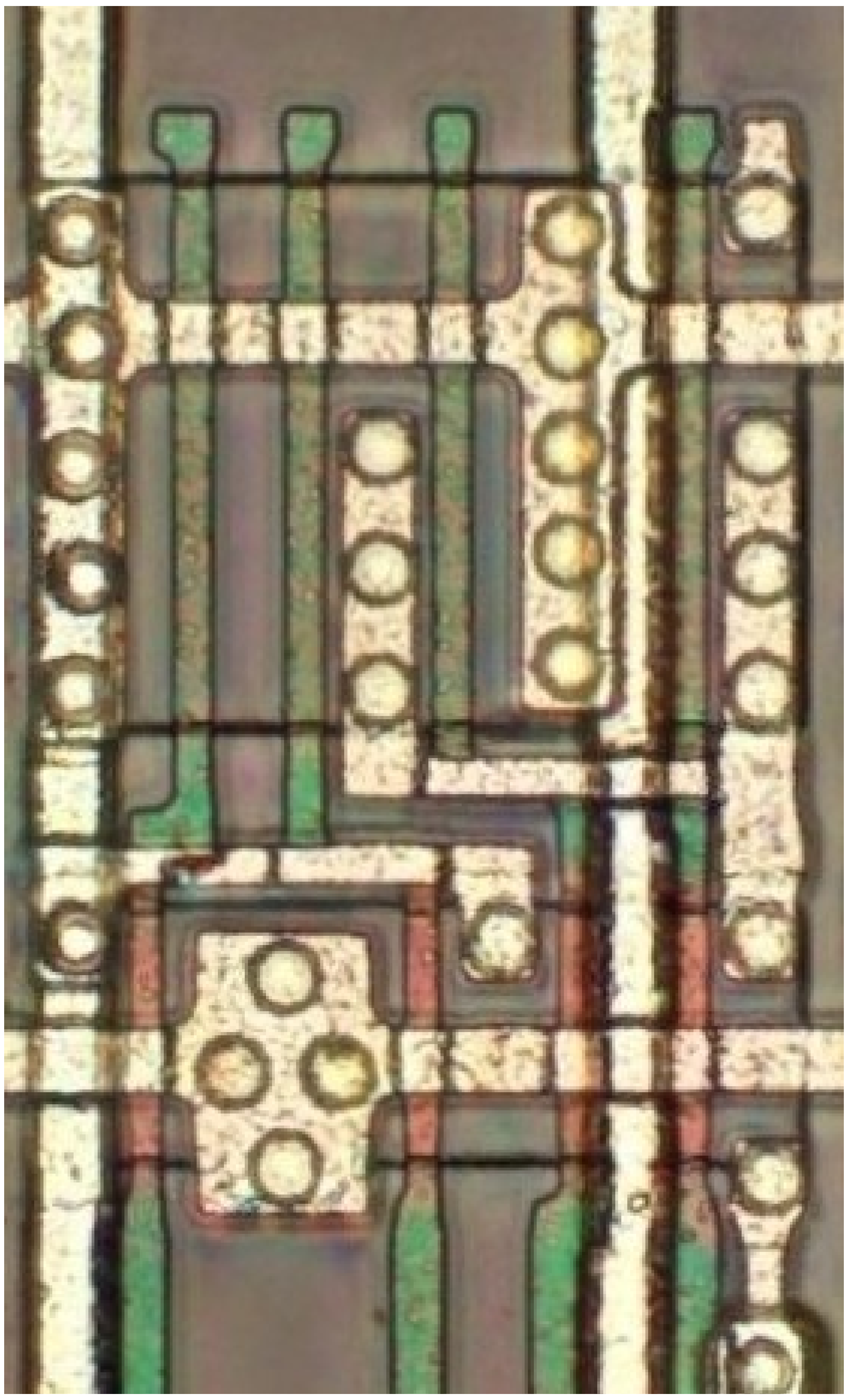




#5d5d5dff

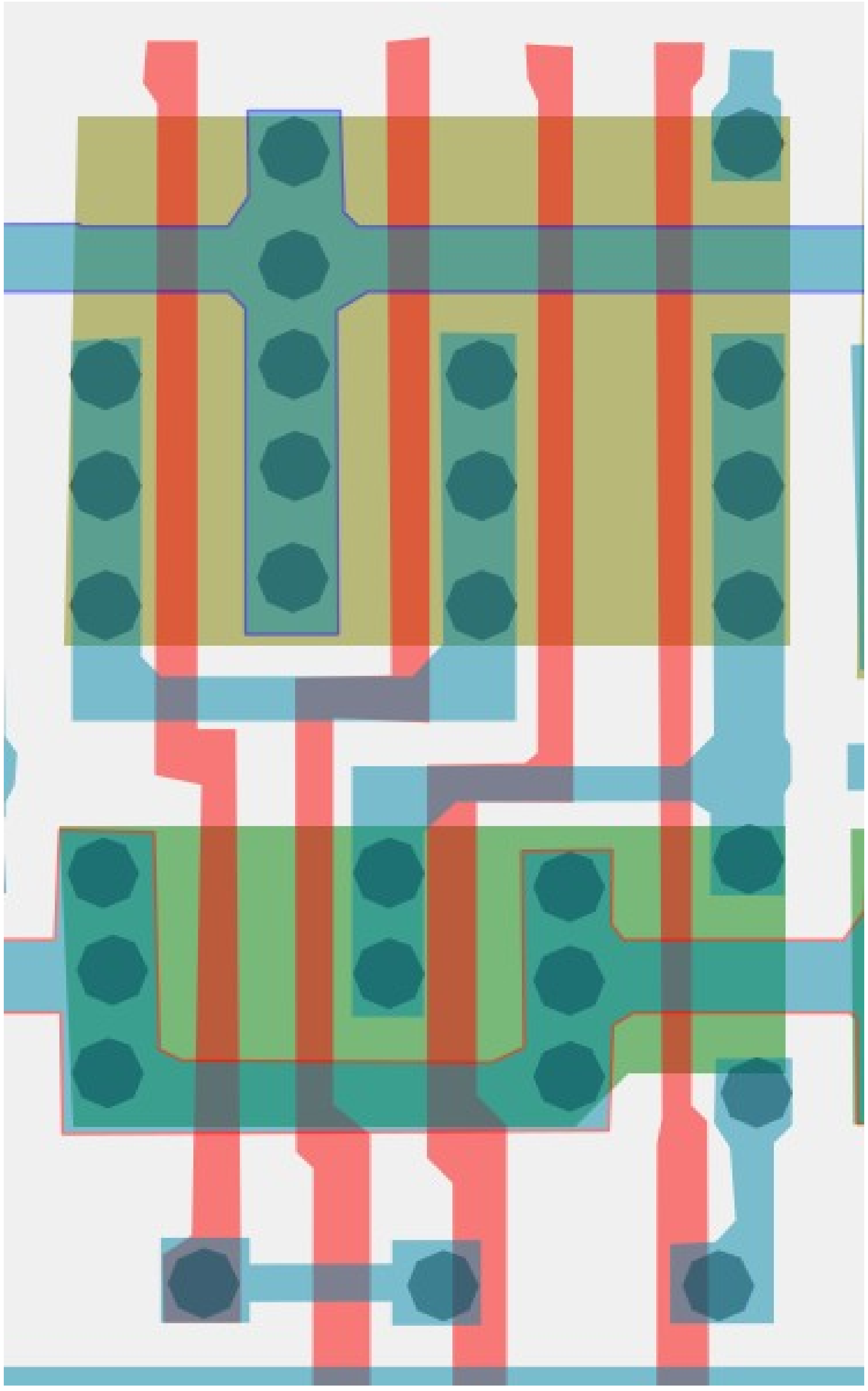
Cell type 27C

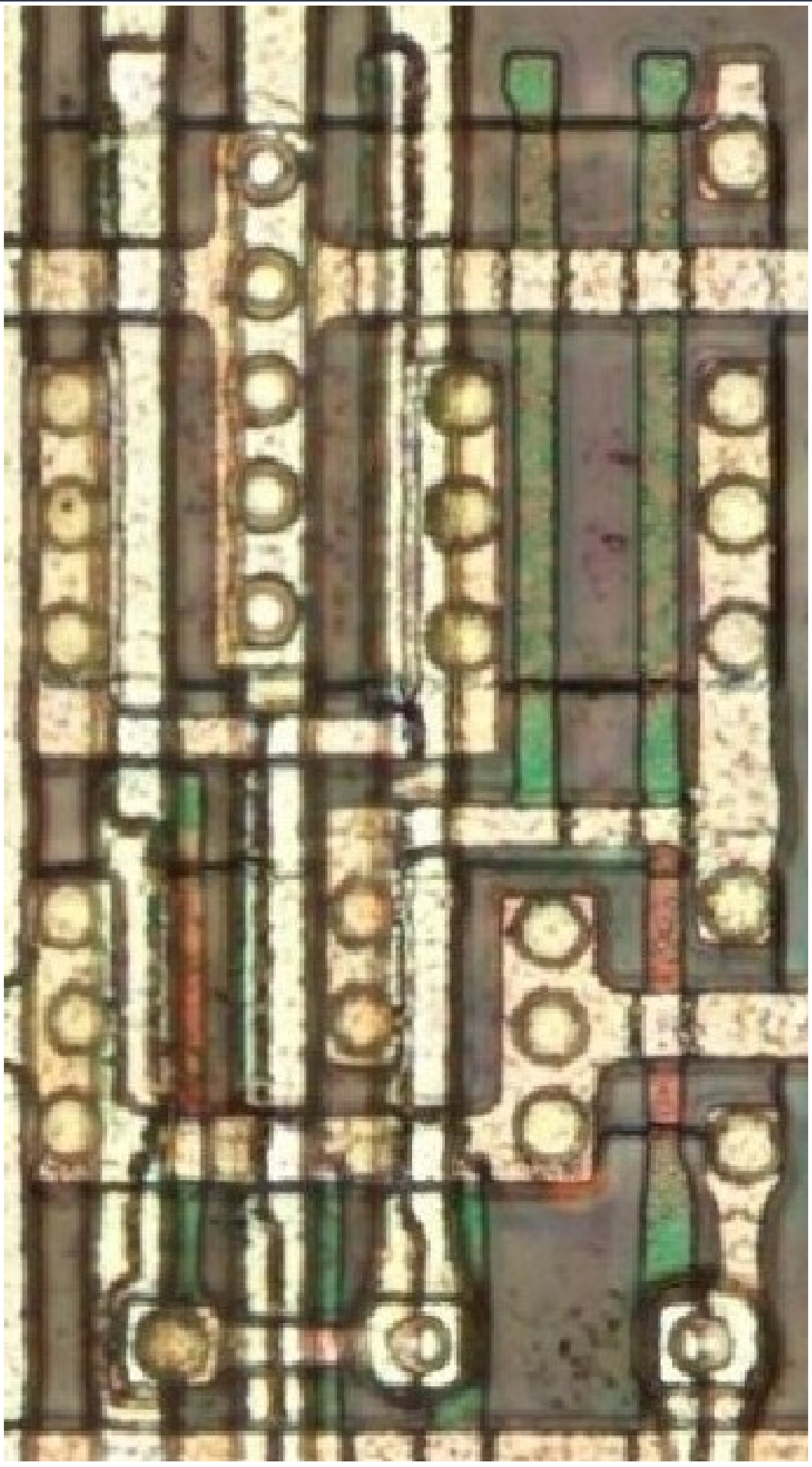




#668573ff

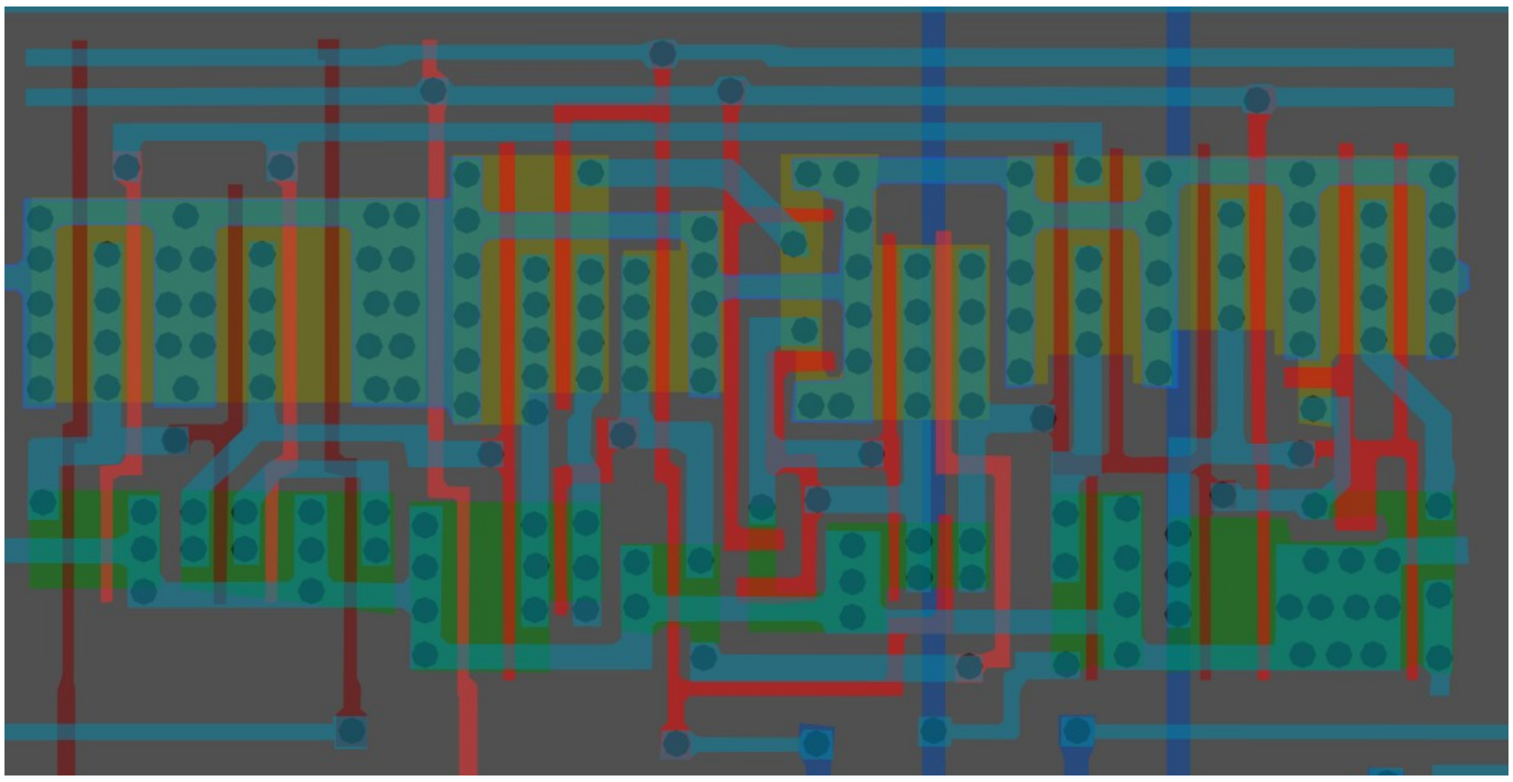
Cell type 27d

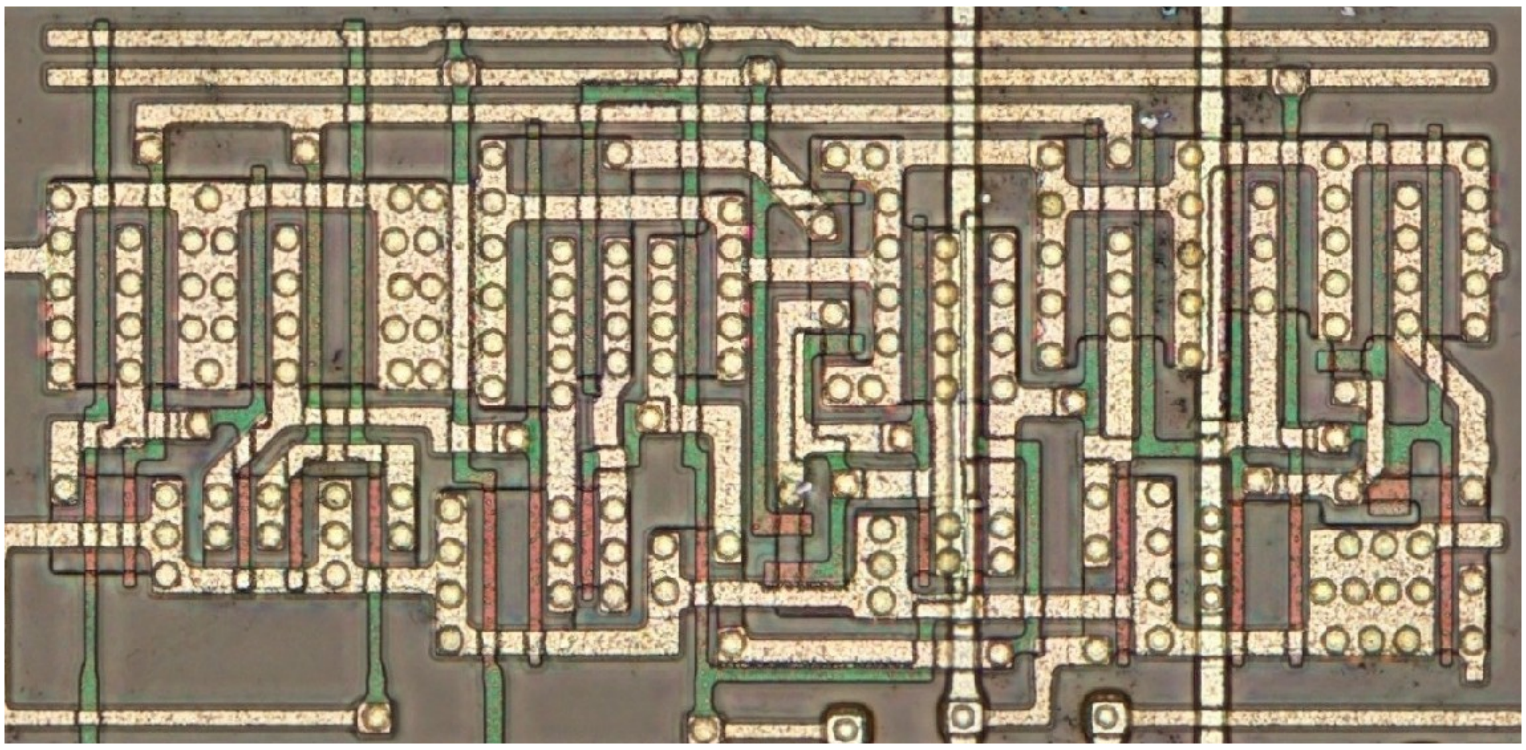




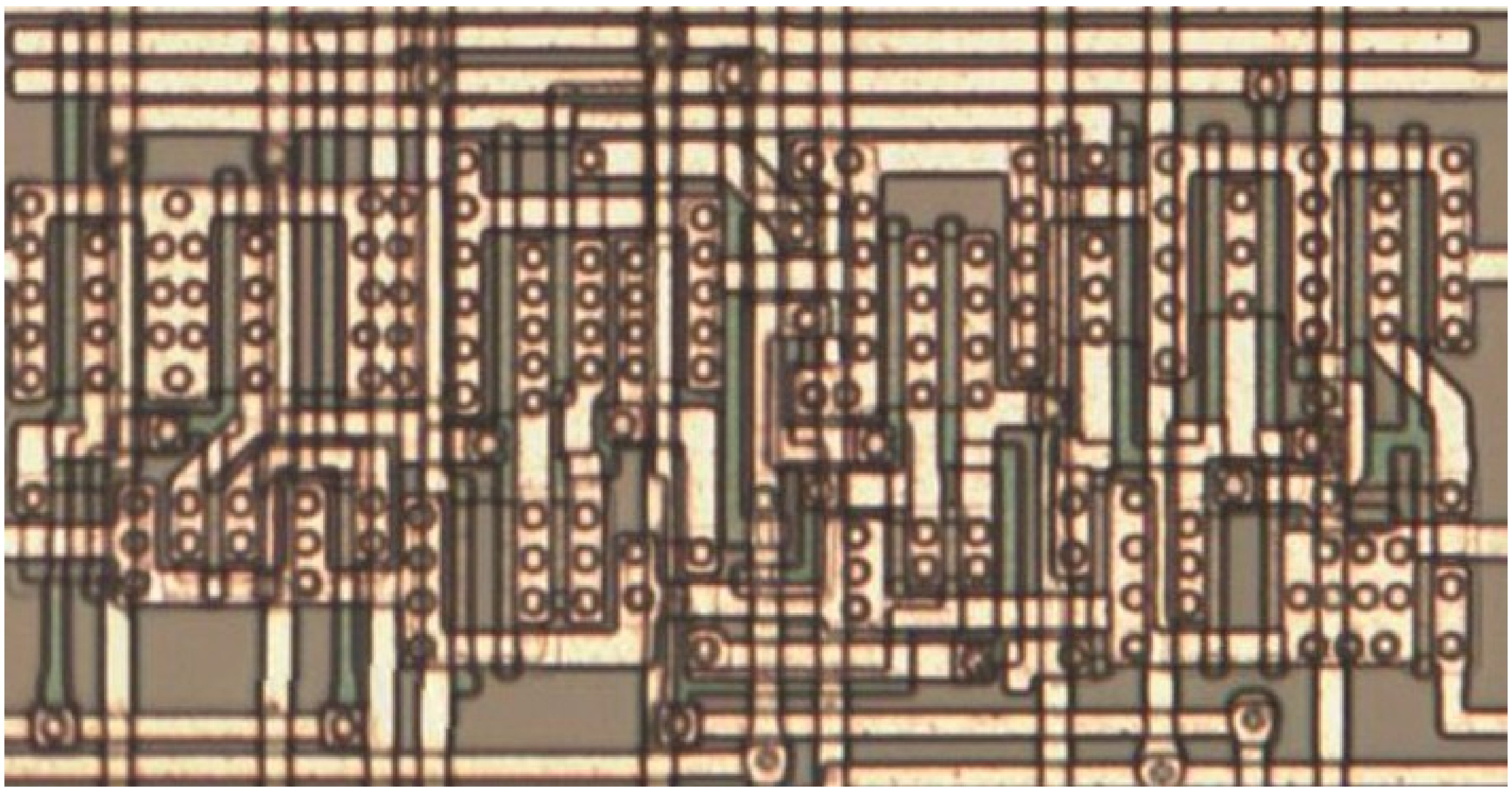
#00b75aff

Cell type 28





Same cell on LA32 (MB87136A)

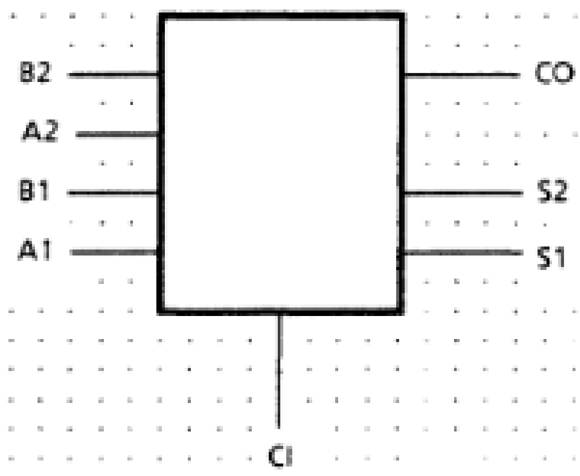


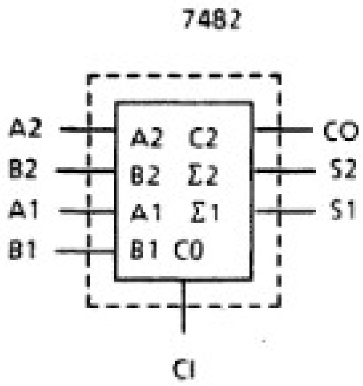
#d5546dff

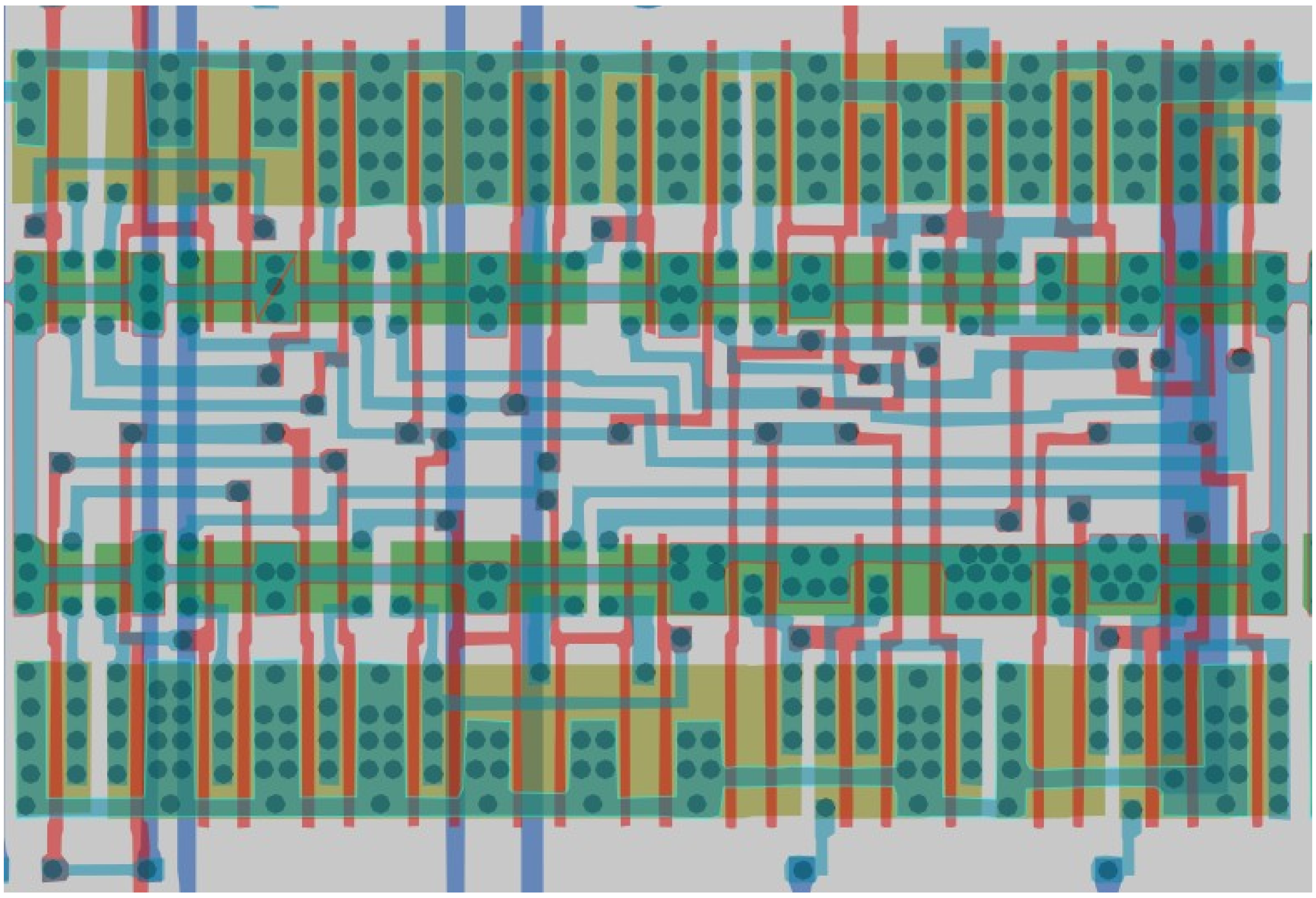
2-bit Full Adder

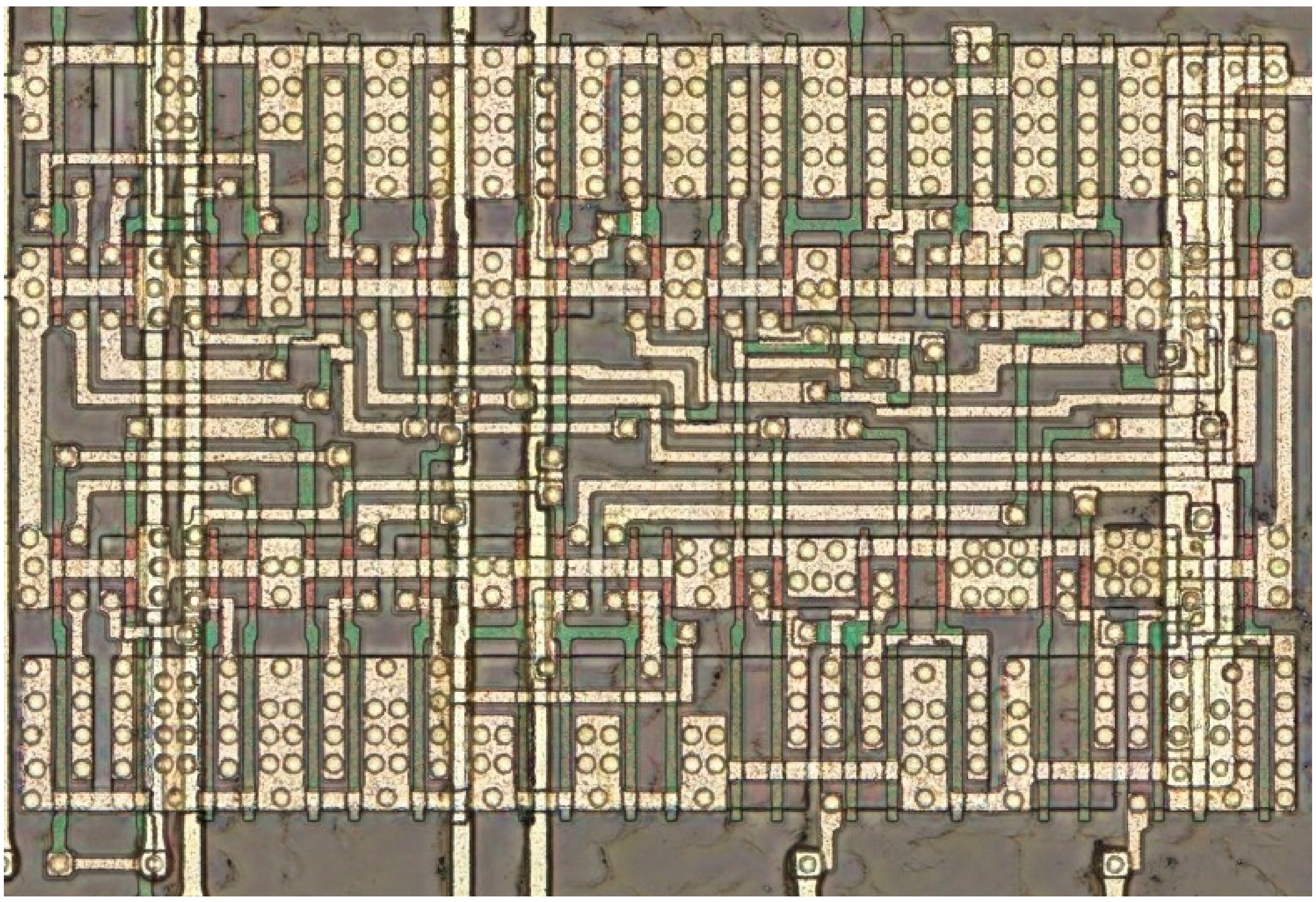
Cell 29

Inputs				Outputs					
				CI = L			CI = H		
A1	B1	A2	B2	S1	S2	$\overline{CO}$	S1	S2	$\overline{CO}$
L	L	L	L	L	L	L	H	L	L
H	L	L	L	H	L	L	L	H	L
L	H	L	L	H	L	L	L	H	L
H	H	L	L	L	H	L	H	H	L
L	L	H	L	L	H	L	H	H	L
H	L	H	L	H	H	L	L	L	H
L	H	H	L	H	H	L	L	L	H
H	H	H	L	L	L	H	H	L	H
L	L	L	H	L	H	L	H	H	L
H	L	L	H	H	H	L	L	L	H
L	H	L	H	H	H	L	L	L	H
H	H	L	H	L	L	H	H	L	H
L	L	H	H	L	L	H	H	L	H
H	L	H	H	H	L	H	L	H	H
L	H	H	H	H	L	H	L	H	H
H	H	H	H	L	H	H	H	H	H









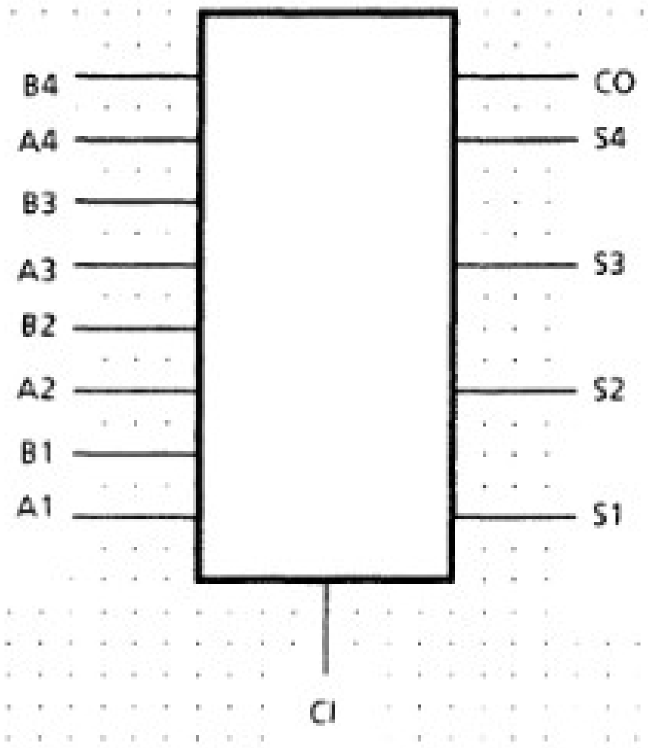
#a9546dff

4-bit Full Adder

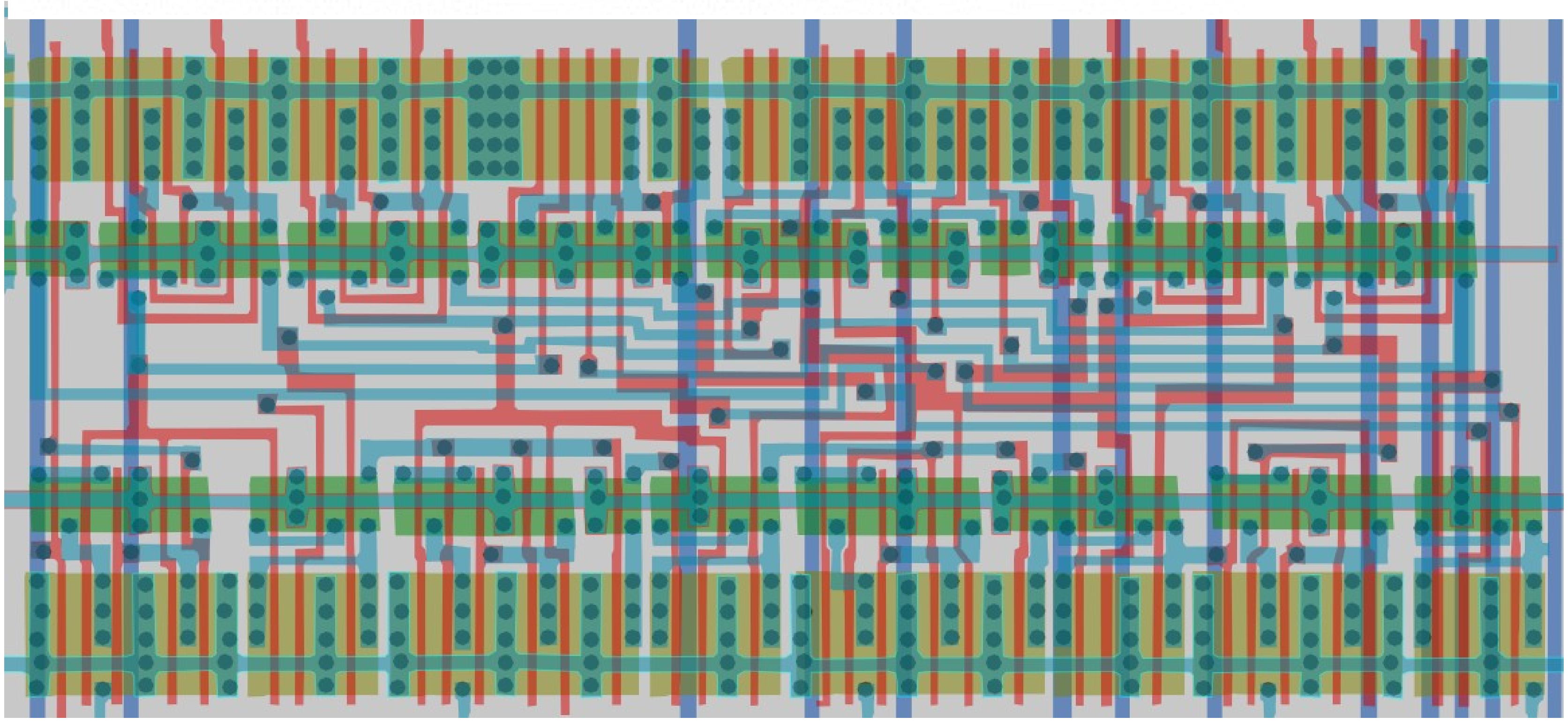
Cell 30

Standard cell only found in adder section of chip (as of now)

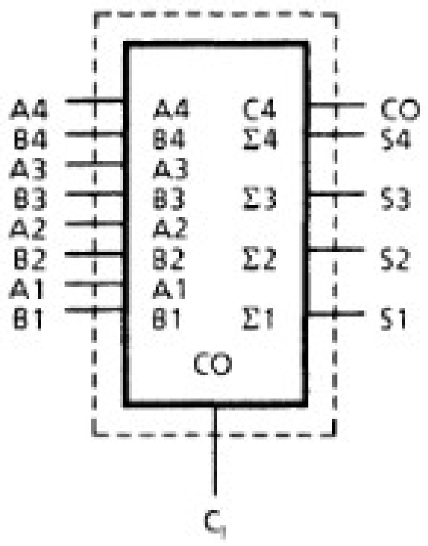
Inputs				Outputs					
				C_I = L / C_2 = L			C_I = H / C_2 = H		
A1 / A3	B1 / B3	A2 / A4	B2 / B4	S1 / S3	S2 / S4	C2 / CO	S1 / S3	S2 / S4	C2 / CO
L	L	L	L	L	L	L	H	L	L
H	L	L	L	H	L	L	L	H	L
L	H	L	L	H	L	L	L	H	L
H	H	L	L	L	H	L	H	H	L
L	L	H	L	L	H	L	H	H	L
H	L	H	L	H	H	L	L	L	H
L	H	H	L	H	H	L	L	L	H
H	H	H	L	L	L	H	H	L	H
L	L	L	H	L	H	L	H	H	L
H	L	L	H	H	H	L	L	L	H
L	H	L	H	H	H	L	L	L	H
H	H	L	H	L	L	H	H	L	H
L	L	H	H	L	L	H	H	L	H
H	L	H	H	H	L	H	L	H	H
L	H	H	H	H	L	H	L	H	H
H	H	H	H	L	H	H	H	H	H

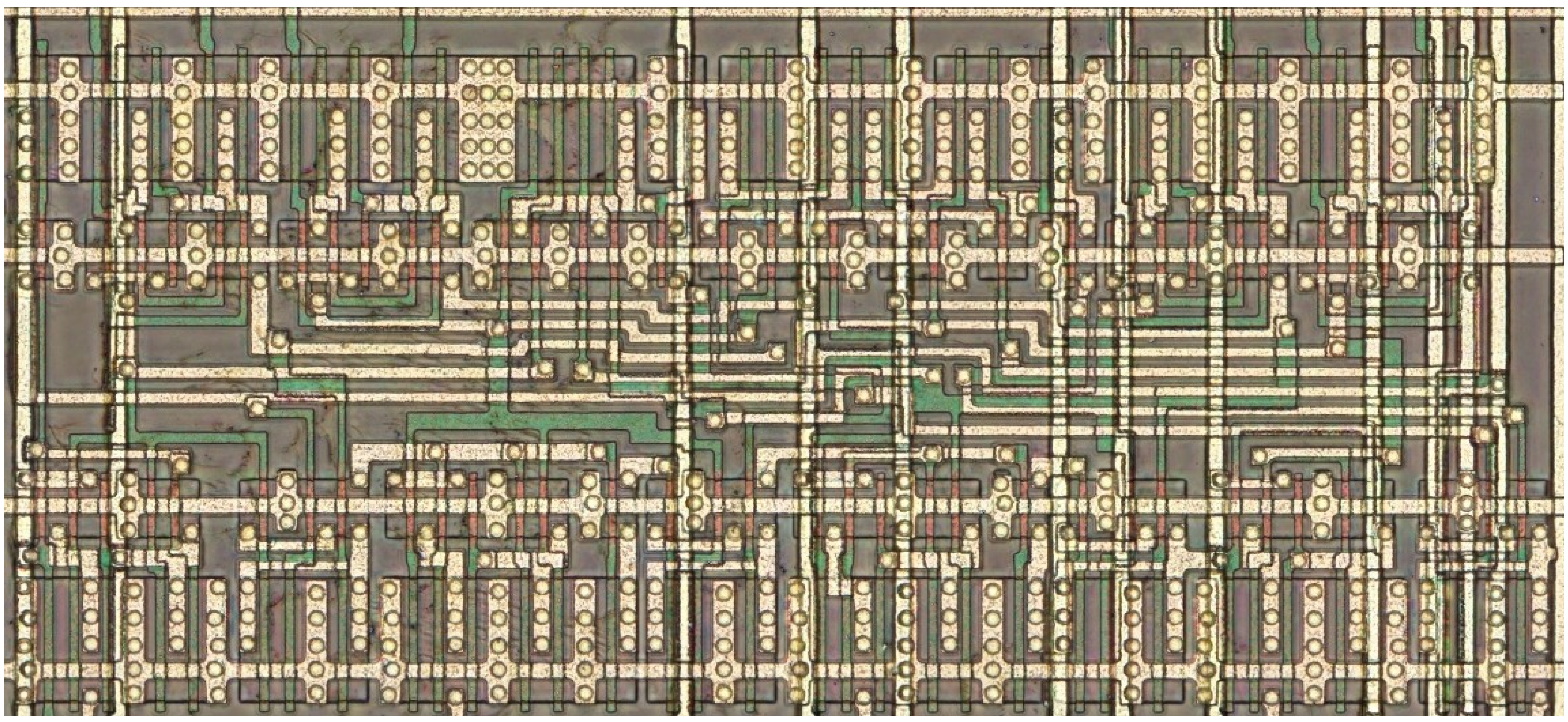


Note:
Input conditions at A1, B1, B2 and CI are used to determine outputs S1 and S2 and the value of the internal carry C2. The values at C2, A3, B3, A4 and B4 are then used to determine outputs S3, S4, and CO.



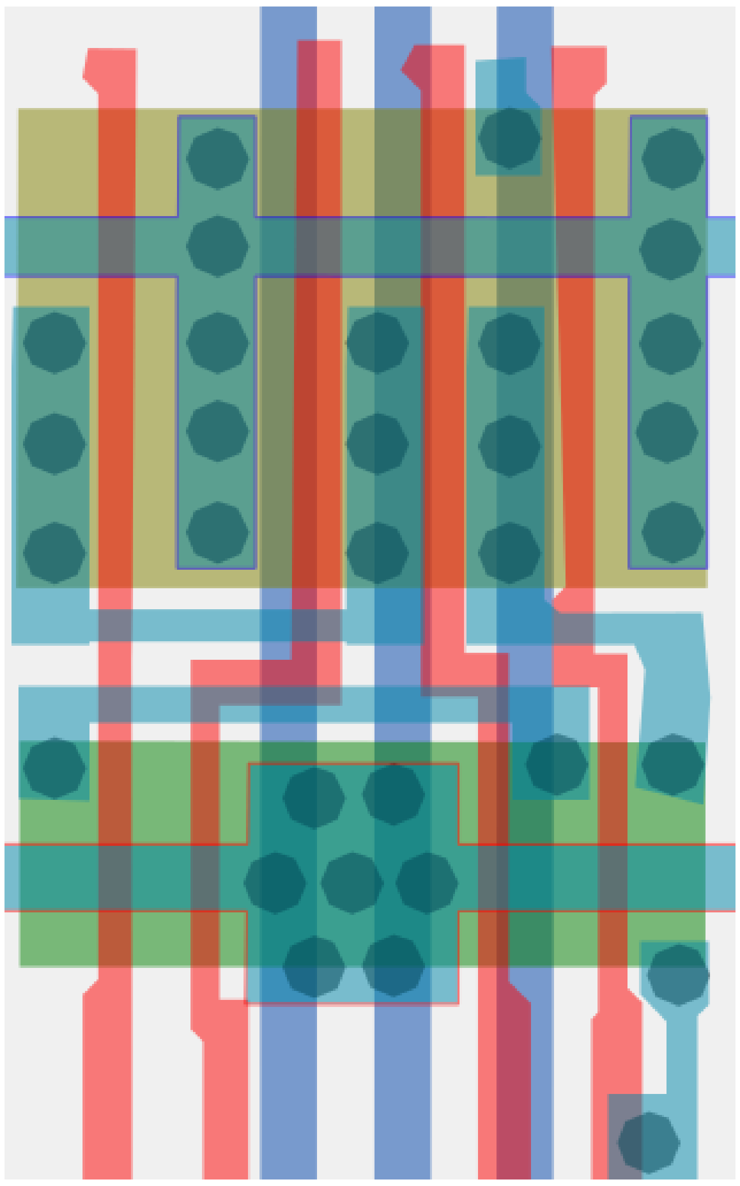
74LS83A

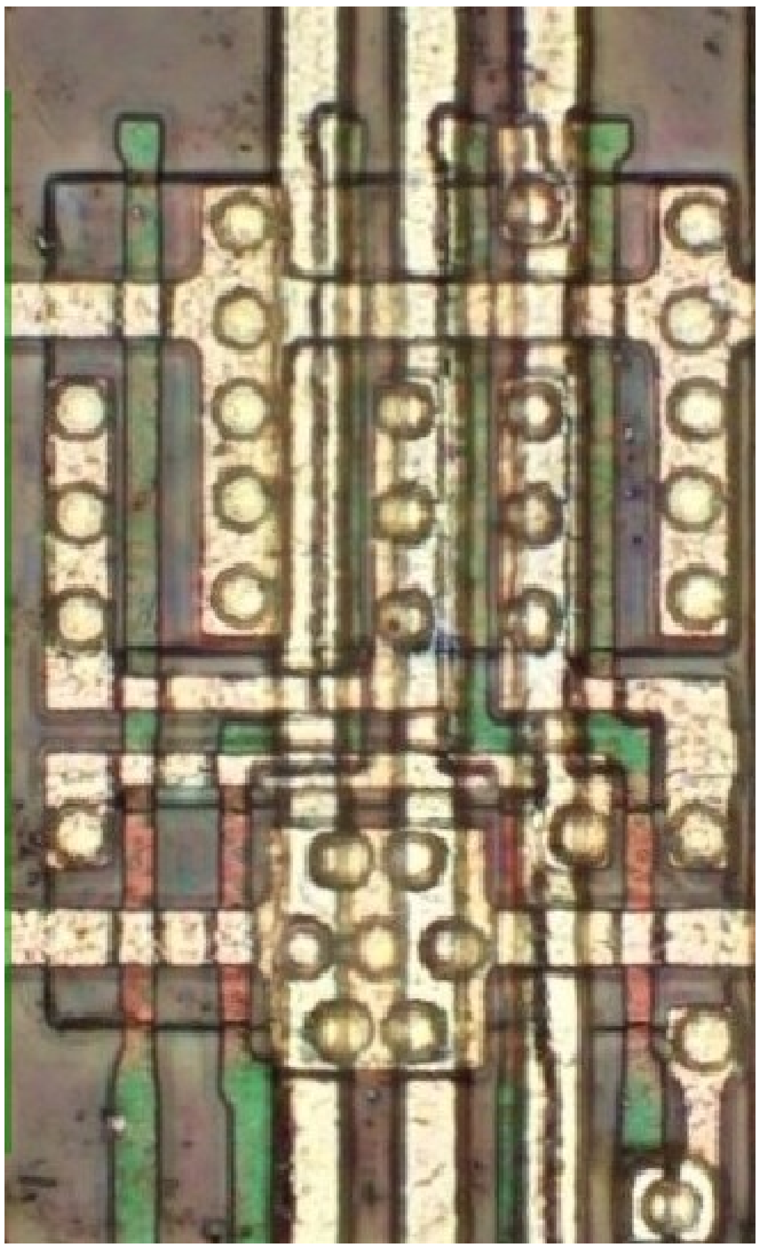




#ff4af8ff

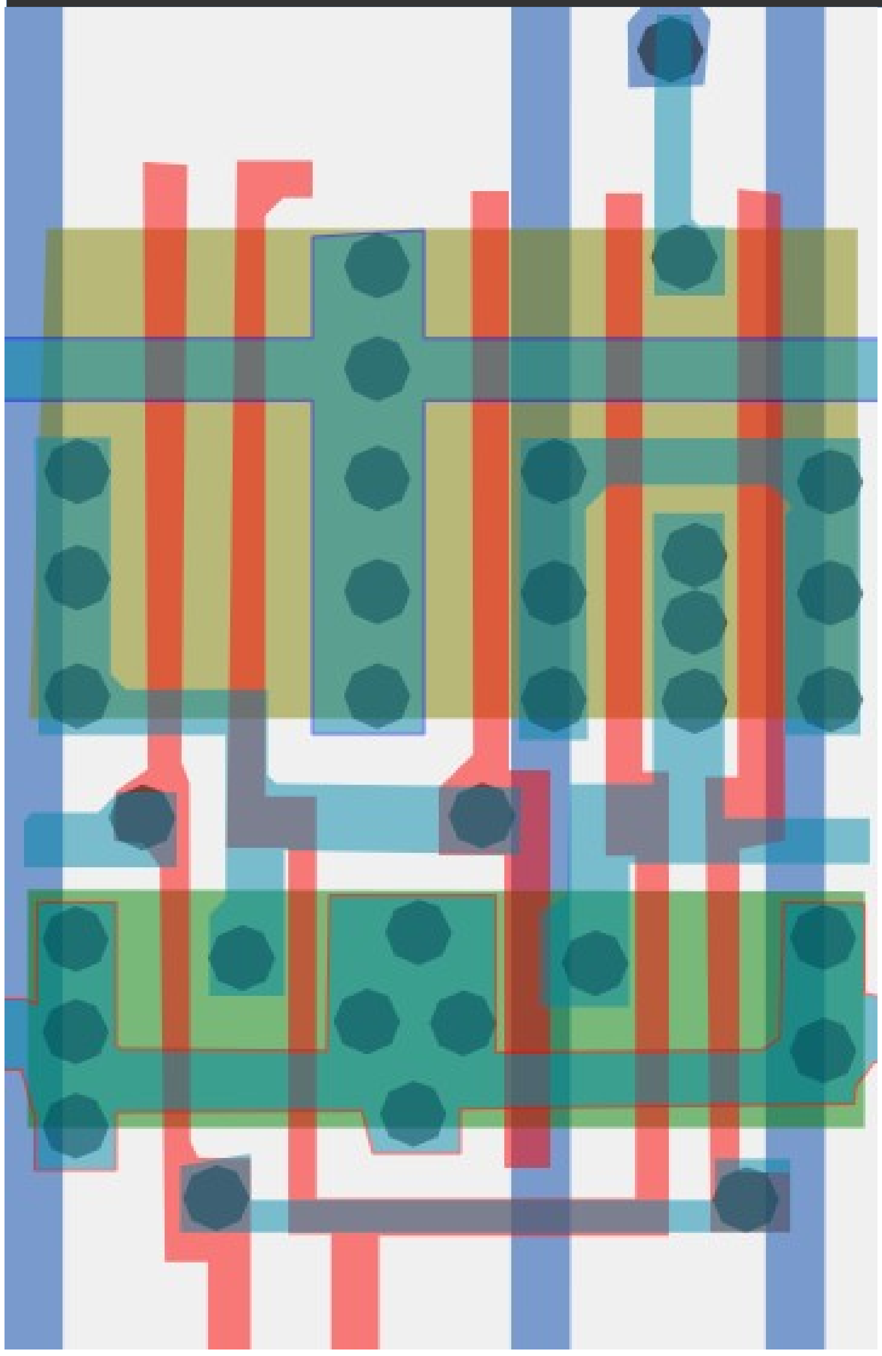
Cell type 31

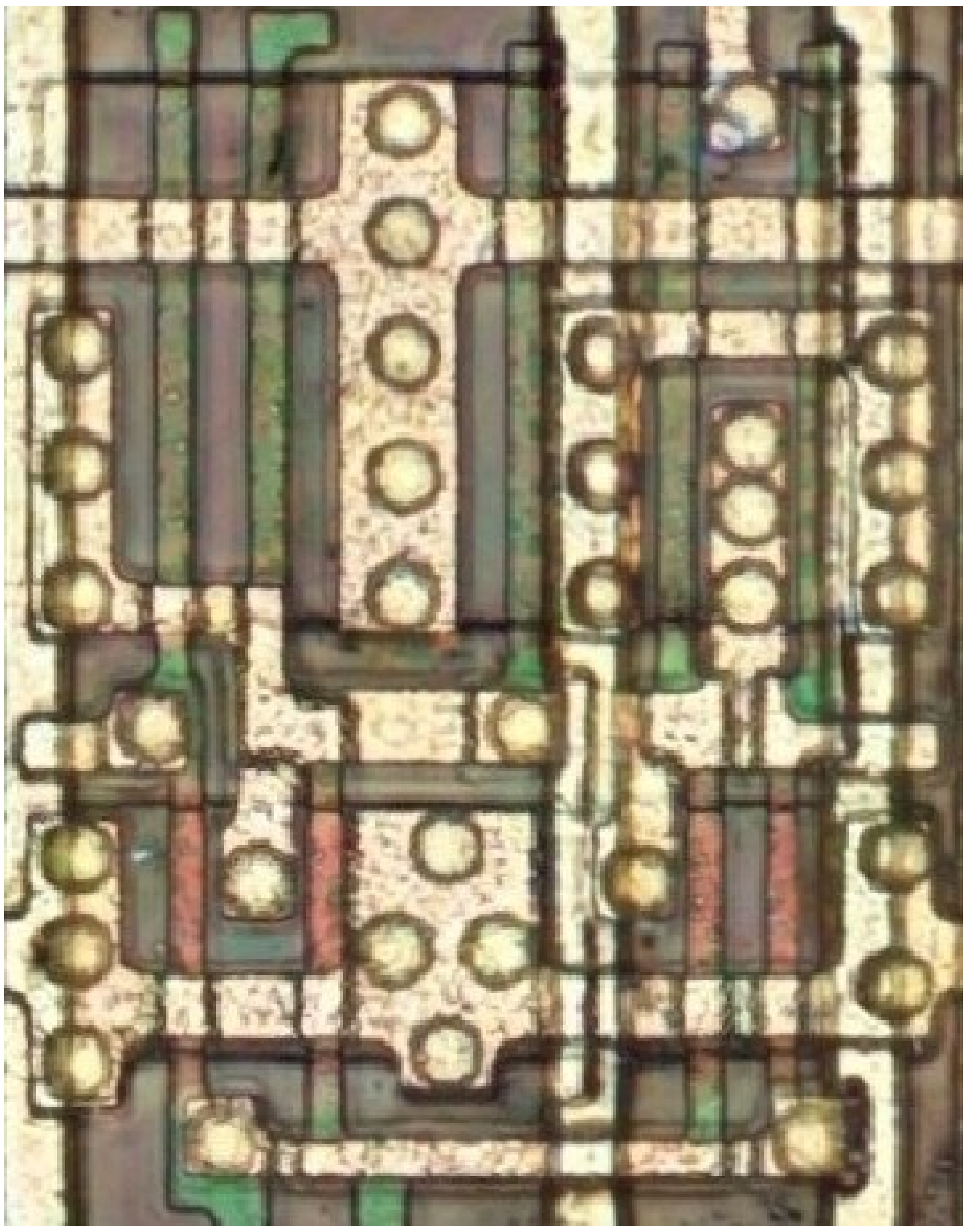




#90a05aff

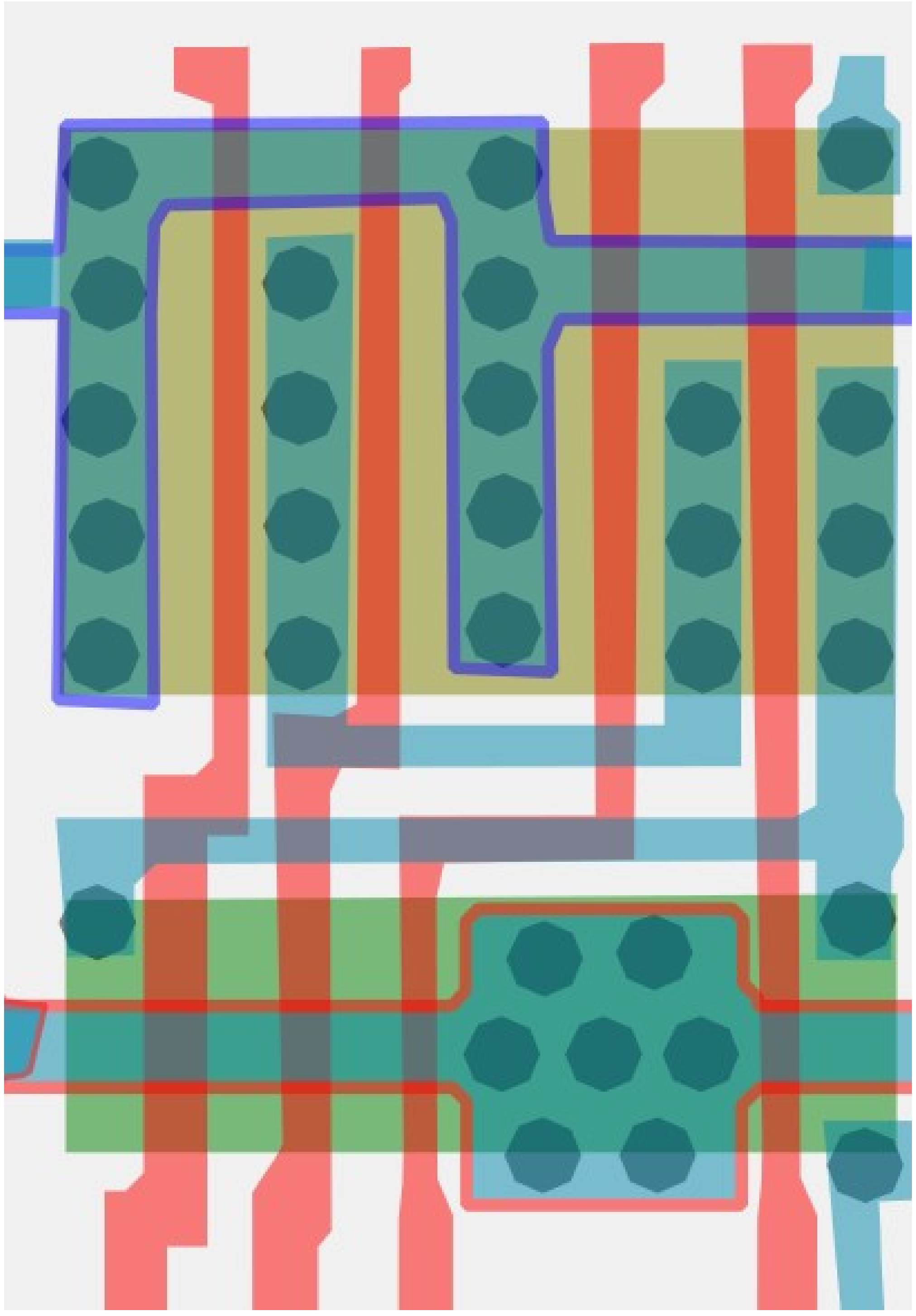
Cell type 32

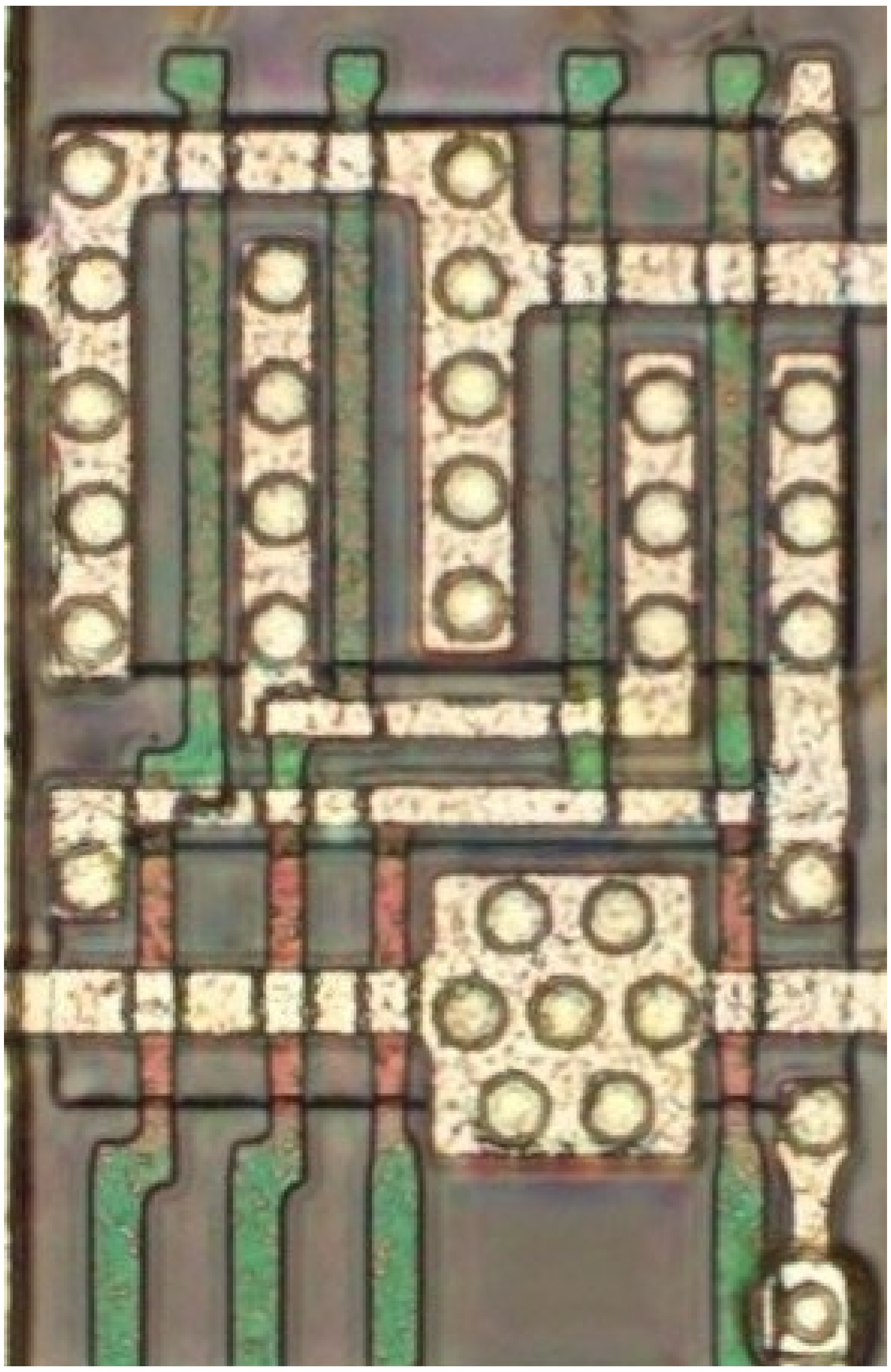




#57f330ff

Cell type 33a

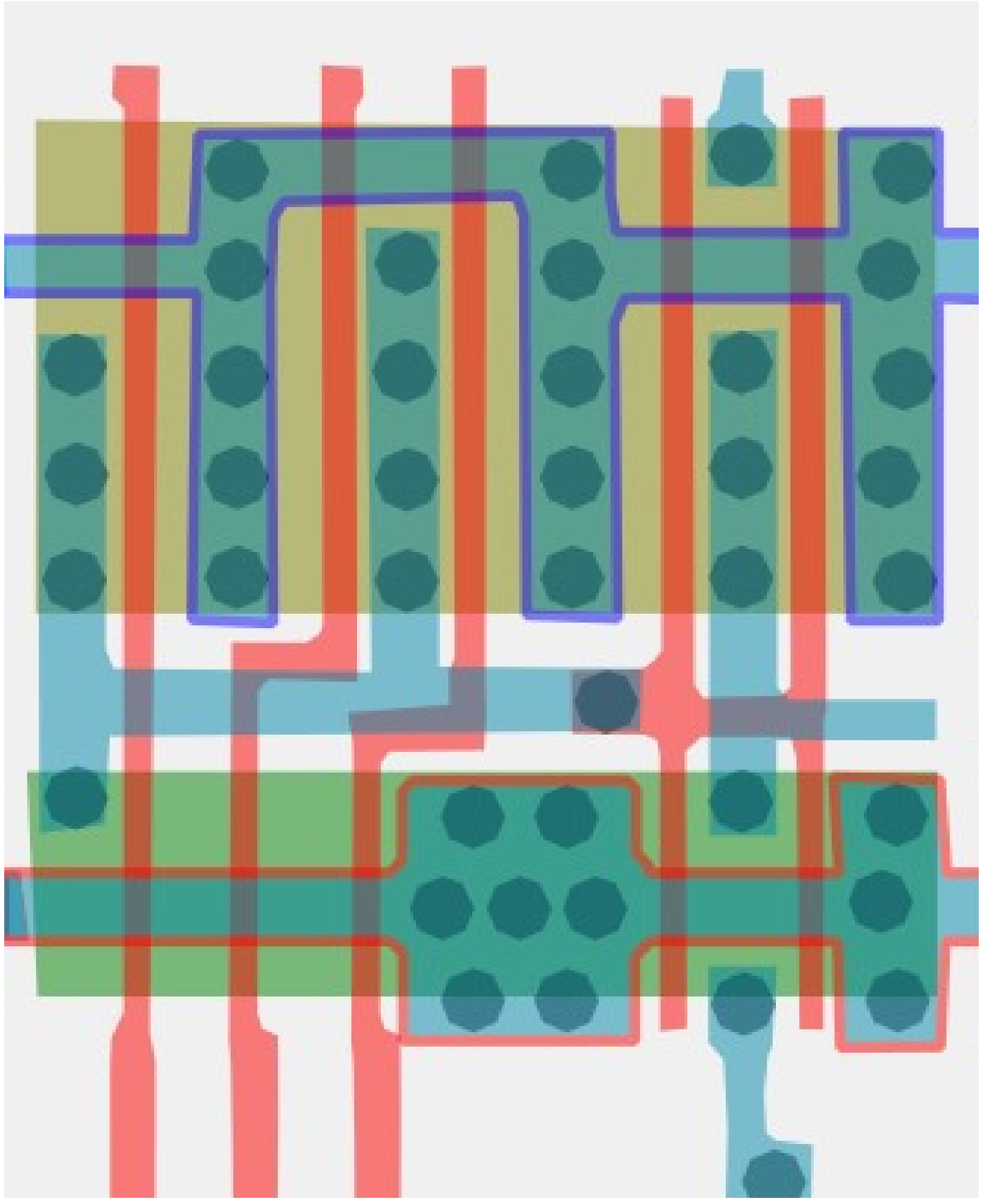


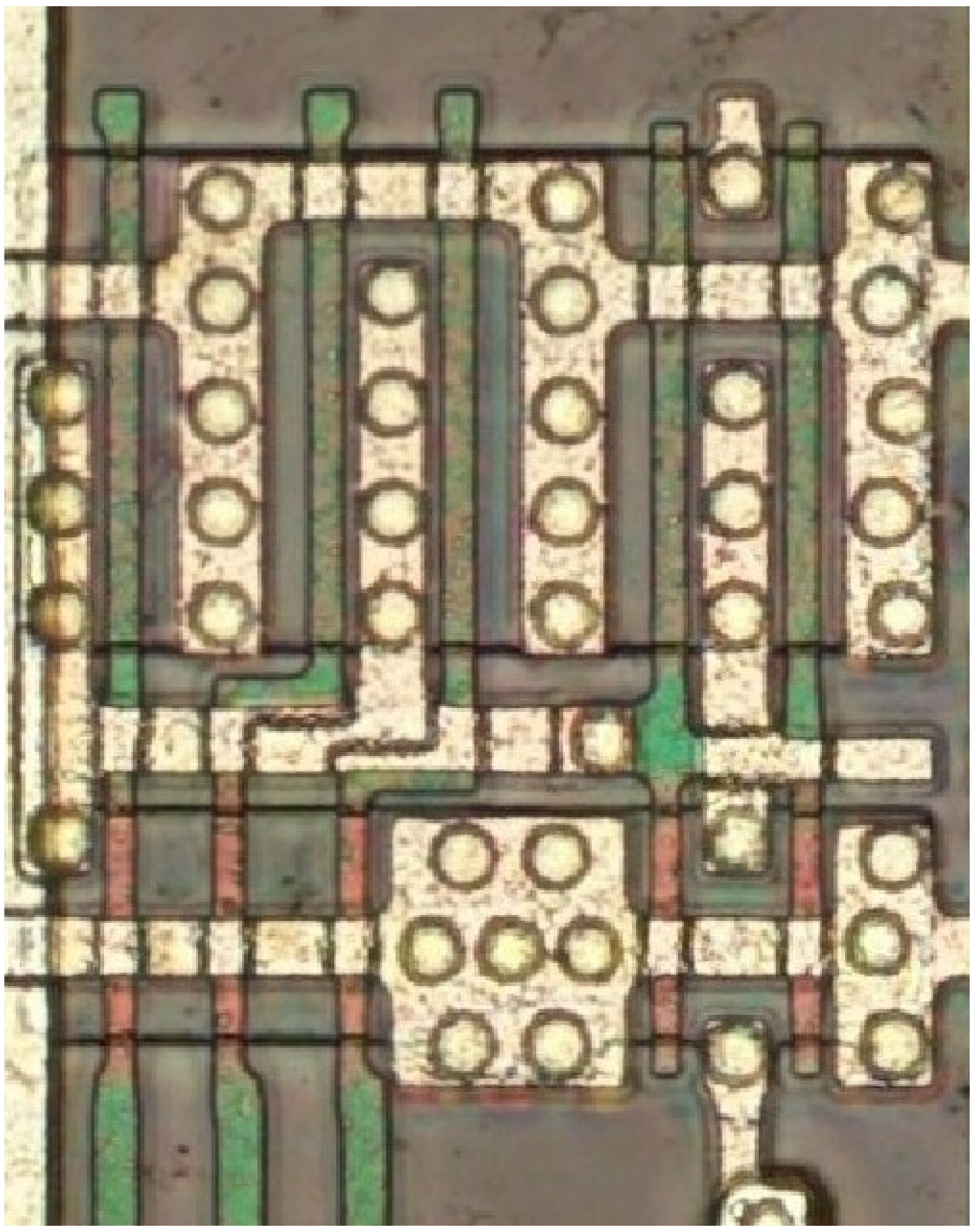


#77bd31ff

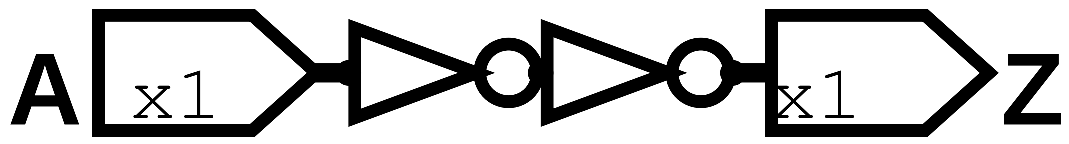
3-Input AND

Cell 33b

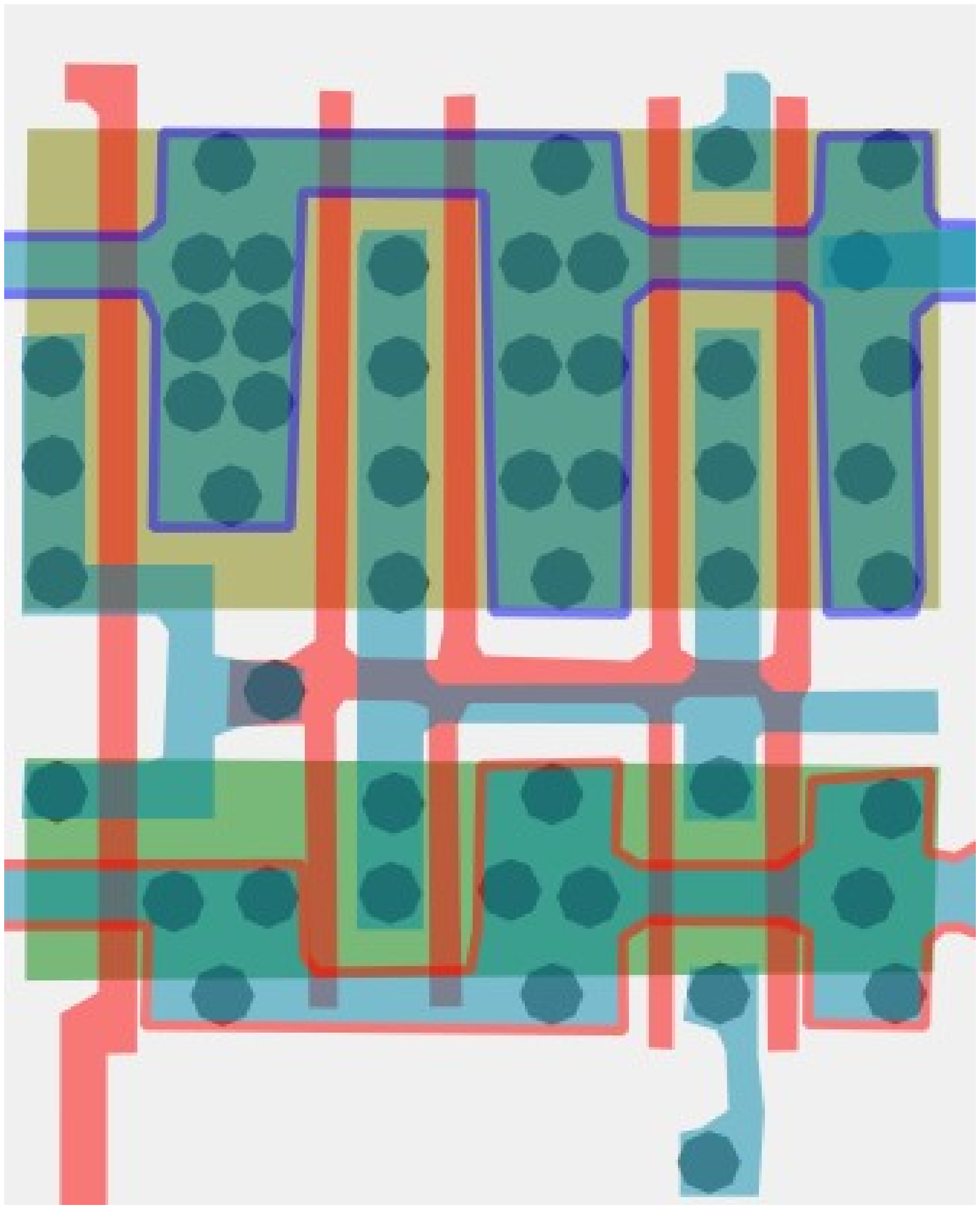


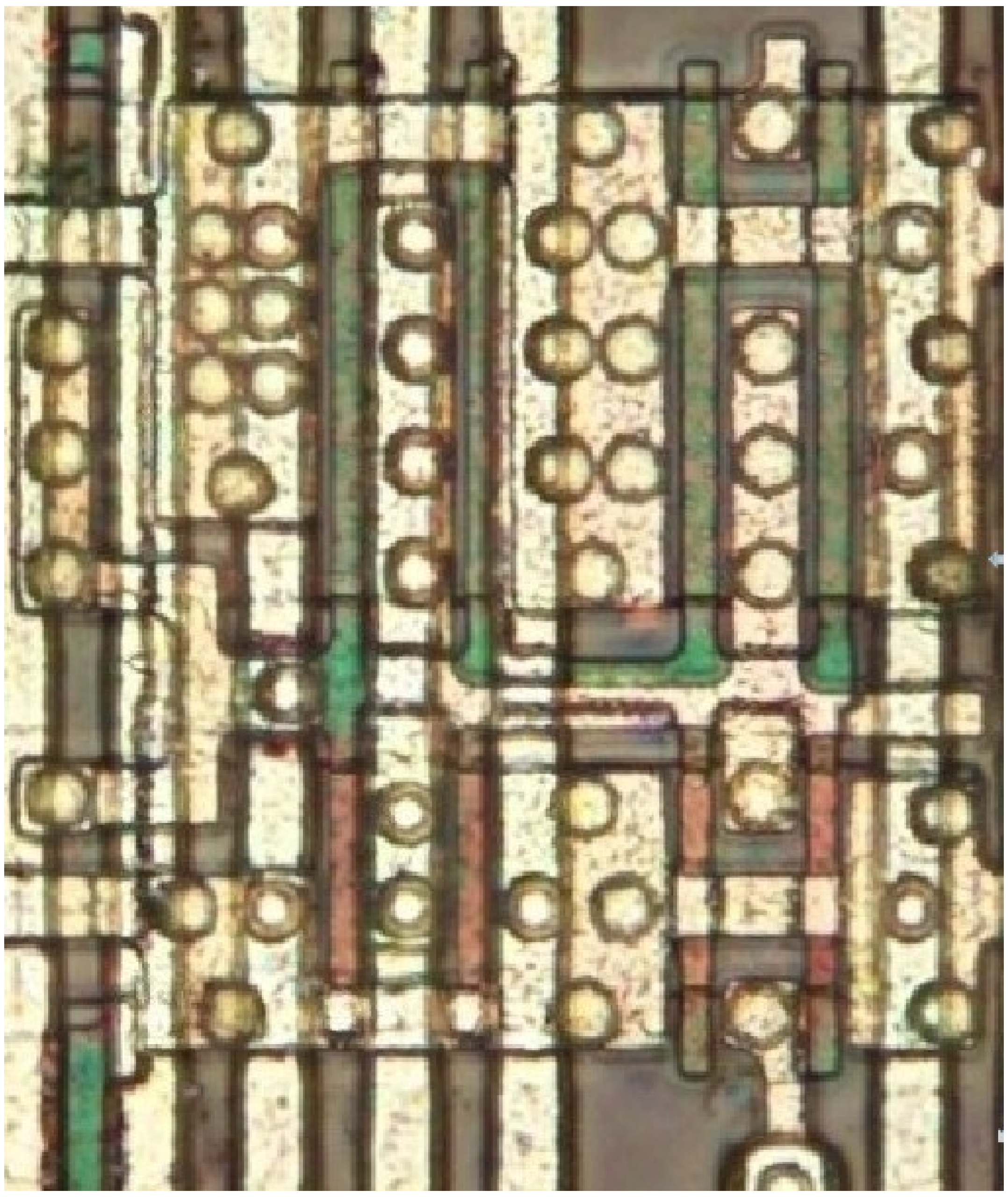


#5e58adff **Buffer** Cell 34



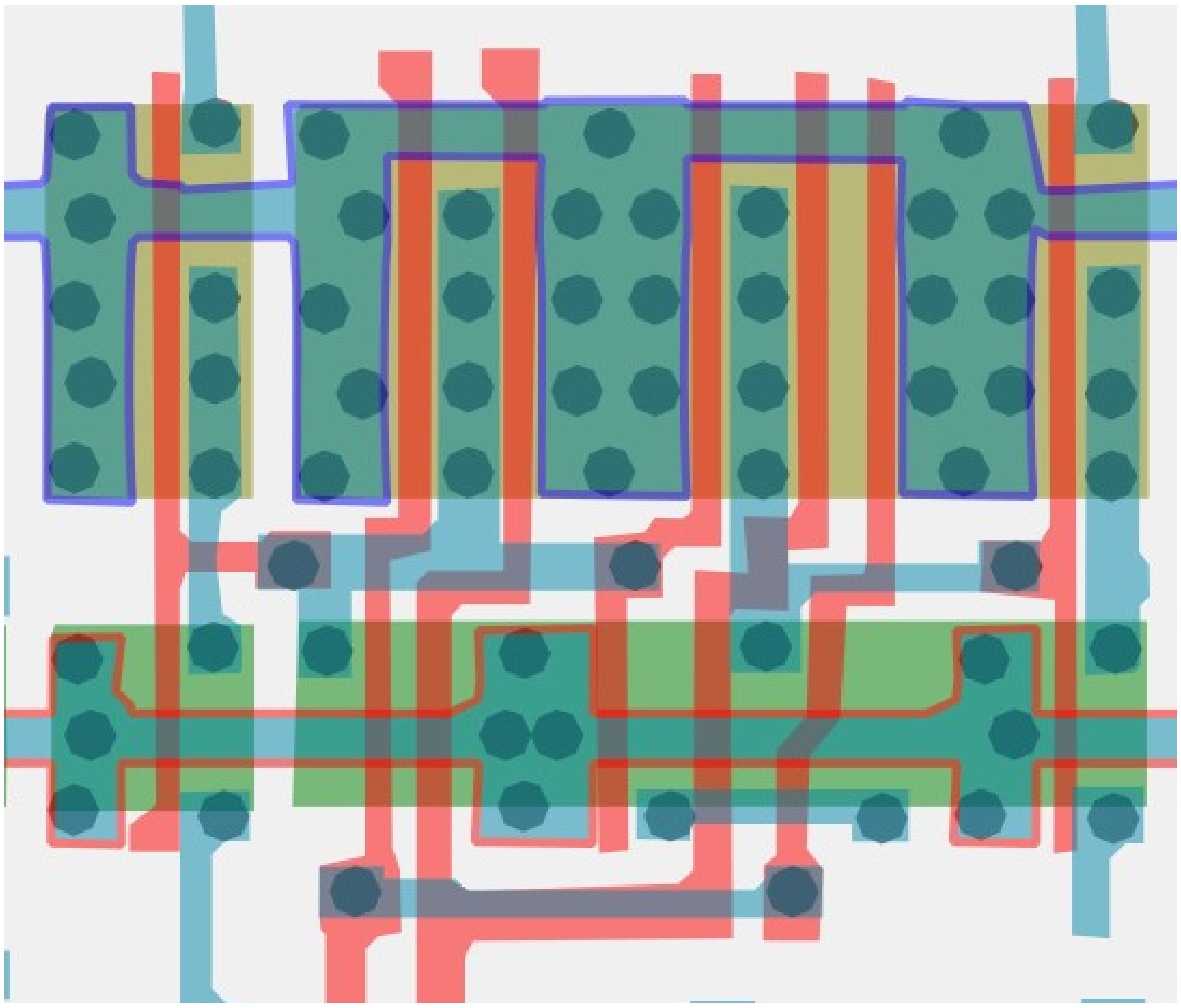
A	Z
0	0
1	1

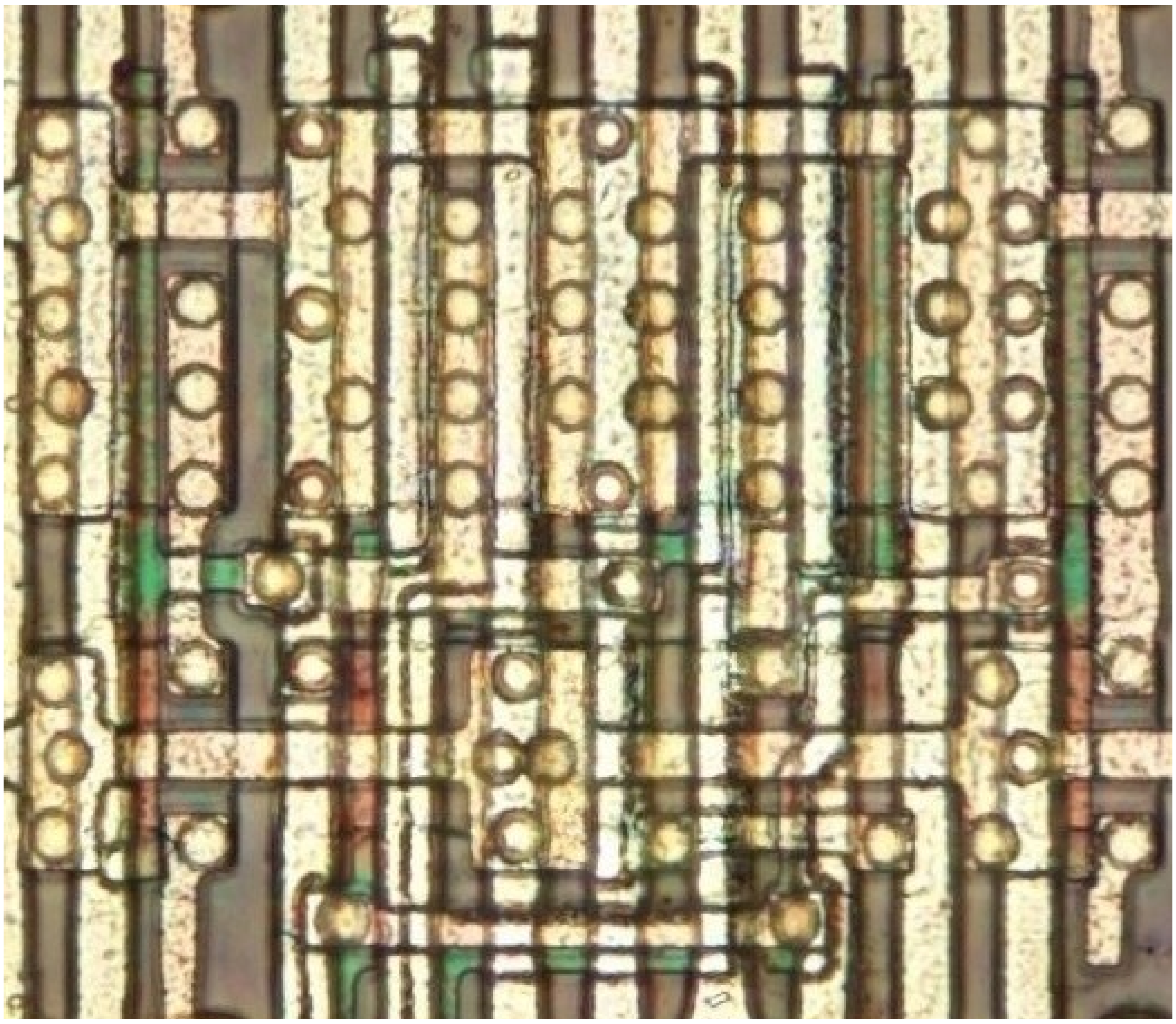




#8581c6ff

Cell type 35

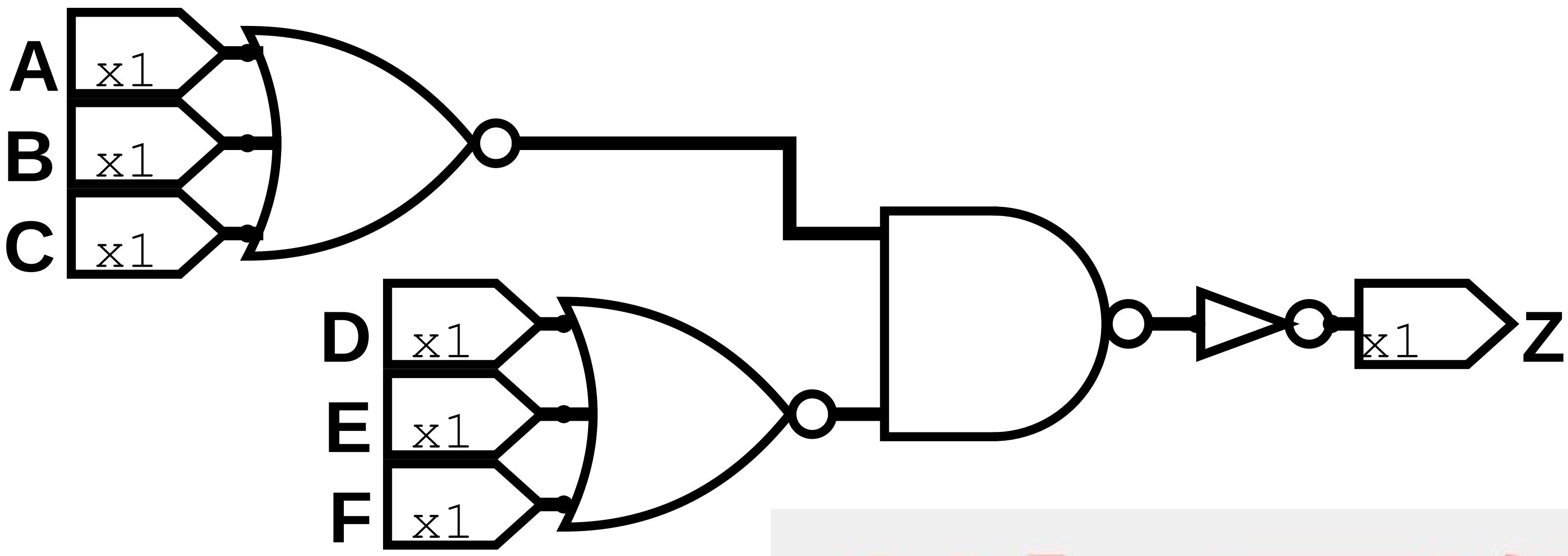


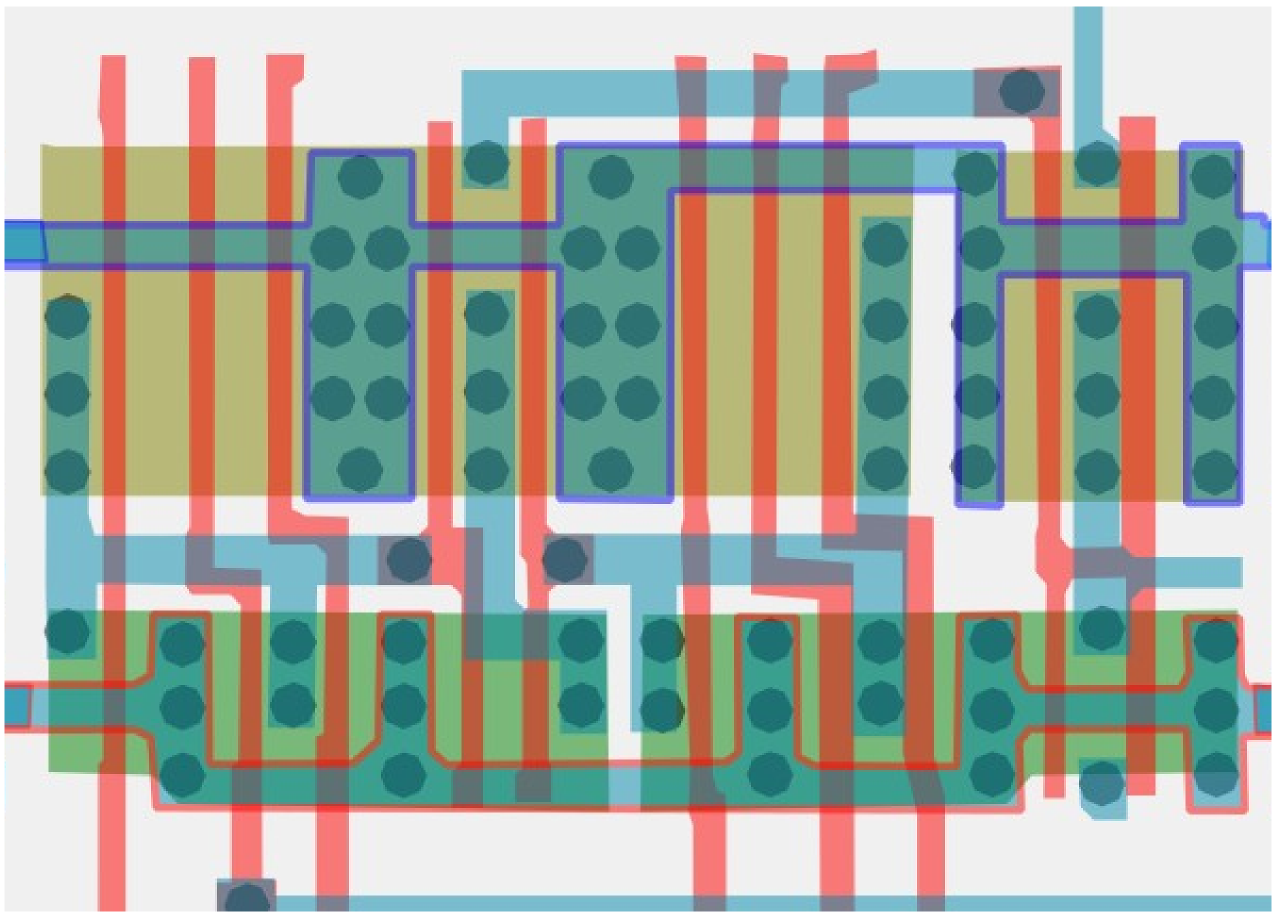


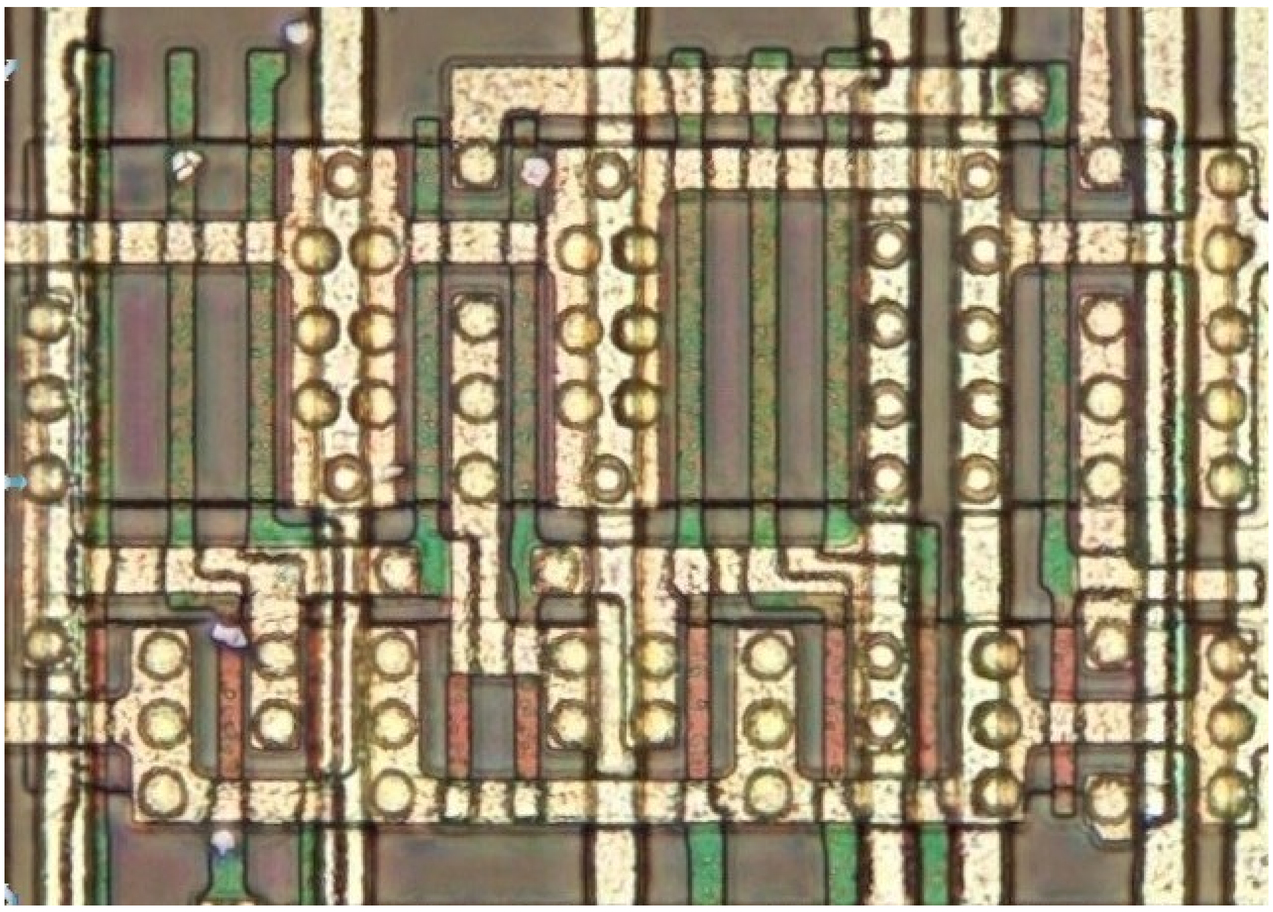
#a09dd4ff

6-Input NOR

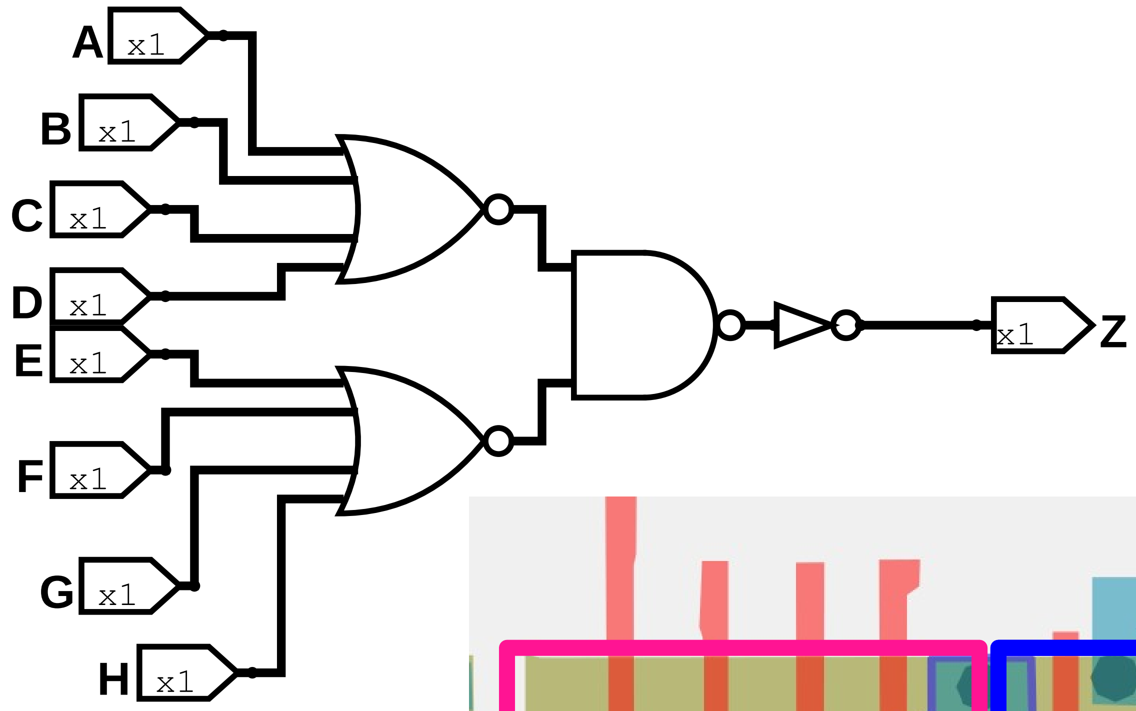
Cell 36a







#b0aedbff 8-Input NOR Cell 36b



I have verified this one in logisim